

PRIME DNA

The Odd Weil Form on a Window

The sliding-window identity, the logarithmic edge law,
and the bounded-relay conjecture

Revised after independent review

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Abstract

Let $\lambda(a)$ be the floor of the odd-sector Weil form over test functions supported in $[-a, a]$. Weil's criterion, in Yoshida's odd-test form, says that $\lambda(a) \geq 0$ for every a is equivalent to the Riemann Hypothesis; the floor is certified positive to $a = 0.8$ (Zhu) and numerically decays doubly exponentially in a . We study the floor as a function of the window. Two results are exact. First, the archimedean part of the form on a window is one half of the Dirichlet form of the logarithmic Laplacian plus a bounded form, with an explicit confining wall $\frac{1}{2} \log(1/(a - |y|))$ at the edge; by the boundary theory of the logarithmic Laplacian, a bounded minimizer is bounded by $C_0(\log(a/\delta))^{-1/2}$ at distance δ from the edge, and numerically it behaves like $C(a)(\log(a/\delta) + \beta)^{-1/2}$ with an amplitude $C(a)$ that we measure on meshes graded to 10^{-12} of the edge. Consequently the entry of a prime power into the window produces no kink in λ but a soft kink, λ' continuous with an increment of order $C^2/\log(1/\varepsilon)$ at scale ε , which we verify over four decades. Second, between entries the floor obeys the sliding-window identity

$$a \lambda'(a) = - \left[D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f) \right],$$

proved for the K -mode form and, for the exact form, as a two-sided Dini-derivative statement holding at every point of differentiability, where f is the minimizer, D_∞ is the archimedean dilation form, whose symbol tends to 1, $D_P(f) = - \sum_n 2\Lambda(n)n^{-1/2} \log n g'_f(\log n)$ is the Euler product differentiated against the derivative of the autocorrelation of f at the prime-power positions, and P, M are the polar moments. We evaluate its three terms in ball arithmetic on the K -mode minimizer and confirm the identity independently on the Fourier side; three order-one quantities cancel to the size of λ . The floor is continuous and non-increasing, and under a bounded-variation hypothesis on near-minimizers it is locally Lipschitz; every minimizer computed, on meshes graded to 10^{-12} of the edge and with up to 160 modes, has total variation exactly $4 \sup |f|$, being unimodal on each half window, so the regularity clause of the conjecture below reduces to that hypothesis. A third statement is numerical: on seven supports, with and without primes, $\lambda'(a) = -2C(a)^2$ to a part in a thousand (an eighth, at an entry, to 1.7%), the Hadamard formula one expects for this window problem, so that $\Phi' = 2C^2/\lambda$ with $\Phi = -\log \lambda$. We prove this boundary law under the edge law with two derivatives, as a Pohozaev identity whose boundary term is the overlap of the leak of the dilation generator with the singular force outside the edge; and for the archimedean form, without primes, we prove the edge law itself from an exact Wiener–Hopf representation of the critical point: the edge of its transform is

$A_0/(t\sigma_-(t))$ with σ_- the outer factor of the symbol, the profile is $|A_0|(\Psi_\infty(1/\delta) - \lambda - \gamma)^{-1/2}$ with Euler's constant γ , verified to a part in a thousand with nothing fitted, and the law $\Phi' = \pi T_{\text{eff}}$ stated below is exact for that form; with the prime 2 inside the window the constants are unchanged, the edge is accompanied by its echo, and the law of the zeros is a theorem under RH for $a_2 < a < a_3$. At every support below $a_\infty = 0.843$, where the walks that the entries generate inside the window are confined, the echoes form a finite tree, the symbol of the edge is the Schur complement of the tree's adjacency, a finite continued fraction along each chain, the constants of the edge are unchanged and the law of the zeros is a theorem under RH; beyond a_∞ the echo set is dense, the representation with the window's symbol remains exact, and the onset of the law is set by the band of the confined echo lattice, whose spectral radius is a third of the periodic limit's: the law holds from five horizons on inside the region where the symbol of the window still changes sign. By the explicit formula, the defect of the zeros above a height against their mean density is the prime sum at the entries, the oscillating part of the symbol, equal to twice the square of the echoes' amplitudes averaged over the tail and verified on the zeros to a few parts in a million; so the two sampling hypotheses reduce to one statement about the minimizer alone, that the sharp tail law holds at a bounded multiple of the horizon. The derivative of the floor is a property of the ground state alone: the pencil $(-Q', Q)$ has eigenvalues of size $6 \times 10^{10} T^*$ at $a = 1$ while the minimizer's direction gives $3.7 T^*$, so no inequality between the two forms can hold. With the horizon $T^*(a) = 2\pi e^{2a}$, finite-mode upper bounds on the floor from height 17 to height 130 (down to $\lambda \approx 10^{-93}$, with up to 230 modes and rigorous enclosures of the truncated matrix's lowest eigenvalue beyond $a = 1$) give Φ'/T^* rising from 3.2 to 3.9 on coarse grids, and the Hellmann–Feynman derivative resolves a sawtooth at every prime-power entry: a rise into the entry, of 10 to 45% over the last 0.003, as the form without the entering prime heads toward indefiniteness, a drop across it, and a relaxation after, with peaks 3.3 at the entry of 2 and 4.1–4.3 at the entries of 3, 4, 5, 7. We state the bounded-relay conjecture, λ locally absolutely continuous with $\lambda' \geq -cT^*\lambda$ almost everywhere, which implies the Riemann Hypothesis by Grönwall's inequality from Zhu's certified support, and the decay law $\Phi'/T^* \rightarrow c_0$; the data exclude the raw phase-space count, disfavour the logarithmic law and the derivative of Zhu's even-sector Landau–Widom law, and favour a constant near 3.8. Evaluated on the first 6000 zeros of ζ , the sum of $|F(\gamma)|^2$ over the zeros returns the floor of the K -mode minimizer to 3×10^{-4} or better at $a = 0.6, 0.8, 1.0$, down to $\lambda = 10^{-26}$, and the zero-side dilation sum returns its derivative to 3×10^{-3} or better: the explicit formula closes on the actual zeros for both, a third and independent measurement of the identity after the matrices and the Fourier symbol. The zeros below the horizon carry at most 10^{-5} of the floor, the first zero 10^{-24} of it at $a = 1$, and the floor is made by the zeros between one and a half and three horizons. The fraction of the floor carried by the zeros above height T is $\Phi'/(\pi T)$ from a few horizons on, on every minimizer computed (edge-FEM to 1% over two decades of height, K -mode to 1–3% down to $\lambda = 10^{-93}$): the rate of decay of the floor is π times the effective height of the zeros that carry it, $\Phi' = \pi T_{\text{eff}}$, and the two conjectures say that this height is tied to the horizon. Under the Riemann Hypothesis the floor is a sampling constant for the zeros above the horizon, and a prolate-type count bounds Φ below by about half of what is observed. Nothing in this paper proves the Riemann Hypothesis. Theorems, computations with their precision, and conjectures are distinguished throughout.

Standalone research paper. Companion to Primes, Folds, and the One Dot and to Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas. Revised after an independent referee report; the report's corrections are incorporated and acknowledged in Section 9.

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Status conventions

A **Theorem**, **Proposition** or **Lemma** without citation is proved in the text. A statement with a citation is taken from the literature and its hypotheses are stated. A **Computation** is a numerical result with its method, precision and cross-checks; a finite-mode minimum is a minimum over a subspace and hence an upper bound on the floor, and every such value is labelled by its mode count K ; derivatives of finite-mode floors are Hellmann–Feynman derivatives of the K -mode form and are called so, never “exact”. A **Conjecture** is a precise statement with the evidence for it. A **Heuristic** or **Remark** makes no mathematical claim. The hypothesis that a minimizer is bounded, needed for the edge law, is written (H_∞) ; the hypothesis that near-minimizers have bounded sup norm times total variation, which gives Lipschitz continuity of the floor, is written (H_{BV}) ; both are carried explicitly wherever they are used.

1 The form, the floor and the horizon

For $f \in L^2(\mathbb{R})$ real with compact support write $F = \hat{f}$, $F(t) = \int f(x)e^{itx} dx$, and $g = f \star \tilde{f}$, $g(u) = \int f(x)f(x-u) dx$, so that $\hat{g} = |F|^2$. The Weil form in the normalization used throughout is

$$Q(f) = W_\infty(g) - \sum_{n \geq 2} \Lambda(n)n^{-1/2}(g(\log n) + g(-\log n)) + (\hat{g}(i/2) + \hat{g}(-i/2)), \quad (1)$$

$$W_\infty(g) = -(\gamma + \log \pi)g(0) + 2 \int_0^\infty \frac{g(0)e^{-2x} - g(x)e^{-x/2}}{1 - e^{-2x}} dx. \quad (2)$$

By the explicit formula [1], $Q(f) = \sum_\rho \hat{g}((\rho - \frac{1}{2})/i)$ over the nontrivial zeros; on the geometric side, in Fourier form,

$$Q(f) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \Psi(t) dt + \text{polar}, \quad \Psi(t) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi - \sum_n \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n), \quad (3)$$

where the polar term is $2F(i/2)^2$ for even f and $-2P(f)^2$, $P(f) = \int f(x) \sinh(x/2) dx$, for odd f ; the multiplier $\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ is that of W_∞ by the integral representation of ψ . This

is the normalization of Zhu [6] and of Checkpoints 15–20 of [20]; Computation 2.1 confirms the identification numerically.

Definition 1.1. For $a > 0$ the *odd floor* is

$$\lambda(a) = \inf\{Q(f) : f \text{ odd, } \text{supp } f \subset [-a, a], \|f\|_2 = 1, Q(f) < \infty\},$$

$\Phi(a) = -\log \lambda(a)$ wherever $\lambda(a) > 0$, and the *horizon* is $T^*(a) = 2\pi e^{2a}$. $N(T)$ denotes the number of zeros with $0 < \gamma \leq T$.

Since g is supported in $[-2a, 2a]$, only the prime powers with $\log n < 2a$ enter $Q(f)$; the *entries* are $a_n = \frac{1}{2} \log n$. The form Q itself does not depend on a , so λ is non-increasing.

Theorem 1.2 (Weil [1]; Yoshida [2], Prop. 1, as recorded in Suzuki [4]). *The Riemann Hypothesis holds if and only if $Q(f) \geq 0$ for every compactly supported smooth f ; and $Q(f) \geq 0$ for every odd such f already implies it. Hence $\text{RH} \iff \lambda(a) \geq 0$ for all $a > 0$.*

Theorem 1.3 (Zhu [6], Theorem 6.2). *For $a = 0.8$, in the odd sector, $8.206 \times 10^{-15} \leq \lambda(0.8) \leq 2.347 \times 10^{-14}$, certified by interval arithmetic (and in the even sector $8.9 \times 10^{-18} \leq \lambda^{\text{even}}(0.8) \leq 2.523 \times 10^{-16}$). In particular $\lambda(a) > 0$ for $0 < a \leq 0.8$.*

The horizon is the height at which the zeros become too dense for the window. A function of exponential type a has zeros of upper density at most a/π (Jensen’s formula; see Boas [12]); the zeros of ζ have density $(2\pi)^{-1} \log(t/2\pi)$ at height t , which equals a/π exactly at $t = T^*(a)$. Zhu’s resolution threshold, $2\pi e^{(4+o(1))e^a}$ for a one-stroke positivity certificate at half-width a [6], is a different and far larger scale. Everything below is organized around the pair $(\lambda(a), T^*(a))$.

Lemma 1.4 (existence, monotonicity, continuity, differentiability a.e.). *For every $a > 0$ the infimum defining $\lambda(a)$ is attained, the infimum over odd $C_c^\infty(-a, a)$ functions is the same number, λ is non-increasing and continuous, and λ is differentiable at almost every a .*

Proof sketch. By Proposition 3.1 and the discussion after it, Q on $L_{\text{odd}}^2(-a, a)$ is one half of the Dirichlet form of L_Δ plus a form bounded on L^2 ; the form domain $\{f : \int \log(2 + |t|)|F|^2 < \infty\}$ embeds compactly into $L^2(-a, a)$ [8, Thm. 2.1 ff.], so Q is bounded below and lower semicontinuous on the unit sphere and a minimizing sequence has a convergent subsequence. Density of C_c^∞ in the form domain gives the second claim. Monotonicity is inclusion of the admissible sets, and a monotone function is differentiable almost everywhere. Continuity: for $a < a'$ and a minimizer f' at a' , the dilate $f'_{a/a'}$ (notation of Section 5) is admissible at a , so $0 \leq \lambda(a) - \lambda(a') \leq Q(f'_{a/a'}) - Q(f')$. Minimizers at supports in $[a, a+1]$ have form norm bounded in terms of a and $\lambda(a)$ alone (the wall is positive, the prime and polar terms are bounded on the unit sphere), and on a form-bounded set $Q(f_b) - Q(f) \rightarrow 0$ uniformly as $b \rightarrow 1$: in the archimedean term because $|\Psi_\infty(s/b) - \Psi_\infty(s)| \leq w_* |\log b|$ with $w_* = \sup_t |t\Psi'_\infty(t)| < \infty$, in the prime terms because $|g_f(u) - g_f(u')| \leq \|f\| \|f(\cdot - u) - f(\cdot - u')\|$ and $\|f(\cdot - h) - f\|^2 \leq h^2 T^2 \|f\|^2 + 4 \int_{|t|>T} |F|^2$ is uniformly small on a form-bounded set, in the polar term because P is a bounded functional. Letting $a' \downarrow a$ gives right continuity; left continuity is $\lambda(a') \leq Q(f'_{a'/a}) \rightarrow Q(f) = \lambda(a)$ as $a' \uparrow a$, for a minimizer f at a . In particular the floor cannot jump. $\square$

2 Three engines

All computations use the orthonormal odd Legendre basis on the window, $f(y) = a^{-1/2} \sum_{i < K} c_i N_i P_{2i+1}(y/a)$, $N_i = \sqrt{(4i+3)/2}$, in which the matrix $Q_K(a)$ of (1) is assembled from the exact cross-correlations

$g_{ij}(x) = \int \phi_i(u)\phi_j(u-x) du$ by Gauss–Legendre rules exact for the polynomial products. The K -mode floor $\lambda_K(a)$ is the lowest eigenvalue of $Q_K(a)$; it is an upper bound on $\lambda(a)$, and λ_1 below denotes the second eigenvalue. Three implementations are used, all in the repository accompanying this paper [22].

1. `rh_weil_odd.py`: mpmath, 50–60 digits, $K \leq 72$, 80 outer quadrature nodes. Reproduces the floors and the failure points of [19] to 4–6 digits. Its outer rule is exact only to polynomial degree 159, which costs 10^{-4} in relative accuracy at $K = 72$.
2. `rh_weil_arb.py`: ball arithmetic (python-flint 0.9.0 over FLINT/Arb), $K \leq 230$, 500–800 bits, with certified Gauss–Legendre rules, $2K + 8$ inner and $2K + 100$ outer nodes, and a rigorous enclosure of the lowest eigenvalue of the assembled matrix. The Legendre recurrence is evaluated with a midpoint reset at each degree, since its ball version doubles the radius per degree; the enclosure therefore covers the assembly and the eigenvalue computation but not the polynomial evaluations, which are 500-bit floating point, nor the truncation of the outer rule. It reproduces the mpmath floors to twenty digits where both are converged and runs a 180-mode point in six minutes.
3. `rh_edge_fem.py`: odd-extended hat functions on a mesh of the half window graded geometrically to $10^{-12}a$ (350–490 nodes; the “two-stage” mesh has ratio 0.9 from $0.05a$ to $10^{-4}a$ and $\frac{1}{2}$ below), exact cross-correlations, closed-form archimedean tails, double precision. Built to read the minimizer at the edge, where a Legendre truncation resolves only $\delta \gtrsim a/K^2$. Its amplitudes carry a mesh bias of 10^{-3} (prime-free) to 10^{-2} (full form).

Computation 2.1. $\lambda_{40}(0.8) = 1.5935 \times 10^{-14}$ and $\lambda_{64}(0.8) = 1.5659 \times 10^{-14}$, both inside Zhu’s certified odd interval of Theorem 1.3, which fixes the normalization and shows the truncation error at that support. At $a = 0.3$, where no prime is inside the window, $\lambda_{64}(0.3) = 0.2225569$ and the edge-FEM floor 0.2225566 agree to 1.3×10^{-6} ; at $a_3 = \frac{1}{2} \log 3$ the 64-mode profile and the FEM profile agree to 10^{-4} for $|y| \leq 0.99a$.

3 The archimedean wall and the edge law

Proposition 3.1 (decomposition on the window). *Let $J_\Gamma(s) = e^{-|s|/2}/(1-e^{-2|s|})$, $\mathcal{J}(x) = \int_x^\infty J_\Gamma = \operatorname{artanh}(e^{-x/2}) + \arctan(e^{-x/2})$, and $c_1 = -(\gamma + \log \pi) - \frac{\pi}{2} - 3 \log 2$. For f supported in $[-a, a]$,*

$$W_\infty(g_f) = c_1 \|f\|^2 + \frac{1}{2} \iint_{[-a,a]^2} J_\Gamma(u-v) |f(u)-f(v)|^2 du dv + \int_{-a}^a V_a(y) |f(y)|^2 dy, \quad V_a(y) = \mathcal{J}(a-y) + \mathcal{J}(a+y),$$

and $V_a(y) = \frac{1}{2} \log \frac{1}{a-|y|} + \kappa_a + o(1)$ as $|y| \rightarrow a$, with $\kappa_a = \frac{1}{2} \log 4 + \frac{\pi}{4} + \mathcal{J}(2a)$.

Proof. With $g(0)-g(x) = \frac{1}{2} \int |f(u)-f(u-x)|^2 du$, the integral in (2) is $\frac{1}{2} \iint_{\mathbb{R}^2} J_\Gamma(u-v) |f(u)-f(v)|^2$, and $2 \int_0^\infty (e^{-2x} - e^{-x/2})/(1 - e^{-2x}) dx = \psi(\frac{1}{4}) + \gamma = -\pi/2 - 3 \log 2$ by the integral representation of ψ . Splitting the double integral according to whether v lies in the window gives the three terms, the two cross-regions contributing $\frac{1}{2} \int V_a |f|^2$ each. The substitution $w = e^{-s/2}$ gives $\int_x^\infty J_\Gamma = \int_0^{e^{-x/2}} 2 dw/(1-w^4) = \mathcal{J}(x)$, and $\operatorname{artanh}(e^{-x/2}) = \frac{1}{2} \log(4/x) + o(1)$ gives the constant; at $a = a_3$, $\kappa_a = 2.660623$, and the direct evaluation of $V_a - \frac{1}{2} \log(1/\delta)$ at $\delta = 10^{-9}$ gives 2.6606231. $\square$

The symbol of the whole-line form $\frac{1}{2} \iint J_\Gamma |f(u) - f(v)|^2$ is $\sigma(\xi) = \int J_\Gamma(s)(1 - \cos s\xi) ds = \log |\xi| + O(1)$, so on the window the archimedean form is one half of the Dirichlet form $\int \log(\xi^2) |\hat{f}|^2$

of the logarithmic Laplacian L_Δ of Chen–Weth [8] plus a bounded quadratic form; the prime terms are bounded translation operators and the polar term has rank one. Write $Q(f) = \frac{1}{2}\mathcal{E}_L(f) + \langle Bf, f \rangle$ with B bounded and self-adjoint on $L^2(-a, a)$ (its kernel is bounded on the window). The weak Euler–Lagrange equation of a minimizer is then

$$L_\Delta f = 2(\lambda - B)f \quad \text{in } (-a, a), \quad f = 0 \text{ outside}, \quad (4)$$

tested against $C_c^\infty(-a, a)$.

Hypothesis (H_∞). The minimizer f is bounded. *This is not known to follow from the form domain, which embeds in no L^p , $p > 2$; an L^∞ bound of De Giorgi–Moser type for L_Δ with a bounded lower-order perturbation would supply it. Numerically $\max |f| = 1.37$ at a_3 and 1.61 at $a = 0.3$.*

Theorem 3.2 (edge law; Chen–Weth [8], Thm. 1.11; Hernández-Santamaría–López Ríos–Saldaña [9], Thm. 1.1). *Assume (H_∞). Then f is continuous on $\mathbb{R}$ and there is C_0 with*

$$|f(y)| \leq C_0 \left(\log \frac{1}{a-|y|} \right)^{-1/2} \quad (|y| < a, \ a - |y| < 0.1).$$

In particular $f(\pm a) = 0$.

Proof sketch. By (4) and (H_∞), $f \in H(\Omega) \cap L^\infty(\mathbb{R})$ is a weak solution of $L_\Delta f = h$ with $h = 2(\lambda - B)f \in L^\infty(\Omega)$, $\Omega = (-a, a)$, which is a Lipschitz domain with the uniform exterior sphere condition. [8, Thm. 1.11] gives continuity and the rate $(\log(1/\delta))^{-\tau}$ for every $\tau < \frac{1}{2}$; [9, Thm. 1.1] gives $\tau = \frac{1}{2}$, with $\ell(r) = -1/\ln \min\{0.1, r\}$. $\square$

The exponent $\frac{1}{2}$ is sharp for the torsion function of a small interval [9, Thm. 1.2], and a Hopf-type lemma gives the rate from below for nonnegative supersolutions [9, Thm. 1.4]; the minimizer here is odd, so the lemma does not apply to it directly and the lower rate is a numerical statement (Computation 3.4).

Heuristic 3.3. In $t = \log(1/\delta)$, $\delta = a - y$, equation (4) near the edge has the shape $th(t) - \frac{1}{2} \int_0^t h(\tau) d\tau = O(h) + O(1)$, the coefficient t being $\frac{1}{2}$ from the wall and $\frac{1}{2}$ from the interior kernel, and the nonlocal term coming from the long-range part of the kernel, which in the variable t is the constant $\frac{1}{2}$ on one side; its solutions are $h = Ct^{-1/2}(1 + O(1/t))$. We have not derived the equation in this form.

Checkpoint 17 of [20] applied Theorem 3.2 to the forced solutions of the same operator and left open the existence of the normalized trace $\lim \sqrt{\log(1/\delta)} f(a - \delta)$; the following computation is the numerical answer for the ground state.

Computation 3.4 (the edge amplitude). On every mesh and with or without primes, $1/f(a - \delta)^2$ is linear in $t = \log(a/\delta)$ with relative residual below 10^{-4} over $14 \leq t \leq 20$; the local exponent $-d \log f / d \log t$ descends toward $\frac{1}{2}$ exactly as $\frac{1}{2}t/(t + \beta)$ does. Writing $f(a - \delta) = C(t + \beta)^{-1/2}(1 + o(1))$ on the two-stage mesh:

form	a	λ (FEM)	C	β	mesh dependence of C
prime-free	0.3	0.2225566	1.3089	-1.49	10^{-3}
full	$a_3 = 0.5493$	1.4914×10^{-5}	0.02412	-1.87	10^{-2}

The six K -mode endpoint values of [19] at a_3 ($f_K(a_3) = 0.009471, \dots, 0.008207$ for $K = 24, \dots, 64$) are this law read at the truncation’s resolution: with the fitted C, β , the effective $t_K = C^2/f_K^2 - \beta$ equals $2 \log K + 2.0$ within 0.3 for all six, with and without primes, so a degree- $2K$ expansion resolves $\delta \approx e^{-2}a/K^2$ at the edge and $f_K(a)^2 = C^2/(2 \log K + \beta')$ with $\beta' = 2 + \beta$.

4 The relay is a soft kink

At an entry a_n the new term $-2\Lambda(n)n^{-1/2}g(\log n)$ vanishes and grows with a .

Proposition 4.1 (*K-mode kink*). *For the K-mode form, λ_K is continuous at a_n and $\lambda'_K(a_n^+) - \lambda'_K(a_n^-) = 4\Lambda(n)n^{-1/2}f_K(a_n)^2$, where f_K is the normalized K-mode minimizer at entry.*

Proof. For $a = a_n + \varepsilon$ and odd f , $g(\log n) = \int_{a-2\varepsilon}^a f(y)f(y-\log n) dy = -\int_0^{2\varepsilon} f(a-\delta)f(a-2\varepsilon+\delta) d\delta$, which is $-2\varepsilon f(a)^2 + O(\varepsilon^2)$ for a function continuous at the edge; the K-mode minimizer is a polynomial. Hellmann–Feynman with the minimizer frozen gives the jump. $\square$

Computation 4.2 (the K-mode kink). With 24 modes at the entry of 3, the measured jump is 0.948 of the prediction 2.276×10^{-4} at step 5×10^{-4} and 0.9998 at step 10^{-5} (`rh_kink_steps.py` in [22]); the deficit at coarser steps is the reorganization of the minimizer within the scale gap/ $\|Q'_K\| \approx 3 \times 10^{-5}$, the ratio of the spectral gap $\lambda_1 - \lambda_K$ to the norm of the entering term's derivative.

For the exact form, under (H_∞) , Theorem 3.2 gives $f(a_n) = 0$ and the proposition's jump is a quantity of the truncation. What replaces it follows from the fitted profile of Computation 3.4: with $f(a - \delta) = C(\log(a/\delta) + \beta)^{-1/2}$, the product integral is

$$-2\Lambda(n)n^{-1/2}g(\log n) = \frac{4\Lambda(n)C^2}{\sqrt{n}} \frac{\varepsilon}{\log(a/\varepsilon) + \beta + 1 - \log 2} \left(1 + O(\log^{-2}(1/\varepsilon))\right), \quad (5)$$

so λ' is continuous across the entry and its increment at scale ε is $4\Lambda(n)n^{-1/2}C^2/(\log(a/\varepsilon) + \beta + 1 - \log 2)$: a soft kink, vanishing only logarithmically. (An earlier version omitted the constant $1 - \log 2$ in the denominator, which is an 8% effect at $\varepsilon = 10^{-3}$.)

Computation 4.3 (the soft kink). On the edge FEM (134-node mesh, for which $C = 0.0241$, $\beta = -1.93$) at the entry of 3, with $D(\varepsilon) = [\lambda(a_3 + \varepsilon) - 2\lambda(a_3) + \lambda(a_3 - \varepsilon)]/\varepsilon$, all entries $\times 10^{-4}$:

ε	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
measured $D(\varepsilon)$	2.45	1.90	1.48	1.21	1.01
first-order energy of the entering term on the a_3 minimizer	3.42	2.13	1.59	1.27	1.06
law (5)	3.15	2.11	1.59	1.27	1.06

The law reproduces the first-order energy from $\varepsilon = 10^{-4}$ down, as it must, since it is that quantity evaluated on the fitted profile; the measured increment sits 3–5% below both, which is the second-order reorganization of the minimizer, and at 10^{-3} the smooth curvature $\lambda''\varepsilon \approx 0.9 \times 10^{-4}$ is still inside D . The increment falls by about 22% per decade, as the law does. The 24-mode form gives the constant 2.28×10^{-4} .

At the resolution of the grids of Section 6 (steps 0.004 to 0.025) the entries beyond $n = 3$ leave no visible trace in λ'_K . The Hellmann–Feynman derivative resolves what those steps average over.

Computation 4.4 (the relay resolved). `rh_dilation.py` gives $\lambda'_K(a) = \langle c, Q'_K(a)c \rangle$ at $a_n + \text{offset}$ for the first five entries ($K = 48, 48, 64, 80, 96$ for $n = 2, 3, 4, 5, 7$; ball arithmetic, central differences of the matrix at step 10^{-30}). Entries are Φ'/T^* ; data in `relay_scan.csv`.

n	a_n	-0.01	-0.003	-0.001	-0.0001	+0.0001	+0.001	+0.01
2	0.3466	2.42	2.96	3.16	3.27	2.80	2.59	2.30
3	0.5493	2.31	3.02	3.65	4.08	3.41	3.17	3.08
4	0.6931	3.61	3.32	3.70	4.08	3.73	3.63	3.51
5	0.8047	3.96	2.98	3.32	4.32	3.63	3.48	3.51
7	0.9730	3.18	3.66	3.06	4.12	3.59	3.54	3.84

Over the last 0.003 before each entry the ratio rises by 10%, 35%, 23%, 45%, 13% ($n = 2, 3, 4, 5, 7$), monotonically for $n = 2, 3$ and after a dip at -0.003 for $n = 4, 5, 7$; across the entry, between offsets ∓ 0.0001 , it drops by 14%, 16%, 9%, 16%, 13%, which at this scale is the K -mode kink of Proposition 4.1 and the soft kink (5) nearly coincide, since $1/K^2 \approx 10^{-4}$; then it relaxes. The peaks at the entries are 3.27, 4.08, 4.08, 4.32, 4.12 and the dips between them reach 2.3. The values are stable in K : at $a_3 - 0.001$, $\Phi'/T^* = 3.6404, 3.6485, 3.6505$ for $K = 32, 48, 64$; at $a_5 - 0.003$, 3.001 at $K = 64$ against 2.98 at $K = 80$. The rise is the form without the entering prime power heading toward its failure point a_{fail} , the deleted-form failure of [19] (0.5574 for $n = 3$): λ_K falls roughly linearly there ($\lambda'_K \approx -1.14 \times 10^{-3}$ at $a = 0.549$, reaching zero at 0.562 by linear extrapolation) while λ'_K barely moves, so $\Phi' = -\lambda'_K/\lambda_K$ grows like $1/(a_{\text{fail}} - a)$ until the entry rescues the floor. Each value near a_3 was checked against a finite difference of λ_K with step 2×10^{-4} to four digits. The coarse grids average this sawtooth into the plateau of Computation 6.1.

The relay is therefore the sawtooth described in [19], measured on the K -mode form: not a kick at the entry but a rise into it, cut off by the entering prime power, whose first-order action at the entry is the soft kink.

5 The sliding-window identity

For f supported in $[-a, a]$ and $b > 0$ let $f_b(x) = b^{-1/2}f(x/b)$, supported in $[-ab, ab]$ with $\|f_b\| = \|f\|$. Define, with $F = \hat{f}$ and $g = g_f$,

$$D_\infty(f) = \frac{1}{2\pi} \int |F(t)|^2 t \partial_t \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) dt, \quad D_P(f) = - \sum_{\log n < 2a} \frac{2\Lambda(n)}{\sqrt{n}} \log n g'(\log n),$$

$P(f) = \int f(x) \sinh(x/2) dx$, $M(f) = \int f(x) x \cosh(x/2) dx$, and

$$\mathcal{D}(f) = D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f). \quad (6)$$

Lemma 5.1 (dilation derivative of the form). *Let a lie strictly between two consecutive entries and let f be supported in $[-a, a]$ with $Q(f) < \infty$ and g_f differentiable at every $\log n < 2a$. Then $b \mapsto Q(f_b)$ is differentiable at $b = 1$ and $\frac{d}{db}Q(f_b)|_{b=1} = -\mathcal{D}(f)$.*

Proof. $\hat{f}_b(t) = \sqrt{b}F(bt)$ exactly, jumps included. In (1), $g_{f_b}(u) = g_f(u/b)$, so the prime terms are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, differentiable at $b = 1$ with derivative $2\Lambda(n)n^{-1/2} \log n g'_f(\log n) =$ the n -th term of $-D_P(f)$; for b near 1 no new prime enters. The archimedean term is $\frac{1}{2\pi} \int b|F(bt)|^2 \Psi_\infty(t) dt = \frac{1}{2\pi} \int |F(s)|^2 \Psi_\infty(s/b) ds$ with $\Psi_\infty = \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$; since $t\Psi'_\infty(t)$ is bounded on $\mathbb{R}$, differentiation under the integral is justified by dominated convergence and gives $-\frac{1}{2\pi} \int |F|^2 t\Psi'_\infty(t) dt = -D_\infty(f)$ at $b = 1$. For the polar term, $P(f_b) = \sqrt{b} \int f(y) \sinh(by/2) dy$ gives $\frac{d}{db}P(f_b)|_{b=1} = \frac{1}{2}P(f) + \frac{1}{2}M(f)$, so $\frac{d}{db}(-2P(f_b)^2)|_{b=1} = -2P^2 - 2PM$. $\square$

Theorem 5.2 (sliding-window identity, K -mode form). *Let a lie strictly between two consecutive entries and let $\lambda_K(a)$ be simple with normalized eigenvector c , f the corresponding function. Then λ_K is differentiable at a and*

$$a \lambda'_K(a) = -\mathcal{D}(f). \quad (7)$$

Proof. The basis functions are a dilation family, $\phi_i^{(ab)} = (\phi_i^{(a)})_b$, so $Q_K(ab) = [Q((\phi_i^{(a)})_b, (\phi_j^{(a)})_b)]_{ij}$ is real-analytic in b near 1 (each entry is a finite sum of terms as in Lemma 5.1, with g_{ij} polynomial), Hellmann–Feynman for a simple eigenvalue gives $\frac{d}{db}\lambda_K(ab)|_{b=1} = \langle c, \frac{d}{db}Q_K(ab)|_{b=1}c \rangle = \frac{d}{db}Q(f_b)|_{b=1}$ by bilinearity, and Lemma 5.1 evaluates it. $\square$

Proposition 5.3 (exact form: two-sided Dini derivatives). *Let a lie strictly between two consecutive entries and let f be a minimizer at support a (Lemma 1.4) with g_f differentiable at every $\log n < 2a$. Then*

$$a D^+ \lambda(a) \leq -\mathcal{D}(f) \leq a D_- \lambda(a),$$

where D^+, D_- are the upper-right and lower-left Dini derivatives; hence $a\lambda'(a) = -\mathcal{D}(f)$ at every point of differentiability of λ , which by Lemma 1.4 is almost every a .

Proof. For every $b > 0$, f_b is admissible at support ab , so $\lambda(ab) \leq Q(f_b)$, with equality at $b = 1$. By Lemma 5.1, $Q(f_b) = \lambda(a) - (b-1)\mathcal{D}(f) + o(b-1)$. For $b > 1$, $(\lambda(ab) - \lambda(a))/(b-1) \leq -\mathcal{D}(f) + o(1)$; for $b < 1$ the division reverses the inequality. Since $\frac{d}{db} = a \frac{d}{da}$ at $b = 1$, the two bounds follow. $\square$

Remark 5.4. The hypothesis on g'_f is implied by $f' \in L^1$, which the edge law $C(\log(1/\delta))^{-1/2}$ suggests but which is not proved; it is the only condition beyond existence. The form-domain condition $\int \log(2 + |t|)|F|^2 < \infty$ does not give $\int |t|F|^2 < \infty$, so the prime part of $\mathcal{D}$ is to be read in x -space, as defined.

The hypothesis is natural in the class of functions of bounded variation, and in that class the dilation derivative is bounded, which makes the floor Lipschitz.

Lemma 5.5 (dilation of a function of bounded variation). *Let $f \in L^\infty(\mathbb{R})$ have bounded variation $\text{Var } f$ and support in $[-a, a]$. Then g_f is Lipschitz with constant $\|f\|_\infty \text{Var } f$; the map $b \mapsto Q(f_b)$ is locally Lipschitz on $(0, \infty)$, entries included, with $b \frac{d}{db} Q(f_b) = -\mathcal{D}(f_b)$ at every b at which g_f is differentiable at the points $(\log n)/b$; and for every b*

$$|\mathcal{D}(f_b)| \leq w_* \|f\|^2 + \frac{\|f\|_\infty \text{Var } f}{b} \sum_{\log n < 2ab} \frac{2\Lambda(n) \log n}{\sqrt{n}} + (2\sigma(ab)^2 + 2\sigma(ab)\mu(ab)) \|f\|^2,$$

where $w_* = \sup_t t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) = 2.0807$, attained at $t = 0.520$, $\sigma(a)^2 = \int_{-a}^a \sinh^2(x/2) dx$ and $\mu(a)^2 = \int_{-a}^a x^2 \cosh^2(x/2) dx$. If f is continuous, $g_f \in C^1$ and $b \mapsto Q(f_b)$ is C^1 .

Proof. For f of bounded variation, $\int |f(y) - f(y-h)| dy \leq |h| \text{Var } f$, so $|g_f(u) - g_f(u')| \leq \|f\|_\infty \int |f(x-u) - f(x-u')| dx \leq \|f\|_\infty \text{Var } f |u-u'|$. The prime terms of $Q(f_b)$ are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, locally Lipschitz in b , with $b \frac{d}{db} g_f(\log n/b) = -(\log n/b) g'_f(\log n/b) = -\log n g'_{f_b}(\log n)$ wherever g_f is differentiable at $\log n/b$; a term with $\log n = 2ab$ enters with the value $g_f(2a) = 0$ and continuously, since g_f vanishes on $[2a, \infty)$. The archimedean and polar terms are C^1 in b for every b , with the derivatives computed in Lemma 5.1. The bound uses $|t \partial_t \text{Re } \psi| \leq w_*$, $|g'_{f_b}| \leq \|f_b\|_\infty \text{Var } f_b = b^{-1} \|f\|_\infty \text{Var } f$, and $|P(f_b)| \leq \sigma(ab) \|f\|$, $|M(f_b)| \leq \mu(ab) \|f\|$ by Cauchy-Schwarz. If f is continuous, $g'_f(u) = -\int f(z+u) d(Df)(z)$ with Df the finite derivative measure of f , which is continuous in u by dominated convergence. $\square$

Proposition 5.6 (the floor is Lipschitz under a variation bound). *Let $0 < a_0 < a_1$ and $M > 0$, and assume*

(H_{BV}) for every $a \in [a_0, a_1]$ and $\varepsilon > 0$ there is $f \in L^\infty$ of bounded variation, supported in $[-a, a]$, with $\|f\| = 1$, $Q(f) \leq \lambda(a) + \varepsilon$ and $\|f\|_\infty \text{Var } f \leq M$.

Then for $a_0 \leq a < a' \leq a_1$,

$$0 \leq \lambda(a) - \lambda(a') \leq L \log \frac{a'}{a} \leq \frac{L}{a_0} (a' - a), \quad L = w_* + \frac{a_1}{a_0} M \sum_{\log n < 2a_1} \frac{2\Lambda(n) \log n}{\sqrt{n}} + 2\sigma(a_1)^2 + 2\sigma(a_1)\mu(a_1).$$

In particular λ is Lipschitz on $[a_0, a_1]$, hence absolutely continuous there, with $\lambda' \geq -L/a_0$ almost everywhere.

Proof. The left inequality is Lemma 1.4. For the right one take f' as in (H_{BV}) at a' and put $b_0 = a/a'$. The dilate f'_{b_0} is supported in $[-a, a]$ with unit norm, so $\lambda(a) \leq Q(f'_{b_0})$. By Lemma 5.5, $b \mapsto Q(f'_b)$ is Lipschitz on $[b_0, 1]$ with $|b \frac{d}{db} Q(f'_b)| \leq L$ almost everywhere (the prime sum runs over $\log n < 2a'b \leq 2a_1$, and $b^{-1} \leq a_1/a_0$), so $Q(f'_{b_0}) \leq Q(f') + L \log(1/b_0) \leq \lambda(a') + \varepsilon + L \log(a'/a)$. Let $\varepsilon \rightarrow 0$. $\square$

Remark 5.7. (H_{BV}) concerns near-minimizers, not the minimizer, and it follows from a bound on the K -mode minimizers uniform in K : $\lambda(a) = \inf_K \lambda_K(a)$ by Lemma 1.4 (the odd Legendre truncations of an odd $C_c^\infty(-a, a)$ function converge to it in the form norm, their jumps at $\pm a$ being of vanishing size and costing $O(\text{jump}^2)$ in a norm that tolerates jumps), so the K -mode minimizers are near-minimizers of the exact problem, and they are polynomials, hence of bounded variation. Under (H_{BV}) on every compact subinterval of $[a_0, \infty)$ the regularity clause of Conjecture 7.1 holds and the conjecture is the inequality $\lambda' \geq -cT^*\lambda$ a.e. alone. The trivial bound $\lambda' \geq -L/a$ also shows what the conjecture asks: on $[0.6, 1]$ with $M = 9$ (Computation 5.8) it gives $L = 126$ and $\lambda' \geq -211$, while λ' is -4×10^{-5} at $a = 0.6$ and -3×10^{-24} at $a = 1$; the conjecture is that $\mathcal{D}(f)$, whose three terms are each of order one, cancels to $O(T^*\lambda)$ on the minimizer, as in (8).

Computation 5.8 (bounded variation of the minimizers). `rh_bv_check.py`. On the edge-FEM minimizers of Computation 8.10 (486 nodes, edge resolved to $10^{-12}a$, $\|f\| = 1$): the supremum, the total variation over the window, and the increments of the Hardy integral $H(\delta_0) = \int_{\delta_0}^{a/10} f(a - \delta)^2 d\delta/\delta$ between $\delta_0 = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}$ (finiteness of $H(0)$ is the $H^{1/2}$ condition at the edge).

a	primes	$\sup f $	$\text{Var } f$	$\text{Var } f / (4 \sup f)$	$H(10^{-3})$	increments per three decades
0.3	no	1.70138	6.80553	1.0000	2.64	1.63, 0.93, 0.62
0.4	no	1.47286	5.89146	1.0000	2.13	1.26, 0.69, 0.46
0.5	no	1.31669	5.26678	1.0000	1.83	1.01, 0.58, 0.39
0.6	no	1.20136	4.80545	1.0000	1.66	0.86, 0.52, 0.35
0.4	yes	1.55514	6.22057	1.0000	0.56	0.23, 0.12, 0.08
0.45	yes	1.54288	6.17152	1.0000	0.17	0.049, 0.027, 0.018
0.5	yes	1.49876	5.99842	1.0006	0.037	0.0054, 0.0030, 0.0020
a_3	yes	1.47916	5.91666	1.0000	0.0060	0.0005, 0.0003, 0.0002

On every minimizer $\text{Var } f = 4 \sup |f|$ to four digits (1.0006 at $a = 0.5$ with primes, where the half-window profile has a small interior ripple): each half-window profile rises from 0 at the centre to its maximum and falls monotonically to the edge. Mesh refinement at $a = 0.5$ without primes (124, 237, 471, 486 nodes; $\delta_{\min} = 6 \times 10^{-9}$ and then 7×10^{-13} ; 100, 200, 400, 200 uniform intervals on the inner 90% of the half window; the last mesh with the two-stage grading) gives $\sup |f| = 1.316720, 1.316696, 1.316694, 1.316694$ and $\text{Var } f = 5.266880, 5.266785, 5.266777, 5.266777$. The Hardy increments decrease like those of $C^2 \log(\log(a/\delta_0) + \beta)$ with the C, β of Computation 8.10 (at $a = 0.3$: 1.63, 0.93, 0.62 against 1.66, 0.83, 0.55), i.e. $H(\delta_0) \rightarrow \infty$ doubly logarithmically as the edge law predicts: the minimizers are not in $H^{1/2}$, and $L^\infty \cap BV$ is the right class. On the K -mode minimizers (ball-arithmetic engine, two steps of inverse iteration, evaluated on 44,000 points of the half window graded to $10^{-12}a$; the variation includes the jump $f_K(a)$ at the edge):

a	K	12	24	48	72	96	128
0.6	$\sup f_K $	1.461834	1.461771	1.461786	1.461780	1.461771	
	$\text{Var } f_K$	5.847337	5.847085	5.847145	5.847121	5.847085	
0.8	$\sup f_K $	1.407373	1.421807	1.421859	1.421851 [†]	1.421844	1.421842
	$\text{Var } f_K$	5.629491	5.687230	5.687435	5.687404 [†]	5.687377	5.687368

($\dagger$: $K = 64$.) At $a = 1.0$ (600 bits): $\sup |f_K| = 1.392228, 1.399706, 1.399722, 1.399716, 1.399714$ and $\text{Var } f_K = 5.568912, 5.598825, 5.598888, 5.598865, 5.598858$ for $K = 24, 48, 80, 120, 160$. In every case $\text{Var } f_K / (4 \sup |f_K|) = 1.0000$, and $\|f_K\|_\infty \text{Var } f_K$ converges, to 8.55, 8.09, 7.84 at $a = 0.6, 0.8, 1.0$: (HBV) holds on these supports with $M = 9$, in the form of Remark 5.7.

Since $t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) \rightarrow 1$, $D_\infty(f) = \|f\|^2 + O(\int |F|^2 t^{-2})$; on a minimizer $\|f\| = 1$ while λ is 10^{-90} at $a = 1.5$, so (7) forces

$$D_P(f) = -D_\infty(f) - 2P^2 - 2PM + O(aT^* \lambda) : \quad (8)$$

the Euler product, differentiated and applied to the derivative of the minimizer's autocorrelation at the prime-power positions, cancels the archimedean dilation form to ninety digits. Together with the explicit formula on the same function, $\sum 2\Lambda(n)n^{-1/2}g(\log n) = W_\infty(g) - 2P^2 - \lambda$, both the values and the log-weighted derivatives of g_f at the prime powers are pinned by the archimedean side.

Computation 5.9 (the identity on the K -mode minimizer). `rh_dilation.py` differentiates the archimedean matrix A , the prime matrices Π_n and the polar matrix $ss^\top$ of $Q_K(a)$ by central differences with step 10^{-30} in ball arithmetic and contracts them with the eigenvector; every entry is a ball with the radius shown, and the last column is the sum, which is $a\lambda'_K$ by Theorem 5.2.

a	K	λ_K	$a\langle c, A'c \rangle$	$a\langle c, \Pi'c \rangle$	$-2a\langle c, (ss^\top)'c \rangle$	sum = $a\lambda'_K$	radius
0.6	48	5.9659×10^{-7}	-1.014313	+1.086974	-0.072687	-2.493408×10^{-5}	4×10^{-20}
0.8	64	1.5659×10^{-14}	-1.016605	+1.103966	-0.087361	$-1.295854 \times 10^{-12}$	4×10^{-27}
1.0	80	1.4820×10^{-26}	-1.018057	+1.114966	-0.096909	$-2.516852 \times 10^{-24}$	3×10^{-39}
1.25	140	3.4161×10^{-51}	-1.019178	+1.123585	-0.104407	$-1.233712 \times 10^{-48}$	2×10^{-63}

The three columns are $-D_\infty$, $-D_P$ and $-(2P^2 + 2PM)$ evaluated through the matrices; that their sum is $a\lambda'_K$ is Theorem 5.2, not an independent check. The independent check is on the Fourier side: at $K = 12$, $a = 0.6$ (`rh_fourier_check.py`), with the same coefficient vector, D_∞ evaluated as $(1/2\pi) \int |F|^2 t \partial_t \Re \psi(\frac{1}{4} + \frac{it}{2}) dt$ with F in closed form through spherical Bessel functions and the Parseval tail, D_P from $g'(\log 2)$ and $g'(\log 3)$ of the exact autocorrelation, and P, M by quadrature, gives $D_\infty = 1.01430470102641$, $D_P = -1.08691690953774$, $2P^2 + 2PM = 0.0726373685293583$ and $-D(f) = -2.51600180293 \times 10^{-5}$, against $a\langle c, Q'_K c \rangle = -2.51600180295 \times 10^{-5}$ from the matrices: twelve digits. (The eigenvector enters through one step of inverse iteration from a shift at relative distance 2^{-20} below λ_K , which fixes $a\langle c, Q'_K c \rangle$ to about seven digits: at $K = 12$ a second step moves it by 1×10^{-7} relatively, to $-2.51600203711 \times 10^{-5}$, which is the central difference of λ_K itself to twelve digits. The ball radii in the table bound the arithmetic only.) The archimedean dilation form sits at 1.014–1.019, the excess over $\|f\|^2$ being the finite- t correction; the prime dilation form grows with the primes present and cancels it; the polar term grows from 7% to 10%. The implied Φ'/T^* are 3.339, 3.324, 3.658, 3.774; at $a = 0.8$ this lies inside the sawtooth of the entry of 5 (Computation 4.4).

5.1 The derivative is a property of the ground state

Let $Q'_K(a)$ be the a -derivative of the K -mode matrix, a symmetric form on the K -mode space. Is there an inequality $-Q'_K \leq cT^* Q_K$ between the two forms, which would give the derivative bound for every f at once? Since a form inequality on the exact form would restrict to every K -mode subspace, the pencil $(-Q'_K, Q_K)$ answers this for the exact form too.

Computation 5.10 (the pencil). Largest and smallest eigenvalues of $Q_K^{-1}(-Q'_K)$ divided by T^* , computed from the midpoint matrices with an approximate QR eigenvalue routine at 500 bits (Q_K has condition number up to 10^{26} here, leaving over a hundred digits):

a	minimizer direction	largest	second, third	smallest
0.6	3.34	74.7	4.26, 3.69	-70.5
0.8	3.32	1.46×10^5	332, 6.08	-1.46×10^5
1.0	3.66	6.2×10^{10}	3.1×10^7 , 8.2×10^4	-6.2×10^{10}

The extreme eigenvalues come in $\pm$ pairs and grow explosively. They are near-null directions of Q_K whose dilation derivative is enormous relative to their value; heuristically they are the directions along which the minimizer rotates as a moves, and their size is of the order of the inverse of the scale gap/ $\|Q'_K\|$ of Computation 4.2. The ground state is neither extreme; it is a special interior direction in which the dilation derivative is anomalously small, by ten orders at $a = 1$. Hence the derivative bound of Section 7 is a statement about the ground-state eigenvector alone, $\lambda' = \langle f, Q'f \rangle$, and cannot be an inequality between forms. A proof must pass through the Euler–Lagrange equation (4) of the minimizer.

6 The floor to height 130

Computation 6.1 (the band). Three data sets (`floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`): $K = 40$, step 0.0125 on $[0.30, 1.20]$ with the entries $a_n \pm 0.004$ added; $K = 72$, step 0.025 on $[0.475, 1.05]$; and the ball-arithmetic engine at pairs $a \pm 0.005$ around the centres 1.06, 1.10, 1.15, 1.25, 1.33, 1.40, 1.45, 1.50 with $K = 110, 110, 110, 140, 140, 180, 180, 180$, plus $K = 180$ at 1.25 and $K = 230$ at 1.5. Φ' is a centered difference over the nearest admissible pair (± 0.025 for $K = 72$, ± 0.005 for arb), never across an entry; the pair at 1.10 straddles $a_9 = 1.0986$ and is excluded. Every value is a K -mode upper bound on the floor. Convergence: λ_{40} agrees with λ_{72} to 5% up to $a = 0.925$ and 26% at 1.0; λ_{140} with λ_{180} to 2% at 1.25; at 1.5, λ_{180} and λ_{230} differ by 39% and the ratio by 4%, more modes raising it, so the last ratio is a lower bound. For $a > 1$ each eigenvalue carries a rigorous enclosure of the assembled matrix’s lowest eigenvalue: radius 2×10^{-64} for the four points computed before the midpoint reset of Section 2, $\approx 10^{-143}$ at 500 bits after it, and $\approx 10^{-233}$ at 800 bits; the enclosures do not cover the polynomial evaluations or the outer rule.

a	K	T^*	λ_K	Φ'	Φ'/T^*
0.500	72	17.1	1.938×10^{-4}	55.2	3.23
0.650	72	23.1	1.837×10^{-8}	74.8	3.24
0.775	72	29.6	2.105×10^{-13}	107.5	3.63
0.900	72	38.0	7.176×10^{-20}	138.5	3.64
1.000	72	46.4	1.491×10^{-26}	172.6	3.72
1.060	110	52.4	2.97×10^{-31}	195.8	3.74
1.150	110	62.7	1.04×10^{-39}	240.3	3.83
1.250	180	76.5	3.59×10^{-51}	289.7	3.79
1.330	140	89.8	3.58×10^{-62}	339.2	3.78
1.400	180	103.3	1.73×10^{-73}	398.4	3.86
1.450	180	114.2	1.22×10^{-82}	443.5	3.88
1.500	230	126.2	2.97×10^{-93}	491.1	3.89

For the $K = 72$ rows λ_K is the grid value at the centre; for the arb rows it is the geometric mean of the pair. Means of Φ'/T^* : 3.23 on $[0.5, 0.7]$ (5 grid points), 3.61 on $[0.85, 1.05]$ (5), 3.79 on $[1.06, 1.25]$ (3), 3.85 on $[1.33, 1.5]$ (4). One-parameter least-squares fits of Φ'/T^* against a on the nine tabulated rows with $a \geq 0.85$, residual = rms of the relative deviations: $\Phi' = cT^*$ with $c = 3.79$, 2.1%; $\Phi' = cT^* \log T^*$, 7.5%; the raw count $\Phi \propto N(T^*)$, whose ratio is $\propto 4a - 1$, 18.4%; the derivative of Zhu's even-sector law $\Phi \simeq C N(T^*)/\log N(T^*)$, whose ratio is $\propto 2a(L - 1)/L^2$ with $L = \log N(T^*) = 2a + \log(2a - 1)$ and tends to C/π , 8.1%. With Zhu's own $C = 2\pi^2$ that derivative is 3.14 at $a = 1$ and 3.72 at $a = 1.5$, a 42% rise against the observed 7%. The ratio λ_1/λ_K stays above 5.7 and grows overall to 10^6 at $a = 1$ (with local decreases of up to 18%), so the K -mode ground state is simple throughout.

At step 0.004 the entries beyond $n = 3$ show no net drop in Φ' , because the difference quotient across an entry averages the steep rise before it against the drop at it; Computation 4.4 resolves both. Over the fine scan, Φ'/T^* lies in $[2.3, 4.3]$, and the supremum over all measured points, the quantity Conjecture 7.1 bounds, is 4.32, at the entry of 5.

7 The conjectures

Conjecture 7.1 (bounded relay). *There are $a_0 > 0$ and $c > 0$ such that λ is locally absolutely continuous on $[a_0, \infty)$ and*

$$\lambda'(a) \geq -cT^*(a)\lambda(a) \quad \text{for almost every } a \geq a_0.$$

The statement is meaningful whatever the sign of λ ; absolute continuity is part of the conjecture: Lemma 1.4 gives continuity and differentiability a.e., and Proposition 5.6 gives Lipschitz continuity under (H_{BV}) , a bound on the sup norm times the total variation of near-minimizers, which Computation 5.8 supports on every minimizer computed ($\text{Var } f = 4 \sup |f| \approx 6$, converged in the mesh and in K). Where $\lambda > 0$ it reads $\Phi' \leq cT^*$.

Proposition 7.2. *Conjecture 7.1 with $a_0 \leq 0.8$ implies the Riemann Hypothesis.*

Proof. Put $u(a) = \lambda(a) \exp(\frac{c}{2}T^*(a))$, which is locally absolutely continuous with $u' = e^{cT^*/2}(\lambda' + cT^*\lambda) \geq 0$ a.e., using $\frac{d}{da}T^* = 2T^*$. Hence u is non-decreasing and $\lambda(a) \geq \lambda(a_0) \exp(-\frac{c}{2}(T^*(a) - T^*(a_0)))$ for $a \geq a_0$. With $a_0 = 0.8$, Theorem 1.3 gives $\lambda(0.8) \geq 8.206 \times 10^{-15} > 0$, so $\lambda > 0$ on $[0.8, \infty)$, and $\lambda > 0$ on $(0, 0.8]$ by the same theorem. Theorem 1.2 concludes. $\square$

Conjecture 7.3 (decay law). *For $a \geq a_0$, $\lambda(a) > 0$ and $\Phi'(a)/T^*(a)$ is bounded above and below by positive constants. In its sharp form, $\Phi'(a)/T^*(a) \rightarrow c_0$, i.e. $\lambda(a) = \exp(-(c_0/2 + o(1))2\pi e^{2a})$.*

Computation 6.1 supports the weak form from height 17 to 130 and, on the odd sector, favours a constant $c_0 \approx 3.8$ (residual 2.1%) over every law with a logarithmic factor; the derivative of Zhu's even-sector law, whose constant $C = 2\pi^2$ would give the asymptote 2π , predicts a rise six times larger than observed. Whether the ratio saturates or continues to creep up beyond height 130 is not decidable from these data. Conjecture 7.1 is the one that implies the Riemann Hypothesis; the constant is not its content.

By Theorem 5.2 and Proposition 5.3, Conjecture 7.1 is, at almost every a , the inequality

$$\mathcal{D}(f) \leq caT^*(a)Q(f) \quad \text{on the minimizer } f, \tag{9}$$

and by Computation 5.10 it holds only there.

Remark 7.4 (what a proof cannot be). Positivity of Q_{24} on a window is destroyed by displacing $\log 2$ by 3×10^{-14} at $a = 0.8$ and by displacements of order $\exp(-cT^*(a))$ in general [19, 22], so no statement about the primes that is invariant under such displacements can imply positivity; and Checkpoint 20 of [20] shows that bounding the reflected prime terms separately from the polar term yields $-\infty$ on translated tests. A proof of Conjecture 7.1 must therefore use the exact multiplicative structure of Λ , and by Computation 5.10 it must use that f minimizes Q .

7.1 Under the Riemann Hypothesis

Let PW_a be the Paley–Wiener space of transforms $F = \hat{f}$ of $f \in L^2(-a, a)$, here with f odd. If all zeros are real, $Q(f) = \sum_{\gamma} |F(\gamma)|^2$ (the sum over γ with multiplicity, of both signs; it converges when $Q(f) < \infty$), and the dilation computation of Lemma 5.1 on the zero side, $Q(f_b) = \sum_{\gamma} b |F(b\gamma)|^2$, gives at a point of differentiability

$$a\lambda' = \lambda + \sum_{\gamma} \gamma \partial_t |F|^2(\gamma)$$

whenever the last sum converges absolutely, so Conjecture 7.1 reads $\sum_{\gamma} \gamma \partial_t |F|^2(\gamma) \geq -(1 + caT^*) \sum_{\gamma} |F(\gamma)|^2$: the minimizer is small at the zeros to the same order in its derivative as in its value. And

$$\lambda(a) = \inf \{ \sum_{\gamma} |F(\gamma)|^2 : F \in \text{PW}_a \text{ odd, } \|f\|_2 = 1 \}$$

is the lower sampling constant of the zeros for PW_a . The zeros have density $(2\pi)^{-1} \log(t/2\pi)$, which exceeds a/π exactly above $T^*(a)$, so only the zeros above the horizon can sample PW_a , and the floor measures how well a band-limited function can hide from them.

Both statements can be evaluated on the zeros themselves. The explicit formula is unconditional, and every zero used below lies on the line, so for the K -mode minimizer the sum over the zeros of $|F_K(\gamma)|^2$ must return $Q(f_K) = \lambda_K$ up to the contribution of the zeros above the last one used; this closes the loop on the object the form was built from.

Computation 7.5 (the floor and the identity on the zeros). `rh_zero_side.py`, with the first 6000 zeros ($\gamma \leq 6365.85$; `data/zeros_6000.txt`, from `mpmath.zetazero`, the first 400 to 36 digits and the rest to 21). For the K -mode minimizers of Computation 5.9, F_K is evaluated in closed form through spherical Bessel functions at 35 digits; the sums run over $\gamma > 0$ with the factor 2; the tail above γ_{6000} is the Riemann–von Mangoldt density integrated against $|F_K|^2$, with $|F_K(t)|^2$ replaced by its mean $2f_K(a)^2/t^2$ beyond $\gamma_{6000} + 60$, and the derivative tail is the corresponding $-2\times$ that; $a\lambda'_K$ is the matrix-side value of Computation 5.9, the sums run over $\gamma > 0$.

a	K	λ_K	$2 \sum F_K(\gamma) ^2 + \text{tail}$	ratio	$a\lambda'_K$	$\lambda_K + 2 \sum \gamma \partial_t F_K ^2 + \text{tail}$	ratio
0.6	48	5.96594×10^{-7}	5.96554×10^{-7}	0.99993	-2.49341×10^{-5}	-2.49310×10^{-5}	0.99988
0.8	64	1.56587×10^{-14}	1.56568×10^{-14}	0.99988	-1.29585×10^{-12}	-1.29685×10^{-12}	1.00077
1.0	80	1.48201×10^{-26}	1.48157×10^{-26}	0.99971	-2.51685×10^{-24}	-2.52433×10^{-24}	1.0030

At $a = 0.6$ the value ratio with $N = 10, 30, 100, 300, 1000, 3000, 6000$ zeros (tail included) is 0.977, 0.952, 0.973, 0.989, 0.9974, 0.99967, 0.99993; the residual is the fluctuation of the zeros about their mean density, of the size of the tail. The derivative sum is first-order sensitive to the positions of the zeros, by $4\gamma |F'_K(\gamma)|^2 \varepsilon$ per zero for an error ε in γ , while the value sum is not; with all zeros at 21 digits the derivative sum at $a = 1$ came out $+7 \times 10^{-22}$ against -2.5×10^{-24} , stable from $N = 30$ on, and the 36-digit zeros cure it. The residuals of the derivative sums, 1.2×10^{-4} , 8×10^{-4} ,

3×10^{-3} , grow with a as this sensitivity does (the zeros beyond the 400th are still at 21 digits); the value sums are insensitive, and their residuals are the fluctuation of the zeros about their mean density in the tail. The explicit formula, which built the form from the primes and the archimedean symbol, returns on the actual zeros both the floor and its rate of change.

Where the floor sits among the zeros is the second reading of the same computation. At $a = 0.6$ ($T^* = 20.9$) one zero lies below the horizon and carries 1.0×10^{-5} of $2 \sum |F_K(\gamma)|^2$; at $a = 0.8$ ($T^* = 31.1$) four zeros lie below and carry 1.0×10^{-7} ; at $a = 1$ ($T^* = 46.4$) eight lie below and carry 1.6×10^{-13} , the first zero alone 9×10^{-25} . The largest single terms are at $\gamma_7 = 40.9$ (15% of the floor), $\gamma_{13} = 59.3$ (19%) and $\gamma_{27} = 94.7$ (8%), at 2.0, 1.9 and 2.0 horizons, and at each support the mass sits between 1.5 and 3 horizons. So the minimizer hides from the zeros below the horizon to an accuracy far beyond the floor itself, and the floor is made by the zeros between the horizon and about three times the horizon, as the sampling picture says. This is the shadow in its most literal form: $2 \sum_{\gamma > 0} |F_K(\gamma)|^2 = \lambda_K$ says that the worst test function is small at every zero, to 10^{-13} at $a = 1$, and the derivative column says that it is small there in its dilation derivative too.

The distribution has a law. Write $\vartheta(T)$ for the fraction of $2 \sum_{\gamma > 0} |F(\gamma)|^2$ carried by the zeros above height T .

Computation 7.6 (the tail law). `rh_tail_law.py`, same zeros. (i) The edge-FEM minimizers with primes of Computation 8.10 (486 nodes, edge resolved to $10^{-12}a$), with F of the odd piecewise-linear function in closed form, the value at the last node included as a jump, and λ' the matrix-side derivative of the same computation. The FEM floor is recovered from the zeros: $2 \sum_{\gamma \leq \gamma_{6000}} |F(\gamma)|^2 / \lambda_{\text{FEM}}$ plus the tail $\Phi' / (\pi \gamma_{6000})$ is 0.999995, 0.999997, 0.999999 at $a = 0.4, 0.45, 0.5$, a check of the FEM engine on the zeros independent of everything else. Then $T\vartheta(T)/T^*$ against $\Phi' / (\pi T^*)$, with $\Phi' = -\lambda' / \lambda_{\text{FEM}}$:

a	$\Phi' / (\pi T^*)$	$T/T^* = 2$	3	5	7	10	20	50	100
0.4	0.7213	0.674	0.732	0.742	0.718	0.721	0.728	0.724	0.724
0.45	0.8676	0.956	0.836	0.857	0.856	0.872	0.865	0.868	0.867
0.5	1.0811	1.105	1.267	1.085	1.103	1.060	1.081	1.083	1.081
a_3	1.2948	0.889	0.967	1.059	1.103	1.135	1.160	1.184	1.197

From seven horizons on, $T\vartheta(T)/T^*$ equals $\Phi' / (\pi T^*)$ to 1% at $a = 0.4, 0.45, 0.5$ (0.995–1.009 of it at $T/T^* = 7, 10, 20, 50, 100$), with no logarithmic drift over the decade and a half covered. At the entry a_3 the ratio rises from 0.69 at two horizons to 0.92 at a hundred: there λ' is still moving at the scale $1/T$ that a zero at height T probes (Computation 4.3), and the tail follows the derivative at that scale. The law is a property of the transform sampled by the density of zeros: at $a = 0.5$ the density integral $2 \int_T^\infty |F|^2 \rho$, $\rho(t) = \log(t/2\pi)/2\pi$, is 1.068, 1.060, 1.032, 1.020, 1.012, 1.006 times $-\lambda' / (\pi T)$ at $T/T^* = 5, 7, 10, 20, 50, 100$, the $O(1/\log T)$ correction of Proposition 7.8 below, while the zero sum is 1.004, 1.020, 0.981, 1.000, 1.002, 1.000 times it. (ii) The K -mode minimizers, per-zero shares from `data/zeros_6000.txt` for $a \leq 1.25$ and from the first 1500 zeros at 50 digits for $a = 1.5$ (`data/zeros_1500_50digits.txt`; in the plunge region $|F_K(\gamma)|$ is far below $|F'_K(\gamma)|$ times the error of a 21-digit zero), ϑ at fixed T/T^* :

a	K	$\Phi' / (\pi T^*)$	$\vartheta(2T^*)$	$\vartheta(2.5T^*)$	$\vartheta(3T^*)$	$\vartheta(5T^*)$	$\vartheta(10T^*)$	$T\vartheta/T_{\text{eff}}$ at 3	5	10	median
0.6	48	1.063	0.594	0.481	0.370	0.215	0.1065	1.045	1.011	1.002	2.39
0.8	64	1.058	0.596	0.480	0.415	0.226	0.1009	1.176	1.067	0.953	2.32
1.0	80	1.164	0.810	0.526	0.407	0.240	0.1170	1.048	1.032	1.005	2.65
1.25	140	1.201	0.781	0.528	0.433	0.249	0.1210	1.082	1.038	1.007	2.67
1.5	230	1.239	0.858	0.554	0.434	0.251	0.1230	1.052	1.015	0.993	2.74

($T_{\text{eff}} = \Phi'/\pi$; the median is the height, in horizons, below which half of the floor sits. At $a = 1.5$ the Hellmann–Feynman contraction failed, two inverse-iteration steps at 800 bits leaving the $K = 230$ eigenvector short of the 10^{-90} relative accuracy the contraction needs, and $\Phi' = 491.1$ is the central difference of the $K = 230$ floor between 1.495 and 1.505 from Computation 6.1; the tail law returns it, $T\vartheta/T^* = 1.233, 1.230$ at 7 and 10 horizons against $\Phi'/(\pi T^*) = 1.239$. The λ_K of the $a = 1.25$ row of Computation 5.9 was corrected on this occasion to 3.4161×10^{-51} , the value the zeros return; its Φ'/T^* was already the correct one.) The tail law holds on the K -mode minimizers to 1–3% from five horizons on. Below that, the plunge region $[T^*, 3T^*]$ carries more than the law, $T\vartheta/T^*$ reaching 1.4–1.5 at $1.5T^*$; the median height of the floor’s mass is 2.39, 2.32, 2.65, 2.67, 2.74 horizons at $a = 0.6, 0.8, 1.0, 1.25, 1.5$ (and 1.8, 2.0, 2.4 on the FEM minimizers at 0.4, 0.45, 0.5), rising with a and, at the last point, at $e = 2.72$ The curves $\vartheta(T/T^*)$ at $a = 0.6$ and 0.8 coincide to 0.01 at every point of the grid, those at 0.4, 0.45, 0.5 approach them from below as a grows, and those at 1.0, 1.25 and 1.5 lie above them in the plunge region ($\vartheta(2T^*) = 0.81, 0.78, 0.86$ against 0.59) and rejoin them in the tail. Measured in units of T_{eff} rather than T^* , that is as $T\vartheta(T)/T_{\text{eff}}$, all five agree to 5% from 2.5 horizons on (1.10–1.14 at 2.5, 1.05–1.18 at 3, 1.01–1.07 at 5, 0.99–1.01 at 10), across floors spanning 86 orders of magnitude

Conjecture 7.7 (tail law). *Assume RH. For every $a > 0$ that is not an entry, with f the minimizer at support a and λ' the derivative of the floor,*

$$\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\frac{\lambda'(a)}{\pi}; \quad \text{equivalently} \quad \Phi'(a) = \pi T_{\text{eff}}(a), \quad T_{\text{eff}}(a) = \lim_{T \rightarrow \infty} T\vartheta(T).$$

Under (H_∞) this is a theorem for $a_2 < a < a_3$ (Corollary 8.8), at every support below $a_\infty = 0.843$ where the echo set is finite (Corollary 8.14), and for the archimedean form at every support (Corollary 8.5).

The heuristic behind it is the edge law: a shape with $f(a - \delta) \sim C(\log(a/\delta))^{-1/2}$ has $|F(t)|^2 \approx 4C^2 \cos^2(ta)/(t^2 \log t)$ for large t , the zeros have density $\log(t/2\pi)/2\pi$, and the product integrated from T gives $C^2/(\pi T)$, which is $-\lambda'/(2\pi T)$ by Conjecture 8.1. The asymptotic expansion carries a ratio of logarithms, $\log(T/2\pi)/(\log(aT) + \beta)$, that would be 1.1 to 1.4 at the heights of Computation 7.6 and that the data do not show; the law is sharper than its derivation, which suggests a sum rule behind it. In the form $\Phi' = \pi T_{\text{eff}}$ the two conjectures of Section 7 read: Conjecture 7.1 is $T_{\text{eff}}(a) \leq (c/\pi)T^*(a)$, the effective height of the zeros that carry the floor is a bounded multiple of the horizon; Conjecture 7.3 is $T_{\text{eff}}/T^* \rightarrow c_0/\pi$, and the measured c_0 between 3.2 and 3.9 is T_{eff} between 1.0 and 1.24 horizons.

The limit in Conjecture 7.7 is a theorem once the edge law is assumed with two derivatives.

Proposition 7.8 (edge law implies tail law). *Assume RH. Let f be real, odd, supported in $[-a, a]$ and C^2 on $(-a, a)$, and suppose that for some $C > 0$, $\beta \in \mathbb{R}$ and $\delta_0 > 0$,*

$$f(a - \delta) = C(\log(a/\delta) + \beta)^{-1/2 + \psi(\delta)}, \quad 0 < \delta < \delta_0, \quad \psi^{(k)}(\delta) = o(\delta^{-k} \log^{-3/2}(1/\delta)) \quad (k = 0, 1, 2) \text{ as } \delta \rightarrow 0^+.$$

Then

$$\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = \frac{2C^2}{\pi}.$$

Proof. Step 1: the transform. Put $L(\delta) = \log(a/\delta) + \beta$, $\varphi(\delta) = CL(\delta)^{-1/2}$, so $\varphi'(\delta) = \frac{C}{2}L^{-3/2}\delta^{-1} > 0$ and $|\varphi''(\delta)| \leq C'L^{-3/2}\delta^{-2}$ on $(0, \delta_0)$. Let χ be smooth, 1 on $[0, \delta_0/2]$ and 0 beyond δ_0 ; put

$f_+(x) = (\varphi + \psi)(a - x)\chi(a - x)$ near the right edge, $f_-(x) = -f_+(-x)$, and $f_0 = f - f_+ - f_-$, which is C^2 with compact support in $(-a, a)$, so $F_0(t) = O(t^{-2})$. With $u = (\varphi + \psi)\chi$, $u(0^+) = u(\delta_0) = 0$, one integration by parts gives $F_+(t) = e^{ita} \int_0^{\delta_0} u(\delta) e^{-it\delta} d\delta = e^{ita} (it)^{-1} \int_0^{\delta_0} u'(\delta) e^{-it\delta} d\delta$. In $u' = \varphi'\chi + \psi'\chi + (\varphi + \psi)\chi'$ the last term is smooth and supported away from 0, contributing $O(1/t)$. For $J(t) = \int_0^{\delta_0} \varphi'\chi e^{-it\delta} d\delta$ split at $\delta = 1/t$: on $[0, 1/t]$, $e^{-it\delta} = 1 + O(t\delta)$ gives $\varphi(1/t) + O(t \int_0^{1/t} \varphi'\delta d\delta) = \varphi(1/t) + O(L(1/t)^{-3/2})$; on $[1/t, \delta_0]$ one more integration by parts gives $O(\varphi'(1/t)/t) + O(t^{-1} \int_{1/t}^{\delta_0} |(\varphi'\chi)'|) = O(L(1/t)^{-3/2})$. Hence $J(t) = CL(1/t)^{-1/2}(1 + O(1/\log t))$. The same splitting applied to $\int \psi'\chi e^{-it\delta}$, with $\int_0^{1/t} |\psi'| = o(\log^{-1/2} t)$ and the hypothesis on ψ'' , gives $o(\log^{-1/2} t)$. Therefore $F_+(t) = e^{ita} (it)^{-1} CL(1/t)^{-1/2}(1 + O(1/\log t))$, and since f is real and odd, $F_-(t) = -\overline{F_+(t)} = e^{-ita} (it)^{-1} CL(1/t)^{-1/2}(1 + O(1/\log t))$. So

$$F(t) = \frac{2C \cos(ta)}{it\sqrt{L(1/t)}} (1 + O(1/\log t)) + O(t^{-2}), \quad |F(t)|^2 = \frac{4C^2 \cos^2(ta)}{t^2 L(1/t)} (1 + O(1/\log t)) + O(t^{-3}).$$

Step 2: the sum over the zeros. Under RH the zeros are real and $N(t) = \frac{t}{2\pi} \log \frac{t}{2\pi e} + \frac{7}{8} + S(t) + R(t)$ with R smooth, $R = O(1/t)$, $R' = O(t^{-2})$, and, by Littlewood [5, Ch. XIV, §14.13], $S(t) = O(\log t / \log \log t)$ and $S_1(t) = \int_0^t S = O(\log t / (\log \log t)^2)$. Put $w(t) = t^{-2} L(1/t)^{-1}$, $\rho(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$, so $\rho/L(1/t) \cdot t^2 w = 1 + O(1/\log t)$. Since $\sum_{|\gamma|>T} = 2 \sum_{\gamma>T}$ and $4C^2 \cos^2(ta)w = 2C^2 w(1 + \cos 2ta)$, the claim is $\sum_{\gamma>T} w(\gamma) = \frac{1}{2\pi T}(1 + o(1))$, $\sum_{\gamma>T} \cos(2\gamma a)w(\gamma) = o(1/T)$, and $\sum_{\gamma>T} (w(\gamma)/\log \gamma + \gamma^{-3}) = o(1/T)$. For a positive decreasing g , $\sum_{\gamma>T} g(\gamma) = \int_T^\infty g dN = \int_T^\infty g \rho dt - g(T)S(T) - \int_T^\infty S g' dt + \int_T^\infty g R' dt$, and with $|S| \ll \log t$ the last three terms are $O(g(T) \log T + \int_T^\infty |g'| \log t + \int_T^\infty g/t^2)$: for $g = w$ this is $O(\log T/T^2) = o(1/T)$ while $\int_T^\infty w \rho = \frac{1}{2\pi T}(1 + O(1/\log T))$; for $g = w/\log t$ and $g = t^{-3}$ everything is $o(1/T)$. For the oscillatory sum put $G(t) = \cos(2ta)w(t)$: $\int_T^\infty G \rho dt = O(w(T)\rho(T)/a) = o(1/T)$ by one integration by parts against $\sin(2ta)/2a$, $\int_T^\infty G R' = o(1/T)$, and

$$\int_T^\infty G dS = -G(T)S(T) + G'(T)S_1(T) + \int_T^\infty S_1 G'' dt,$$

where $|G(T)S(T)| = O(\log T w(T))$, $|G'(T)S_1(T)| = O(a \log T w(T))$, and $|G''| \leq 4a^2 w + O(w/t)$ gives $\int_T^\infty |S_1 G''| = O(a^2 \int_T^\infty \frac{\log t}{(\log \log t)^2 t^2 L(1/t)} dt) = O(\frac{a^2}{T(\log \log T)^2})$, all $o(1/T)$. Hence $\sum_{|\gamma|>T} |F(\gamma)|^2 = 4C^2 \cdot \frac{1}{2\pi T}(1 + o(1)) = \frac{2C^2}{\pi T}(1 + o(1))$. $\square$

With Conjecture 8.1, $-\lambda' = 2C^2$, the proposition is Conjecture 7.7; its hypothesis is the edge law of Theorem 3.2 strengthened to an asymptotic equality with two derivatives, which is what Computation 3.4 measures and (H_∞) does not give; for the prime-free form, Proposition 8.4 derives that strengthening, with its constants, from (H_∞) and the Wiener–Hopf representation. RH enters twice, to read Q as the sum over real zeros and through Littlewood’s bound on S_1 ; without the latter the sum over the zeros is still $O(1/T)$, only the constant is lost. The $O(1/\log T)$ in the proof is the correction seen in the density integral of Computation 7.6; that the zero sums show none of it at one percent is the sum rule still to be found. The Euler–Lagrange equation gives an exact identity for the tail at every height, in which the logarithms cancel.

Proposition 7.9 (tail identity). *Let f be a minimizer at support a and let K be the operator with symbol Ψ , so that $Q(f) = \langle f, Kf \rangle - 2P(f)^2$ and f satisfies $Kf = \lambda f + 2P(f)s$ on $(-a, a)$ with $s(y) = \sinh(y/2)\mathbf{1}_{[-a, a]}(y)$. Put $g_{\text{out}} = Kf \cdot \mathbf{1}_{|y|>a}$, the force outside the window, and let f_T be the low-pass of f at height T , $\widehat{f_T} = F\mathbf{1}_{|t|\leq T}$. Then for every $T > 0$*

$$\frac{1}{2\pi} \int_{|t|>T} |F(t)|^2 (\Psi(t) - \lambda) dt = \frac{2P(f)}{2\pi} \int_{|t|>T} \overline{F(t)} \Sigma(t) dt - \int_{|y|>a} f_T(y) g_{\text{out}}(y) dy, \quad \Sigma = \hat{s}.$$

Explicitly, for $|y| > a$, $g_{\text{out}}(y) = -\int J_\Gamma(y-z)f(z)dz - \sum_n \Lambda(n)n^{-1/2}(f(y-\log n) + f(y+\log n))$.

Proof. On the whole line $Kf = \lambda f + 2Ps + g_{\text{out}}$, since the window equation holds inside and g_{out} is by definition the value of Kf outside. In Fourier form, $(\Psi - \lambda)F = 2P\Sigma + G$ with $G = \widehat{g_{\text{out}}}$. Multiply by $\overline{F}\mathbf{1}_{|t|>T}$ and integrate: the term with G is $\frac{1}{2\pi} \int_{|t|>T} \overline{F}G = \langle f - f_T, g_{\text{out}} \rangle$ by Parseval, and $\langle f, g_{\text{out}} \rangle = 0$ because the supports are disjoint. The formula for g_{out} is the kernel of K off the diagonal: from (2), $(K_\infty f)(y) = c_0 f(y) + \int_0^\infty [2f(y)e^{-2x} - (f(y-x) + f(y+x))e^{-x/2}](1 - e^{-2x})^{-1}dx$, which for $f(y) = 0$ is $-\int J_\Gamma(y-z)f(z)dz$, and the prime part of Q is the operator $-\sum_n \Lambda(n)n^{-1/2}(\tau_{\log n} + \tau_{-\log n})$. Both sides converge: g_{out} is smooth away from $\pm a$, decays exponentially, and by the edge law grows like $-C(\log(1/\delta))^{1/2}$ at distance δ outside the edge, while f_T is bounded and $O(1/|y|)$. $\square$

The identity is the mechanism of the tail law. Just outside the edge, $g_{\text{out}}(a+\delta) = -CL(\delta)^{1/2}(1+o(1))$ with $L(\delta) = \log(a/\delta) + \beta$ (the kernel $1/(2|s|)$ acting on the edge profile $CL^{-1/2}$), and the low-pass of f leaks outside the edge as $f_T(a+\delta) = CL(1/T)^{-1/2}\pi^{-1}(\frac{\pi}{2} - \text{Si}(T\delta))(1+o(1))$; their overlap is $2C^2/(\pi T)$ times $\int_0^\infty (\frac{\pi}{2} - \text{Si}) = 1$, with $L^{-1/2}L^{1/2} = 1$: no logarithm, and it is the tail law, since the left side is $\int_{|t|>T} |F|^2 \rho(1+o(1))$. The remaining pieces, the overlap with the smooth part of g_{out} (the regular part of the kernel, the other edge, the reflected primes) and the polar integral, are each of order $CL(1/T)^{-1/2}/T$, larger than C^2/T when C is small, and must cancel.

Computation 7.10 (the tail identity on the FEM minimizers). `rh_sum_rule.py`, on the edge-FEM minimizers with primes at $a = 0.5$ and 0.45 (Computation 8.10; the prime 2 is the only one in the window). The left side is the frequency integral of the closed-form transform against $\Psi - \lambda$ from T to $400T^*$, with the tail beyond by Computation 7.6; g_{out} is the archimedean kernel applied to the piecewise-linear f , the $1/(2s)$ part of J_Γ integrated exactly on every segment and the smooth remainder by Gauss-Legendre, plus the reflected prime; f_T is the inverse sine transform of the closed-form F on $[0, T]$. At $a = 0.5$, $T/T^* = 5, 10, 20$: right side over left side 1.000, 0.990, 0.980; at $a = 0.45$, $T/T^* = 5, 20$: 1.011, 0.989. Beyond twenty horizons the grid in y no longer resolves the oscillation of the leak, whose period is $2\pi/T$, and the ratio drifts to 0.94 and 0.91 at 50 and 100 horizons at $a = 0.5$; the left side itself equals the law of Computation 7.6 to 0.1% there. The polar piece is 0.30, 0.30, 0.28 of the left side at $a = 0.5$, so it is not small. The overlap of f_T with the edge-law force $-CL(\delta)^{1/2}$ alone, on $\delta < a/10$ with the C, β of Computation 8.10, is 0.93, 1.09, 0.98 of the law $-\lambda'/(\pi T)$ at $a = 0.5$ and 1.07, 1.01 at $a = 0.45$, and the rest of the overlap plus the polar piece is +0.09, -0.10, -0.01 and -0.04, -0.02 of the law: the edge overlap is the law and the two order- $CL^{-1/2}/T$ pieces cancel, each to about ten percent. What remains of the sum rule is analytic: the overlap of the leak with $-CL^{1/2}$ as an asymptotic expansion in $1/L(1/T)$ (the leading term has no logarithm; the first correction is not yet derived and is at most ten percent at two units of L), and the cancellation of the smooth overlap against the polar piece, which is a statement about the reflected primes and the regular part of the kernel at the edge.

The first of these is settled by a symmetry.

Lemma 7.11 (the edge overlap has no first-order logarithmic correction). *Let $L(\delta) = \log(a/\delta) + \beta$, $\varphi(\delta) = CL(\delta)^{-1/2}$ the edge profile and $\gamma_e(\delta) = -CL(\delta)^{1/2}$ the edge force, both on $0 < \delta < \delta_0$, and let the leak of φ through the sharp low-pass at height T be $\ell_T(\delta) = \int_0^{\delta_0} \varphi(u) \frac{\sin(T(\delta+u))}{\pi(\delta+u)} du$. Then, with $L_T = L(1/T)$,*

$$-2 \int_0^{\delta_0} \ell_T(\delta) \gamma_e(\delta) d\delta = \frac{2C^2}{\pi T} J(L_T), \quad J(L) = \int_0^\infty \int_0^\infty \left[\frac{L + \log(1/v)}{L + \log(1/s)} \right]^{1/2} \frac{\sin(s+v)}{s+v} ds dv = 1 + O(L^{-2}),$$

the integrals taken in the Abel sense and cut where the model ends. The term of order $1/L$ vanishes: the kernel $\sin(s+v)/(s+v)$ is symmetric in (s, v) while $\log[(L + \log(1/v))/(L + \log(1/s))]$ is

antisymmetric, so only even powers of it survive, and the leading one is $\frac{1}{8} \log^2(s/v)/L^2$, which integrates to $\pi^2/(24L^2)$ against $\int_0^1 \log^2 \frac{1-\sigma}{\sigma} d\sigma = \pi^2/3$ and $\int_0^\infty \sin w dw = 1$ after the substitution $s = w\sigma$, $v = w(1-\sigma)$. Numerically $J(L) = 1.020, 1.012, 1.008, 1.005, 1.005, 1.003$ for $L = 3, \dots, 8$ (`rh_sum_rule.py`); the series $1 + \pi^2/(24L^2) - 1.898/L^3 + \dots$ is asymptotic and useful only from $L \approx 7$.

Proof. Substitute $u = s/T$, $\delta = v/T$ in the double integral, so that $\varphi(s/T)\gamma_e(v/T) = -C^2[L_T + \log(1/v)]^{1/2}[L_T + \log(1/s)]^{-1/2}$ and the kernel becomes $T \sin(s+v)/(\pi(s+v))$ with $du d\delta = T^{-2} ds dv$. Write the ratio as $\exp(u)$ with $u = \frac{1}{2}[\log(1+x_v) - \log(1+x_s)]$, $x = \log(1/\cdot)/L_T$; u changes sign under $s \leftrightarrow v$ and the kernel does not, so $\int \int e^u k = \int \int \cosh(u) k = \int \int (1 + u^2/2 + \dots) k$. With $s = w\sigma$, $v = w(1-\sigma)$, $ds dv = w dw d\sigma$ and $u = \frac{1}{2} \log \frac{1-\sigma}{\sigma} L_T^{-1} + O(L_T^{-2})$ is independent of w at leading order, which gives the constant and the second-order term. $\square$

So the edge overlap is the tail law with a correction of order $L(1/T)^{-2}$, one to two percent at the heights of Computation 7.6, which is what the zero sums show. What remains of the sum rule is the cancellation of the smooth overlap against the polar piece. At leading order both are $2CL_T^{-1/2}/(\pi T)$ times a constant: the polar piece is $2P\langle f, s_{>T} \rangle$ with the high-pass of $s = \sinh(y/2)\mathbf{1}_{[-a,a]}$ concentrated in the Gibbs layer of width $1/T$ at the edges, giving $4P \sinh(a/2) CL_T^{-1/2}/(\pi T)$, and the smooth overlap is $-2g_{\text{reg}}(a) CL_T^{-1/2}/(\pi T)$ with $g_{\text{reg}}(a)$ the regular part of g_{out} at the edge. They cancel if and only if $g_{\text{reg}}(a) = 2P \sinh(a/2)$, which is the value of Kf just inside the edge by the eigen-equation: the sum rule reduces to the statement that the regular part of Kf is continuous across the edge, the singularity $\mp CL^{1/2}$ being carried entirely by the outside. Computation 7.10 supports it to about twenty percent of $g_{\text{reg}}(a)$. It is a local statement, and it holds.

Lemma 7.12 (the regular part of Kf is continuous across the edge). *Let f satisfy the hypothesis of Proposition 7.8 and put $\varphi(\delta) = CL(\delta)^{-1/2}$. Then*

$$\lim_{\delta \rightarrow 0^+} \left[(K_\infty f)(a - \delta) - (K_\infty f)(a + \delta) - CL(\delta)^{1/2} \right] = 0.$$

Consequently, for the minimizer, $g_{\text{reg}}(a) := \lim_{\delta \rightarrow 0^+} [g_{\text{out}}(a + \delta) + CL(\delta)^{1/2}] = (Kf)(a^-) = 2P(f) \sinh(a/2)$, and the function $h = 2P(f)s + g_{\text{reg}}$ is continuous at $\pm a$.

Proof. Write $k(x) = e^{-x/2}/(1-e^{-2x}) = \frac{1}{2x} + \frac{1}{4} + O(x)$ and $m(x) = e^{-2x}/(1-e^{-2x}) = \frac{1}{2x} - \frac{1}{2} + O(x)$, so that by (2) $(K_\infty f)(y) = c_0 f(y) + \int_0^\infty [2f(y)m(x) - (f(y-x) + f(y+x))k(x)] dx$. Outside, $f(a+\delta) = 0$ gives $(K_\infty f)(a+\delta) = -\int_0^{2a} f(a-u)k(u+\delta) du$. Inside, with $f(a-\delta+x) = \varphi(\delta-x)\mathbf{1}_{x < \delta}$, split the integral at $x = \delta$: on $[0, \delta]$ it is $\int_0^\delta [2\varphi(\delta)(m-k) + (2\varphi(\delta) - \varphi(\delta+x) - \varphi(\delta-x))k] dx = O(\delta\varphi(\delta)) + O(\int_0^\delta x^2 |\varphi''| k) = o(1)$, since $|\varphi''| \leq C'L^{-3/2}\delta^{-2}$; on $[\delta, \infty)$ it is $2\varphi(\delta) \int_\delta^\infty m - \int_\delta^\infty \varphi(\delta+x)k(x) dx = \varphi(\delta) \log \frac{1}{1-e^{-2\delta}} - \int_{2\delta}^{2a} f(a-u)k(u-\delta) du$. Hence

$$(K_\infty f)(a-\delta) - (K_\infty f)(a+\delta) - CL(\delta)^{1/2} = \left[\varphi(\delta) \log \frac{1}{2\delta} - CL(\delta)^{1/2} \right] + \int_0^{2a} f(a-u) [k(u+\delta) - k(u-\delta)\mathbf{1}_{u > 2\delta}] du + o(1).$$

The bracket is $CL(\delta)^{-1/2} [\log \frac{1}{2\delta} - L(\delta)] = -C(\beta + \log 2a)L(\delta)^{-1/2} \rightarrow 0$. In the integral, the part over $u < 2\delta$ is $O(\varphi(2\delta))$, and over $u > 2\delta$ one has $k(u+\delta) - k(u-\delta) = -\delta/(u^2 - \delta^2) + O(\delta)$, whose contribution is $O(\delta \int_{2\delta}^\infty \varphi(u)u^{-2} du) + O(\delta) = O(\varphi(2\delta)) + O(\delta) \rightarrow 0$. The terms with ψ in place of φ are integrable against $1/u$ by the hypothesis on ψ and contribute the same continuous function on both sides. For the consequence, $g_{\text{out}} = K_\infty f + K_P f$ outside and $K_P f$ is continuous across the edge (f is continuous), so $g_{\text{reg}}(a) = (K_\infty f)(a^-) + (K_P f)(a) = (Kf)(a^-) = \lambda f(a) + 2P \sinh(a/2)$, and $f(a) = 0$. $\square$

Corollary 7.13 (the sum rule). *Under the hypotheses of Proposition 7.8, for the minimizer,*

$$\frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi - \lambda) dt = \frac{2C^2}{\pi T} \left(1 + O(L(1/T)^{-1})\right),$$

the edge overlap contributing $\frac{2C^2}{\pi T}(1 + O(L^{-2}))$ by Lemma 7.11 and the polar piece plus the smooth overlap contributing $O(C^2/(TL))$.

Proof. In the identity of Proposition 7.9 write $g_{\text{out}} = -CL^{1/2}\mathbf{1}_{\text{edge}} + g_{\text{reg}}$. The polar piece is $2P\langle f, s_{>T} \rangle$ and the smooth overlap is $-\langle f_T, g_{\text{reg}} \rangle = -\langle f, g_{\text{reg}} \rangle + \langle f, (g_{\text{reg}})_{>T} \rangle = \langle f, (g_{\text{reg}})_{>T} \rangle$, the low-pass being self-adjoint and the supports disjoint; their sum is $\langle f, h_{>T} \rangle$ with $h = 2Ps + g_{\text{reg}}$. Inside the window the eigen-equation reads $2P \sinh(y/2) = K_{\infty}f_{\text{int}} + K_P f + e_{\text{in}} - \lambda f$, where f_{int} is the interior part of f and e_{in} the bounded remainder of K_{∞} on the edge profile; outside, $g_{\text{reg}} = K_{\infty}f_{\text{int}} + K_P f + e_{\text{out}}$ with $e_{\text{out}} = (K_{\infty}\varphi) + CL^{1/2}$. The part $K_{\infty}f_{\text{int}} + K_P f$ is smooth across the edge, so h is a smooth function plus the edge terms $e_{\text{in}} - \lambda f$ and e_{out} , which are $O(C)$, continuous at the edge by Lemma 7.12, and of the form $C \times (\text{a function with an expansion in powers of } L^{-1/2})$. The high-pass of the smooth part against f is $O(T^{-N})$; the high-pass of the $O(C)$ edge terms against f , whose own edge is $CL^{-1/2}$, is $O(C^2/(TL))$ by the computation of Lemma 7.11 with $L^{-1/2}$ in place of $L^{1/2}$. $\square$

So the sum rule is the tail identity with two facts about the edge: the overlap of the leak with the singular force is log-free by symmetry, and the regular force is continuous across the edge by the eigen-equation. The data say that the $O(1/L)$ of the corollary is small in practice (the zero sums sit on the law to one percent from five horizons on, where $L(1/T) \approx 2$); its coefficient is not computed here. The corollary is, however, only the far-field form of something more robust.

Computation 7.14 (the tail law is analytic). `rh_sum_rule.py`. (i) On the FEM minimizers with primes, the edge law $CL(\delta)^{-1/2}$ with the C, β of Computation 8.10 is not yet accurate at the scales that the heights of Computation 7.6 probe: $f(a-\delta)^2 L(\delta)/C^2$ at $\delta = 1/(kT^*)$ is 1.48, 2.04, 3.20, 3.17 for $k = e$ at $a = 0.4, 0.45, 0.5, a_3, 1.20, 1.33, 1.54, 1.43$ at $k = 10$, and reaches 1.02–1.06 only at $k = 100$; yet the tail law holds at one percent from five horizons. (ii) On the prime-free critical points of Computation 8.10 ($a = 0.3, 0.4, 0.5, 0.6$; the form $W_{\infty} - 2P^2$, whose floor is $+0.223, -0.078, -0.322, -0.536$), with λ' the matrix-side derivative and no zero anywhere, $T \cdot \frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi_{\infty} - \lambda) dt$ over $-\lambda'/\pi$ is

a	$T/T^* = 3$	5	10	20	50	100
0.3	0.986	0.980	1.014	1.008	1.002	0.999
0.4	0.978	1.020	1.009	1.001	1.000	1.000
0.5	0.998	1.000	0.997	0.997	1.001	1.001
0.6	1.012	1.007	1.004	1.001	0.999	1.000

The law $T \frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi - \lambda) \rightarrow -\lambda'/\pi$ is therefore a property of critical points of window forms with the logarithmic symbol, holding to one percent from three horizons on whatever the sign of the floor, and the zeros enter only through (S_{κ}) : they sample this analytic tail at their density. (iii) The sampling constant $M(\kappa)$ of (S_{κ}) , the density integral over the zero sum above κT^* on the FEM minimizers with primes, is 8.9, 1.53, 1.39, 1.03, 1.01, 1.01, 1.06, 1.05 at $a = 0.5$, 3.6, 0.91, 1.14, 1.09, 1.08, 1.07, 1.07, 1.03 at $a = 0.45$ and 1.8, 0.73, 1.16, 1.01, 1.05, 1.05, 1.08, 1.03 at $a = 0.4$, for $\kappa = 1, 1.5, 2, e, 3, 4, 5, 10$: above the exhaustion height the zeros sample the tail with constant one to within ten percent, and below two horizons they do not.

Two things follow. The finite-height sum rule is analytic and has an exact form: the dilation derivative is itself an edge pairing, and the tail is its remainder at height T .

Proposition 7.15 (the dilation derivative as an edge pairing). *Let f be a critical point as in Proposition 7.9, $\mathcal{D}(f)$ the dilation form (6), and $(xf)' = f + xf'$ the dilation generator, an L^1 function supported in the window. For $T > 0$ let $((xf)')_T$ be its low-pass at height T and $B_T = \langle g_{\text{out}}, ((xf)')_T \rangle$. Then for every T*

$$\mathcal{D}(f) = \text{tail}(T) + \frac{1}{\pi} T |F(T)|^2 (\Psi(T) - \lambda) + \text{tail}_D(T) - 4P \langle s, ((xf)')_{>T} \rangle + 2B_T,$$

with $\text{tail}(T) = \frac{1}{2\pi} \int_{|t|>T} |F|^2 (\Psi - \lambda)$ and $\text{tail}_D(T) = \frac{1}{2\pi} \int_{|t|>T} |F|^2 t \Psi'$. In particular B_T converges, $\mathcal{D}(f) = 2B_\infty$, and by Theorem 5.2 (Proposition 5.3 for the exact form) and the boundary law (Conjecture 8.1, proved under the edge law with derivatives in Proposition 8.3)

$$-a\lambda' = 2 \lim_{T \rightarrow \infty} \langle g_{\text{out}}, ((xf)')_T \rangle, \quad B_\infty = aC^2.$$

Proof. Write $\mathcal{D} = \frac{1}{2\pi} \int |F|^2 t \Psi' + 2P^2 + 2PM$ and cut the integral at $|t| = T$. On $|t| < T$ integrate by parts: $\frac{1}{2\pi} \int_{|t|<T} |F|^2 t \Psi' = \frac{1}{\pi} T |F(T)|^2 \Psi(T) - \frac{1}{2\pi} \int_{|t|<T} \Psi |F|^2 - \frac{1}{2\pi} \int_{|t|<T} \Psi t (|F|^2)'$. The second term is $-(\lambda + 2P^2) + \frac{1}{2\pi} \int_{|t|>T} \Psi |F|^2$. In the third, $\Psi t (|F|^2)' = 2 \text{Re} [\overline{\Psi F} t F']$ and $\Psi F = \lambda F + 2P\Psi + G$ by the eigen-equation; since $tF' = -(\widehat{xf})'$, Parseval gives $-\frac{1}{2\pi} \int_{|t|<T} \Psi t (|F|^2)' = -\lambda [\frac{1}{\pi} T |F(T)|^2 - 1 + \frac{1}{2\pi} \int_{|t|>T} |F|^2] + 4P \langle s, ((xf)')_T \rangle + 2 \langle g_{\text{out}}, ((xf)')_T \rangle$. Collecting, and using $\langle s, (xf)' \rangle = -M/2$ (integration by parts, $f(\pm a) = 0$) to write $4P \langle s, ((xf)')_T \rangle = -2PM - 4P \langle s, ((xf)')_{>T} \rangle$, the terms $\pm\lambda$, $\pm 2P^2$ and $\pm 2PM$ cancel and the identity follows. As $T \rightarrow \infty$ every term but $2B_T$ tends to 0 (tail and tail_D since $|F|^2 \Psi$ is integrable; the boundary term since $t|F|^2 \Psi \rightarrow 0$; the polar term since $s_{>T} \rightarrow 0$ in L^2 and $(xf)' \in L^1 \cap L^\infty$ away from the edge with the edge singularity $\delta^{-1} L^{-3/2}$ integrable), so $B_T \rightarrow \mathcal{D}/2$. $\square$

Corollary 7.16 (the tail as the remainder of the pairing). $\text{tail}(T) = 2(B_\infty - B_T) + 4P \langle s, ((xf)')_{>T} \rangle - \frac{1}{\pi} T |F(T)|^2 (\Psi(T) - \lambda) - \text{tail}_D(T)$. *The polar term is not small: the high-pass of $(xf)'$ carries the edge singularity $\delta^{-1} L^{-3/2}$, whose Gibbs layer against the jump of s gives $-2aPC \sinh(a/2) L(1/T)^{-1/2} (1 + o(1))$, decaying only logarithmically, and $2(B_\infty - B_T)$ cancels it at that order, as in Corollary 7.13; the tail is the $O(1/T)$ remainder of that cancellation. The identity is the dilation identity in Fourier form and does not by itself make the law easier; its value is the exact expression of $-a\lambda'$ as an edge pairing.*

Computation 7.17 (the pairing identity on the prime-free critical point). `rh_cutoff_identity.py`, on the prime-free FEM critical point at $a = 0.5$, every term computed independently: $\mathcal{D} = D_\infty + 2P^2 + 2PM = 1.129084$ from the profile ($D_\infty = 1.020022$, $P = 0.134363$, $M = 0.271485$), against $-a\lambda' = 1.129075$ from the ball-arithmetic engine; $B_T = \langle g_{\text{out}}, ((xf)')_T \rangle$ from the archimedean kernel in x -space and the inverse sine transform of $-tS'(t)$, $B_T = 0.523, 0.549, 0.550, 0.551$ at $T/T^* = 3, 5, 10, 20$ against $\mathcal{D}/2 = 0.5645$; the right side of Proposition 7.15 over $\mathcal{D}$ is 1.010, 1.020, 1.020, 1.011, the two percent being the resolution of the leak of $(xf)'$, whose edge singularity is $\delta^{-1} L^{-3/2}$, on the y -grid; and $\text{tail}(T)$ is 0.997, 0.998, 0.992, 0.989 of $\mathcal{D}/(\pi a T)$. The polar term is $-0.047, -0.044, -0.039, -0.035$, five times the tail and decaying like $L(1/T)^{-1/2}$ as the corollary says, so $B_\infty - B_T$ is dominated by its cancellation; separating the $O(1/T)$ remainder needs B_T to a part in ten thousand, beyond this computation.

The sharpest form of the sum rule is local in t , and it is exact.

Computation 7.18 (the envelope law). The average of $t^2(\Psi(t) - \lambda)|F(t)|^2$ over a window of width $4\pi/a$ centred at T (two periods of the interference of the two edges), divided by $-\lambda'$, on the FEM critical points (`rh_sum_rule.py`):

	$T/T^* = 1$	2	3	5	10	20	50	100	300
$a = 0.5$, no primes	1.197	1.016	1.012	1.001	1.001	1.000	1.000	1.000	1.000
$a = 0.3$, no primes	-2.81	1.025	1.112	1.016	1.006	1.001	1.000	1.000	1.001
$a = 0.5$, primes	-6.30	2.244	1.270	1.046	1.006	1.021	1.005	1.011	1.008
$a = 0.4$, primes	0.782	1.876	1.035	1.145	1.013	0.957	1.032	0.993	0.975

Without primes the local envelope equals $-\lambda'$ to a part in a thousand from ten horizons on and to 0.1% from twenty, over two decades of height, with no logarithmic drift; with primes it fluctuates by a few percent about 1, the cosines of the symbol beating against the interference of the edges.

Conjecture 7.19 (Wiener–Hopf form of the edge). *Let f be a critical point on $[-a, a]$ and $\sigma_\infty = \Psi_\infty - \lambda$. For a real even symbol σ with finitely many real zeros $\pm t_j$ write $\tilde{\sigma} = \sigma \prod_j (t^2 + t_j^2)/(t^2 - t_j^2) > 0$ and $\tilde{\sigma} = \sigma_+ \sigma_-$ for its canonical factorization, $\sigma_\pm = \tilde{\sigma}^{1/2} e^{\pm i\theta}$ on the line with*

$$\theta(t) = \frac{1}{2} \mathcal{H}[\log \tilde{\sigma}](t) = \frac{t}{\pi} \int_0^\infty \frac{\log \tilde{\sigma}(\tau) - \log \tilde{\sigma}(t)}{t^2 - \tau^2} d\tau, \quad (10)$$

odd in t , so that σ_+ continues analytically without zeros to the upper half-plane and σ_- to the lower. (i) Without primes ($\sigma = \sigma_\infty$) there are a height $T_0(a)$ and a real constant κ such that, with $S(t) = \int_0^a f \sin(tx) dx$ ($F = 2iS$),

$$S(t) = -\frac{|A| \cos(ta + \theta(t) - \kappa/t)}{t \tilde{\sigma}(t)^{1/2}} (1 + O(t^{-2})), \quad |A|^2 = -\frac{1}{2} \lambda', \quad t > T_0; \quad (11)$$

equivalently $F(t) = A(t)e^{ita}/(t\sigma_-(t)) - \overline{A(t)}e^{-ita}/(t\sigma_+(t)) + O(t^{-3})$ with $A(t) = -i|A|e^{-i\kappa/t}$: the amplitude of each edge is purely imaginary up to the phase κ/t , and there is no free phase. (ii) With the entries n inside the window ($\log n < 2a$; $c_n = \Lambda(n)n^{-1/2}$) each edge is accompanied by its echoes, at $a - \log n$, $a - \log n - \log m, \dots$, as long as they stay inside the window. For a single entry with $a < \log n < 2a$ there is one echo; the symbol of the edge is $\sigma_{\text{eff}} = \sigma_\infty - c_n^2/\sigma_\infty$, and with $\rho(t) = c_n/\sigma_\infty(t)$, θ_{eff} the phase (10) of σ_{eff} ,

$$S(t) = -\frac{|A| |1 + \rho e^{-it \log n}| \cos(ta + \theta_{\text{eff}}(t) + \arg(1 + \rho e^{-it \log n}) - \kappa/t)}{t \tilde{\sigma}_{\text{eff}}(t)^{1/2}} (1 + O(t^{-2})). \quad (12)$$

To first order in ρ , (12) is (11) with the symbol $\sigma_a = \Psi_a - \lambda$ of the window, $\Psi_a = \Psi_\infty - \sum_{\log n < 2a} 2\Lambda(n)n^{-1/2} \cos(t \log n)$. In both cases the local envelope of Computation 7.18 is $2|A|^2 = -\lambda'$, exactly in (i) and up to $O(\rho^2)$ in (ii), and the nulls of S sit at $ta = \pi/2 + k\pi - \vartheta(t) + \kappa/t$ with ϑ the phase of the respective form.

The form comes from the two edges as two half-line problems, and the argument is the classical one for a Wiener–Hopf equation with a symbol of order zero; no cutoff is needed to see it. Write the eigen-equation on the whole line, $(K - \lambda)f = 2Ps 1_{[-a, a]} + g_{\text{out}}$ with $g_{\text{out}} = Kf$ outside the window as in Proposition 7.9, transform it, $\sigma F = 2P\Sigma + G$, and multiply by e^{-ita} . The function $\tilde{F} = e^{-ita}F = \int_{-2a}^0 f(a+u)e^{itu} du$ is analytic and bounded in the lower half-plane and $O(1/t)$ (a “−” function); $\tilde{G}_+ = \int_{u>0} g_{\text{out}}(a+u)e^{itu} du$ is analytic in the upper half-plane (a “+” function) and $O(t^{-1} \log^{1/2} t)$ by Lemma 7.12; and $\tilde{H} = e^{-ita}(2P\Sigma + \int_{x<-a} g_{\text{out}} e^{itx} dx)$ is a “−” function,

$\tilde{H} = -2iP \sinh(a/2)/(t-i0) + e^{-2ita}\Phi_-(t) + O(t^{-2})$, the first term the step of the polar term at the edge and $\Phi_- = O(t^{-1} \log^{1/2} t)$ the far edge, its polar step and its outside force. The primes outside the window drop out: for $\log n > 2a$ the shifted copies $f(x \mp \log n)$ lie entirely outside the window, so they are part of g_{out} , and their transforms $e^{\pm it \log n} F$ reproduce the term $-2\Lambda(n)n^{-1/2} \cos(t \log n) F$ of σF . So, exactly,

$$\sigma_a \tilde{F} = \tilde{H} + \tilde{G}_+, \quad (13)$$

with the symbol σ_a of the window, the partial copies of the entries inside the window (the part of $f(x - \log n)$ on $x > a$, of $f(x + \log n)$ on $x < -a$) staying in $\tilde{G}_+$ and $\tilde{H}$ on their sides.

Without primes. Divide (13) by σ_+ and split $\tilde{H}/\sigma_+$ additively into its parts analytic in the two half-planes, $[\cdot]_- + [\cdot]_+$. The function $E = \sigma_- \tilde{F} - [\tilde{H}/\sigma_+]_- = \tilde{G}_+/\sigma_+ + [\tilde{H}/\sigma_+]_+$ is entire and tends to zero in every direction, since $\sigma_{\pm} \sim (\log t)^{1/2}$, so $E \equiv 0$ and $\tilde{F} = [\tilde{H}/\sigma_+]_-/\sigma_-$. For the polar step, $[1/((t-i0)\sigma_+(t))]_- = 1/((t-i0)\sigma_+(0))$; for the far edge, $[e^{-2ita}\Phi_-/\sigma_+]_- = e^{-2ita}\Phi_-/\sigma_+ - \sum_j c_j/(t-z_j)$ by closing the Cauchy integral in the lower half-plane, the sum running over the singularities z_j of $1/\sigma_+ = \sigma_-/\sigma$ there, the complex zeros of σ , with $c_j \propto e^{-2iz_j a}$. Hence

$$\tilde{F}(t) = \frac{A(t)}{t\sigma_-(t)} + \frac{e^{-2ita}\Phi_-(t)}{\sigma(t)} + O(t^{-2}), \quad A(t) = -\frac{2iP \sinh(a/2)}{\sigma_+(0)} - \sum_j \frac{c_j t}{t-z_j} = A_0 + \frac{A_1}{t} + \dots,$$

the first term the right edge and the second the left one ($e^{-2ita}\Phi_-/\sigma = e^{-ita}F_L$). The symmetry $\tilde{F}(-t) = \overline{\tilde{F}(t)}$ of a real f , with $\sigma_-(-t) = \overline{\sigma_-(t)}$, gives $A(-t) = -\overline{A(t)}$: A_0 is purely imaginary and A_1 real, so $A(t) = A_0 e^{-i\kappa/t} (1 + O(t^{-2}))$ with $\kappa = A_1/|A_0|$, the first correction is a phase, and (11) follows. The far edge and the polar term fix $|A_0|$ (the sum over the c_j is the coupling of the two edges, which is the quantization of $\lambda(a)$), and $|A_0|^2 = -\lambda'/2$ is the boundary law $\lambda' = -2C^2$ read as a flux: the x -space asymptotics of $1/\sigma_-$ is the edge profile $L(\delta)^{-1/2}$, so $|A_0| = C$. All the logarithms of the edge profile are inside σ_- , which is why the profile of Computation 7.14(i) can be 1.5–3 times above its asymptotic law at scale $1/T$ while the tail law holds at that T to one percent; Proposition 7.8 and Corollary 7.13 are the far field of this structure. The real zeros $\pm t_j$ of σ are the frequencies of the standing waves $e^{\pm it_j x}$ that solve $(K - \lambda)u = 0$ on the line and make up the interior of the critical point; with them the homogeneous problem has solutions, E is a rational function with simple poles at $\pm t_j$ instead of zero, and the standing-wave amplitudes enter A_0 and κ (the $\arctan(t_j/t)$ of the zeros' factors are part of κ). Proposition 7.23 and Corollary 7.24 below make this rigorous, for every L^2 critical point of the prime-free form and without any hypothesis on the edge: the decay of the three functions needs only $f \in L^2$, and the factor of a symbol growing like $\log t$ is the outer function of $\tilde{\sigma}^{1/2}$.

With primes. An entry inside the window shifts the window onto itself: $(K_{\infty} - \lambda)f = 2Ps + \sum_n c_n(f(x - \log n) + f(x + \log n))$ on the window, K_{∞} the archimedean part of K . The right edge's singularity is thereby copied to $a - \log n$, the copy to $a - \log n - \log m$, and so on while the points stay inside the window: a finite set of echoes, a chain for each entry, and the mirror chains from the left edge. Take one entry with $a < \log n < 2a$, so that each chain has one link, and split f near a and near $a - \log n$ into an edge piece f_R and an echo piece f_e with a partition of unity. At the echo point $(K_{\infty} - \lambda)f_e = c_n f_R(x + \log n) + \text{smooth}$, so $F_e = (c_n/\sigma_{\infty})e^{-it \log n} F_R$ up to rapidly decaying terms. At the edge $(K_{\infty} - \lambda)f_R = 2Ps + c_n f_e(x - \log n) + \text{smooth}$ on $x < a$, and the echo shifted back is a two-sided singularity at a with transform $c_n e^{it \log n} F_e = (c_n^2/\sigma_{\infty})F_R$; moved to the left it makes the edge a half-line problem with the real even symbol $\sigma_{\text{eff}} = \sigma_{\infty} - c_n^2/\sigma_{\infty}$, to which everything above applies: $\tilde{F}_R = A(t)/(t\sigma_{\text{eff},-})$. Since f is odd, $F = 2i \text{Im}(F_R + F_e)$, which is (12). With the symbol $\sigma_a = \sigma_{\infty} - 2c_n \cos(t \log n)$ of the window instead one has $\sigma_a^{-1/2} = \sigma_{\infty}^{-1/2} (1 + \rho \cos(t \log n)) + O(\rho^2)$ and $\theta_a = \theta_{\infty} - \rho \sin(t \log n) + O(\rho^2)$, the same as (12) to first order in ρ : the Hilbert transform of the

cosine is the echo. The difference is at second order and it is the window's: for a constant σ_∞ the factorization $\sigma_\infty - 2c_n \cos \tau = (\alpha - \beta e^{-i\tau})(\alpha - \beta e^{i\tau})$, $\alpha^2 = \frac{1}{2}(\sigma_\infty + \sqrt{\sigma_\infty^2 - 4c_n^2})$, $\beta = c_n/\alpha$, expands $1/\sigma_{a,-}$ into the infinite chain $\alpha^{-1} \sum_k (c_n/\alpha^2)^k e^{-ik\tau}$, with α^2 the fixed point of $\sigma \mapsto \sigma_\infty - c_n^2/\sigma$: the edge of a half-line with all its echoes, while the window cuts the chain where it leaves $[-a, a]$, and σ_{eff} is the continued fraction $\sigma_\infty - c_n^2/(\sigma_\infty - c_n^2/(\dots))$ with as many levels as there are echoes inside. Part (iii) of the following computation measures exactly this.

Computation 7.20 (the Wiener–Hopf phase). On the prime-free critical point at $a = 0.5$ of Computation 7.14 (edge FEM, 486 nodes to $10^{-12}a$, $\lambda = -0.32224$, $\lambda' = -2.25815$) the symbol $\sigma = \Psi_\infty - \lambda$ has the single real zero $t_\lambda = 4.562$, and θ of (10) is computed by the trapezoidal rule on a grid of step 0.01 to 8000 and geometric to 10^{11} (self-test on $\log \frac{t^2+1}{t^2+4}$, whose phase is $\arctan \frac{1}{t} - \arctan \frac{2}{t}$: error 5×10^{-10}); `rh_wiener_hopf.py`, logs in `data/`. (i) *The nulls.* S has 319 nulls t_k between 2.8 and 120 horizons, $t_k a = \pi/2 + k\pi + r_k$. The shift r_k is 0.339, 0.294, 0.239, 0.199, 0.162, 0.141, 0.136 at 3.2, 5, 10, 20, 50, 100, 120 horizons, and $-\theta(t_k)$ is 0.259, 0.243, 0.214, 0.187, 0.157, 0.138, 0.134; the difference $r_k + \theta(t_k) = 0.080, 0.051, 0.025, 0.013, 0.005, 0.0023, 0.0022$ is κ/t_k with $\kappa = 4.34$ (fitted on $t \geq 10T^*$; against $t_\lambda = 4.56$, the $\arctan(t_\lambda/t)$ of the zero's factor), and the residual $r_k + \theta(t_k) - \kappa/t_k$ has root mean square 1.2×10^{-4} over all 319 nulls, with no trend. The alternatives fail: a $1/t$ shift alone leaves 0.19; the adiabatic value of the phase, $-\frac{\pi}{4} t \frac{d}{dt} \log \tilde{\sigma} \approx -\frac{\pi}{4 \log(t/2\pi)}$, fitted as $c/\log(t/2\pi)$ (best $c = 0.79$ against $\pi/4 = 0.785$), leaves 3.5×10^{-3} , thirty times the exact phase, its error being 0.029, 0.018, 0.010 at 3.2, 3.9, 5 horizons, where the Hilbert integral (10) differs from its local value by that much. A second mesh (471 nodes, coarser between $\delta = 10^{-6}$ and 10^{-3}) reproduces r_k to 2×10^{-3} up to 20 horizons and drifts beyond, where its mesh at scale $1/t$ shows. (ii) *Pointwise.* With $|A| = (-\lambda'/2)^{1/2} = 1.0626$ and the fitted κ , nothing else adjusted, the relative L^2 distance over one interference period $4\pi/a$ between S and $-|A| \cos(ta + \theta - \kappa/t)/(t\tilde{\sigma}^{1/2})$ is $4.5 \times 10^{-3}, 1.5 \times 10^{-3}, 3.4 \times 10^{-4}, 8 \times 10^{-5}, 3 \times 10^{-5}$ at 3, 5, 10, 20, 50 horizons, decaying like T^{-2} (2.7×10^{-4} at 100, the mesh); with $\kappa = 0$ it is 0.093, 0.052, 0.026, 0.013, 0.005, the phase κ/t itself, and with $\theta = 0$ it is 0.36, 0.29, 0.24, 0.20, 0.16. (iii) *With the prime 2* (the critical point of $W - 2P^2$ at $a = 0.5$, $\lambda = 1.9 \times 10^{-4}$, $\lambda' = -0.011236$; the entry 2 is the only one inside the window, with $a < \log 2 < 2a$, so each edge has one echo, at $\pm(a - \log 2) = \pm 0.193$). The window's symbol $\Psi_\infty - 0.980 \cos(t \log 2) - \lambda$ has three real zeros, 3.31, 6.95, 10.53, and its phase oscillates at the prime's frequency with amplitude about $\rho = 0.490/\sigma_\infty$: $-\theta_a(t_k) = 0.16, 0.29, 0.05, 0.13, 0.13, 0.21, 0.23, 0.07$ at 9.8, 15, 20, 30, 50, 70, 100, 120 horizons. The nulls follow it: $r_k + \theta_a(t_k) - \kappa/t_k$ has root mean square 0.010 over 3–120 horizons ($\kappa = 62.4$), against 0.25–0.32 for r_k , 0.11–0.20 for $r_k + \theta_a(t_k)$, 0.08–0.13 for the smooth law $c/\log(t/2\pi)$, and 0.070 for the phase of the prime-free symbol at the same λ plus κ/t ; so the wobble of the nulls is the Hilbert transform of the prime's cosine, and the form (11) with σ_a is within 0.14, 0.051, 0.024, 0.016, 0.010 of S at 3, 5, 10, 20, 50 horizons, a percent-level remainder that does not decay. The echo form (12) removes it: with $\sigma_{\text{eff}} = \sigma_\infty - 0.240/\sigma_\infty$ (real zeros 3.86, 6.29, 10.26) and $\kappa = 62.2$, the null residual is 1.1×10^{-3} in root mean square over the 204 nulls between 5 and 80 horizons (largest 3.6×10^{-3} ; the window's symbol leaves 9.8×10^{-3} there, no shift 0.32), rising to 0.02 above 80 horizons where the mesh at scale $1/t$ shows, and the pointwise distance is 0.010, 0.0021, 0.0007 at 10, 20, 50 horizons, decaying like T^{-2} (0.0099 at 100, the mesh; 0.15 and 0.044 at 3 and 5, where the interior of this critical point, with $\lambda \approx 0$ and three standing waves, is still felt). So the second-order remainder of the window's symbol is the echo that the window cuts off, and the few-percent modulation of the envelope in Computation 7.18 is its trace.

The derivation is a theorem for the prime-free form, for every L^2 critical point and with no hypothesis on the edge. We need two facts about the symbol.

Lemma 7.21 (the factor of a logarithmic symbol). *Let $\tilde{\sigma}$ be even, positive and C^2 on $\mathbb{R}$ with $c_1 \leq \tilde{\sigma}(t) \leq c_2 \log(e + |t|)$, $(\log \tilde{\sigma})' \in L^2$ and $(\log \tilde{\sigma})''$ bounded, and let θ be given by (10). Then $\sigma_+ = \tilde{\sigma}^{1/2} e^{i\theta}$ is the boundary value of the outer function of $\tilde{\sigma}^{1/2}$: it extends to an analytic function without zeros in the upper half-plane, continuous up to the line, with $c \leq |\sigma_+(z)| \leq C(\log(e + |z|))^{1/2}$ there; $\sigma_-(z) = \overline{\sigma_+(\bar{z})}$ is analytic without zeros in the lower half-plane with the same bounds; $\sigma_+ \sigma_- = \tilde{\sigma}$ on the line; θ is odd and C^1 with θ, θ' bounded, so that $1/\sigma_{\pm}$ are bounded with bounded derivative on the line.*

Proof. $w = \tilde{\sigma}^{1/2}$ has $\log w \in L^1(d\tau/(1 + \tau^2))$, so the outer function $O(z) = \exp \left[\frac{1}{\pi i} \int \left(\frac{1}{\tau - z} - \frac{\tau}{1 + \tau^2} \right) \log w(\tau) d\tau \right]$ is analytic and zero-free in the upper half-plane, $|O(t)| = w(t)$ on the line and $\log |O(x + iy)| = P_y * \log w(x)$ is the Poisson integral [14, Ch. II]. Its argument on the line is the conjugate function of $\log w$; for even w the term $\tau/(1 + \tau^2)$ drops out by parity and the two half-lines combine into (10), which is odd and vanishes at 0; so $O = \sigma_+$, the unimodular constant of the outer function being fixed by $\theta(0) = 0$. Since $\log w \geq \frac{1}{2} \log c_1$, $|O(z)| \geq c$. Since $\log w(\tau) \leq \frac{1}{2} \log \log(e + |\tau|) + C$ and $\log \log(e + |\tau|) \leq \log \log(e + |z|) + \log(1 + |\tau - x|/(e + |z|))$ for $z = x + iy$, the Poisson integral gives $\log |O(z)| \leq \frac{1}{2} \log \log(e + |z|) + C + \frac{1}{2} \int P_y(s) \log(1 + |s|/y) ds$, and the last integral equals $\int P_1(s) \log(1 + |s|) ds < \infty$. Finally $\theta' = \frac{1}{2} \mathcal{H}[(\log \tilde{\sigma})']$, and the Hilbert transform of a C^1 function in L^2 with bounded derivative is bounded (split the integral at $|\tau - t| = 1$). Continuity up to the line follows from that of $\log w$ and θ , and the statements for σ_- are the reflection $z \mapsto \bar{z}$. $\square$

Lemma 7.22 (the kernel of $1/\sigma_+$ away from the origin). *Let $\sigma = \Psi_{\infty} - \lambda$ with $\lambda > \Psi_{\infty}(0)$, so that σ has exactly one pair of simple real zeros $\pm t_{\lambda}$ (Ψ_{∞} is even and increasing on $t > 0$, as $\frac{d}{dt} \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) = -\frac{1}{2} \operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) > 0$), and let $\tilde{\sigma} = \sigma(t^2 + t_{\lambda}^2)/(t^2 - t_{\lambda}^2)$. Then $\tilde{\sigma}$ satisfies the hypotheses of Lemma 7.21; there is $\eta \in (0, \frac{1}{2})$ such that $\tilde{\sigma}$ is analytic and zero-free in the strip $-\eta \leq \operatorname{Im} z \leq 0$, so that $1/\sigma_+ = \sigma_-/\tilde{\sigma}$ is bounded and analytic there; and the kernel of $1/\sigma_+$,*

$$k_+(\xi) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-it\xi}}{\sigma_+(t)} dt \quad (\xi > 0, \text{ as an improper integral}),$$

is a real C^2 function on $(0, \infty)$ with $|k_+^{(j)}(\xi)| \leq C_{\delta} e^{-\eta\xi}$ on $\xi \geq \delta$, $j \leq 2$, for every $\delta > 0$; $1/\sigma_+$ is the transform of the distribution k_+ supported on $[0, \infty)$.

Proof. σ is real-analytic and even with $\sigma(t) = \log(t/2\pi) - \lambda + O(t^{-2})$ and derivatives $O(t^{-j})$ by Stirling's formula, and $\sigma < 0$ on $|t| < t_{\lambda}$, so $\tilde{\sigma} > 0$, $\tilde{\sigma} \asymp \log(e + |t|)$ and $(\log \tilde{\sigma})^{(j)} = O(t^{-j}/\log t)$: Lemma 7.21 applies. For $z = x - iy$, $0 \leq y \leq \eta < \frac{1}{2}$, both $\frac{1}{4} + \frac{iz}{2}$ and $\frac{1}{4} - \frac{iz}{2}$ have real part at least $\frac{1}{4} - \frac{\eta}{2} > 0$, so Stirling's formula holds uniformly and $\sigma(z) = \log(|x|/2\pi) - \lambda + O(1/|x|)$ there; hence $|\tilde{\sigma}(z)| \geq \frac{1}{2} \log |x|$ for $|x| \geq X_0$, and on the compact rectangle $|x| \leq X_0$ the continuous function $\tilde{\sigma}$, which has no zero on the real segment, has none for η small. So $1/\sigma_+ = \sigma_-/\tilde{\sigma}$ is analytic in the strip and bounded by Lemma 7.21. Its derivatives on the line $\operatorname{Im} z = -\eta$ decay: $\log \sigma_-$ is the Cauchy integral of $\log \tilde{\sigma}^{1/2}$, whose j -th derivative at z is $O(\log(e + |z|)/|z|^j)$ (split the integral at $|\tau - x| = |z|/2$ and use $\frac{1}{2} \log \log$ growth outside and the bound $|(\log \tilde{\sigma})'| \leq C/(|x| \log |x|)$ inside), and $\tilde{\sigma}^{(j)} = O(|x|^{-j})$. The improper integral defining k_+ converges after two integrations by parts, since $(1/\sigma_+)' = O(t^{-2})$ on the line; shifting the contour to $\operatorname{Im} t = -\eta$ (the vertical connectors vanish as $|\operatorname{Re} t| \rightarrow \infty$ because $1/\sigma_+$ is bounded and $e^{-it\xi}$ is) gives $k_+(\xi) = \frac{e^{-\eta\xi}}{2\pi} \int e^{-ix\xi} \sigma_+(x - i\eta)^{-1} dx$, and $j+2$ integrations by parts on that line, with the derivative bounds just stated, give $|k_+^{(j)}(\xi)| \leq C_{\delta} e^{-\eta\xi}$ for $\xi \geq \delta$. That k_+ is real follows from $\sigma_+(-t) = \overline{\sigma_+(t)}$, and that it is supported on $[0, \infty)$ from the analyticity and boundedness of $1/\sigma_+$ in the upper half-plane (Paley–Wiener). $\square$

Proposition 7.23 (exact Wiener–Hopf representation of a prime-free critical point). *Let $f \in L^2(-a, a)$ be odd and a critical point of $W_\infty - 2P^2$: $K_\infty f = \lambda f + 2Ps$ on $(-a, a)$, $P = \langle f, s \rangle$, with $\lambda > \Psi_\infty(0)$, and let $\sigma, t_\lambda, \tilde{\sigma}, \sigma_\pm$ be as in Lemma 7.22. With $g_{\text{out}} = K_\infty f$ outside the window define on the line*

$$\tilde{F} = e^{-ita} F, \quad \tilde{G}_+ = \int_0^\infty g_{\text{out}}(a+u) e^{itu} du, \quad \tilde{H} = e^{-ita} \left(2P\Sigma + \int_{y < -a} g_{\text{out}}(y) e^{ity} dy \right),$$

and $\hat{H}_- = \tilde{H} + t_\lambda \left[\frac{\tilde{H}(t) - \tilde{H}(t_\lambda)}{t - t_\lambda} - \frac{\tilde{H}(t) - \tilde{H}(-t_\lambda)}{t + t_\lambda} \right]$. Then $\hat{H}_-/\sigma_+ \in H^1(\mathbb{R})$ and

$$\tilde{F} = \frac{P_- [\hat{H}_-/\sigma_+]}{\sigma_-} \quad \text{on } \mathbb{R}, \quad (14)$$

where $P_\pm = \frac{1}{2}(I \pm i\mathcal{H})$ are the Riesz projections onto the boundary values of H^2 of the upper and lower half-planes.

Proof. The equation. As in the proof of Proposition 7.9, $K_\infty f = \lambda f + 2Ps + g_{\text{out}}$ on the line, so $\sigma F = 2P\Sigma + G$ with $G = \widehat{g_{\text{out}}}$; multiplying by e^{-ita} and splitting G at the window, $\sigma \tilde{F} = \tilde{H} + \tilde{G}_+$. Since $J_\Gamma(s) = \frac{1}{2|s|} + O(1)$ near 0 and $J_\Gamma(s) = O(e^{-|s|/2})$, $g_{\text{out}}(a+u)$ is $-\frac{1}{2}$ times the Hilbert transform of f at $a+u$ plus a bounded, exponentially decaying function, so $g_{\text{out}} \in L^1 \cap L^2$ outside the window, with $yg_{\text{out}} \in L^2$. Hence: $\tilde{F}(z) = \int_{-2a}^0 f(a+u) e^{izu} du$ is entire and bounded by $\|f\|_1$ in the closed lower half-plane; $\tilde{G}_+$ is analytic in the upper half-plane, continuous on its closure, bounded by $\|g_{\text{out}}\|_1$, and tends to zero at infinity there (Riemann–Lebesgue for the family $g_{\text{out}}(a+\cdot)e^{-yu}$, $y \geq 0$, which is compact in L^1); $\tilde{H}$ is analytic in the lower half-plane, continuous on its closure, bounded, and its restriction to the line is in $H^1(\mathbb{R})$, being the transform of a function h with $h, yh \in L^2$.

The real zeros. $\sigma(\pm t_\lambda) = 0$ forces $(\tilde{H} + \tilde{G}_+)(\pm t_\lambda) = 0$. Multiply the equation by $(t^2 + t_\lambda^2)/(t^2 - t_\lambda^2) = 1 + t_\lambda \left[\frac{1}{t - t_\lambda} - \frac{1}{t + t_\lambda} \right]$ and write each product $t_\lambda \tilde{H}(t)/(t \mp t_\lambda)$ as a difference quotient plus $t_\lambda \tilde{H}(\pm t_\lambda)/(t \mp t_\lambda)$, likewise for $\tilde{G}_+$: the terms with real poles cancel, and $\tilde{\sigma} \tilde{F} = \hat{H}_- + \hat{G}_+$ with $\hat{G}_+$ built from $\tilde{G}_+$ as $\hat{H}_-$ from $\tilde{H}$. The difference quotients are analytic where $\tilde{H}$ (resp. $\tilde{G}_+$) is, continuous up to the line, bounded, and $O(1/t)$ with derivative $O(t^{-2})$; so $\hat{H}_-$ and $\hat{G}_+$ have the properties listed for $\tilde{H}$ and $\tilde{G}_+$.

Liouville. Put $\varphi = \hat{H}_-/\sigma_+$; it is in $H^1(\mathbb{R})$ because $1/\sigma_+$ is bounded with bounded derivative (Lemma 7.21). Then $P_\pm \varphi \in H^2$ of the respective half-planes have boundary values in $H^1(\mathbb{R}) \subset C(\mathbb{R}) \cap L^\infty$, are the Poisson integrals of these boundary values, hence bounded and continuous on the closed half-planes, and $|P_\pm \varphi(x \pm iy)| \leq \|\varphi\|_2 (4\pi y)^{-1/2}$. Define $E = \sigma_- \tilde{F} - P_- \varphi$ on the closed lower half-plane and $E = \hat{G}_+/\sigma_+ + P_+ \varphi$ on the closed upper one. On the line the two agree, since their difference is $(\tilde{\sigma} \tilde{F} - \hat{H}_- - \hat{G}_+)/\sigma_+ = 0$. So E is continuous on $\mathbb{C}$ and analytic off the line, hence entire; it is bounded by $C(\log(e + |z|))^{1/2}$ (below by the bounds on σ_- , $\tilde{F}$ and $P_- \varphi$; above by those on $\hat{G}_+$, $1/\sigma_+$ and $P_+ \varphi$), so it is constant by Cauchy's estimate, and $E(iy) \rightarrow 0$ as $y \rightarrow \infty$. Hence $E \equiv 0$, which is (14). $\square$

Corollary 7.24 (the form of the edge). *In the setting of Proposition 7.23, fix $b > 0$ and collect the far edge in*

$$\Phi_-(t) = \int_0^\infty g_{\text{out}}(-a-v) e^{-itv} dv - \frac{2iP \sinh(a/2)}{t - ib},$$

so that $e^{-2ita} \Phi_-$ is the transform of the part $\text{far}(u) = (g_{\text{out}}(a+u) + 2P \sinh(a/2) e^{b(u+2a)}) \mathbf{1}_{u < -2a}$ of the source of $\tilde{H}$. Then, exactly on the line,

$$\tilde{F}(t) = \frac{A(t)}{t \sigma_-(t)} + \frac{e^{-2ita} \Phi_-(t)}{\tilde{\sigma}(t)}, \quad A(t) = t P_- \left[\frac{\hat{H}_- - e^{-2ita} \Phi_-}{\sigma_+} \right](t) - t P_+ \left[\frac{e^{-2ita} \Phi_-}{\sigma_+} \right](t), \quad (15)$$

with $A(t) = A_0 + O(\log t/t)$, A_0 purely imaginary, and $A(-t) = -\overline{A(t)}$. Moreover

$$F(t) = \frac{A(t)e^{ita}}{t\sigma_-(t)} - \frac{\overline{A(t)}e^{-ita}}{t\sigma_+(t)} + O(t^{-2}(\log t)^{-1/2}), \quad S(t) = \operatorname{Im} \frac{A(t)e^{ita}}{t\sigma_-(t)} + O(t^{-2}(\log t)^{-1/2}), \quad (16)$$

so that, writing $A(t) = -i|A_0|e^{-i\kappa(t)/t}(1+\varepsilon(t))$ with κ real, $S(t) = -|A_0|\cos(ta+\theta(t)-\kappa(t)/t)/(t\tilde{\sigma}(t)^{1/2}) + O(t^{-2}(\log t)^{-1/2}) + O(\varepsilon/t)$, $\kappa(t) = O(\log t)$ and $\varepsilon(t) = O(1/t)$.

Proof. (15) is (14) with $P_-[e^{-2ita}\Phi_-/\sigma_+] = e^{-2ita}\Phi_-/\sigma_+ - P_+[e^{-2ita}\Phi_-/\sigma_+]$ and $\sigma_+\sigma_- = \tilde{\sigma}$; the function $e^{-2ita}\Phi_-/\sigma_+$ is in L^2 since $\text{far} \in L^1 \cap L^2$. The symmetry $A(-t) = -\overline{A(t)}$ follows from $\tilde{F}(-t) = \tilde{F}(t)$, $\Phi_-(-t) = \overline{\Phi_-(t)}$, $\sigma_-(-t) = \overline{\sigma_-(t)}$ and the parity of $\tilde{\sigma}$.

The near part. Integrating $2P\Sigma$ by parts, $\hat{H}_- - e^{-2ita}\Phi_- = A'/(t-ib) + \Psi_2$ with $A' = -2iP \sinh(a/2) - t_\lambda(\tilde{H}(t_\lambda) - \tilde{H}(-t_\lambda))$, purely imaginary because $\tilde{H}(-t) = \overline{\tilde{H}(t)}$, and Ψ_2 analytic in the lower half-plane, $O(t^{-2})$ on the line. Now $P_-[1/((t-ib)\sigma_+)] = 1/((t-ib)\sigma_+(ib))$ exactly, because $[1/\sigma_+(t) - 1/\sigma_+(ib)]/(t-ib)$ is in H^2 of the upper half-plane; and $\sigma_+(ib) > 0$ (the outer function of an even weight is real and positive on the imaginary axis). Split the source of Ψ_2 with a smooth partition of unity into a part supported in $(-a, 0]$ and a part supported in $u < -a/2$. The first is smooth on $(-a, 0]$ and continuous at 0 with a kink (the step of the polar term at the edge has been removed), so its transform ψ satisfies $\psi = O(t^{-2})$, $\psi' = O(t^{-3})$, and for such ψ one has $tP_-[\psi/\sigma_+](t) = -\frac{i}{2\pi} \int \psi/\sigma_+ + O(\log t/t)$: indeed $P_- = \frac{1}{2}(I - i\mathcal{H})$, $t\psi = O(1/t)$, and $t\mathcal{H}[\psi/\sigma_+](t) = \frac{1}{\pi} \int \psi/\sigma_+ + \frac{1}{\pi} \int \frac{\psi(\tau)\tau}{\sigma_+(\tau)(t-\tau)} d\tau$, where the last integral is $O(\log t/t)$ on $|\tau| < t/2$, $O(1/t)$ on $|\tau - t| < t/2$ (the principal value of $[m(\tau) - m(t)]/(t-\tau)$ with $m = \tau\psi/\sigma_+$, $m' = O(\tau^{-2})$) and $O(1/t)$ on $|\tau| > 3t/2$. The integral $\int \psi/\sigma_+$ is real, as $\psi(-\tau) = \overline{\psi(\tau)}$ and $\sigma_+(-\tau) = \overline{\sigma_+(\tau)}$.

The far parts. Every remaining piece of $\hat{H}_-$ (the far source of Φ_- and the second part of the source of Ψ_2) is the transform $\tilde{\Phi}$ of a real function $\text{src} \in L^1 \cap L^2$ supported in $u \leq -\delta$ for some $\delta > 0$. By Lemma 7.22, $\tilde{\Phi}/\sigma_+$ is the transform of $\text{src} * k_+$, and $P_+[\tilde{\Phi}/\sigma_+]$ is the transform of its restriction to $u > 0$, $r(u) = \int_{u' \leq -\delta} \text{src}(u')k_+(u-u') du'$, which uses k_+ only on $[\delta, \infty)$: r is real, C^2 on $[0, \infty)$, and $r, r', r'' \in L^1$. Integrating by parts, $\int_0^\infty r(u)e^{itu} du = ir(0^+)/t - r'(0^+)/t^2 + o(t^{-2})$, so $-tP_+[\tilde{\Phi}/\sigma_+](t) = -ir(0^+) + O(1/t)$ with $r(0^+)$ real; and $tP_-[\tilde{\Phi}/\sigma_+] = t\tilde{\Phi}/\sigma_+ - tP_+[\tilde{\Phi}/\sigma_+]$, where $t\tilde{\Phi}/\sigma_+ = O(1/t)$ for the piece of Ψ_2 ($\tilde{\Phi} = O(t^{-2})$ there). Collecting, $A(t) = A_0 + O(\log t/t)$ with $A_0 = A'/\sigma_+(ib) - \frac{i}{2\pi} \int \psi/\sigma_+ - i \sum r(0^+)$ purely imaginary.

The mirror. From $\sigma\tilde{F} = \tilde{H} + \tilde{G}_+$ and (15), $\tilde{G}_+ = \sigma\tilde{F} - \tilde{H} = \frac{\sigma}{\tilde{\sigma}} \frac{\sigma_+ A}{t} + \frac{\sigma}{\tilde{\sigma}} e^{-2ita}\Phi_- - 2P\tilde{\Sigma} - e^{-2ita}(\Phi_- + \frac{2iP \sinh(a/2)}{t-ib})$, where $\tilde{\Sigma} = e^{-ita}\Sigma = -i \sinh(a/2)(1 + e^{-2ita})/t + O(t^{-2})$ and $\sigma/\tilde{\sigma} = 1 - 2t_\lambda^2/(t^2 + t_\lambda^2)$; the terms in e^{-2ita} cancel up to $O(t^{-2})$ (Φ_- is bounded), and $\tilde{G}_+(t) = \sigma_+(t)A(t)/t + 2iP \sinh(a/2)/t + O(t^{-2}(\log t)^{1/2})$. Since f is odd, $g_{\text{out}}(-a-v) = -g_{\text{out}}(a+v)$, so $\Phi_- = -\tilde{G}_+ - 2iP \sinh(a/2)/(t-ib) = -\sigma_- \overline{A}/t + O(t^{-2}(\log t)^{1/2})$, and the second term of (15) is $-e^{-2ita}\overline{A(t)}/(t\sigma_+(t)) + O(t^{-2}(\log t)^{-1/2})$. Multiplying by e^{ita} gives (16); $F = 2iS$ is purely imaginary, and the two terms are conjugate up to sign, so S is the imaginary part of the first. The last statement is $A(-t) = -\overline{A(t)}$ read on $A(t) - A_0 = \beta(t)/t$: $\beta(-t) = \overline{\beta(t)}$, so the even part of β is real, a phase to first order, while its odd part, which is $O(1/t)$ by the expansions above, is imaginary. $\square$

Corollary 7.24 is Conjecture 7.19(i) except for three things it does not decide: the value $|A_0|^2 = -\lambda'/2$, which is the boundary law (Propositions 8.3 and 8.4 prove it under (H_∞) , Corollary 8.5); the constancy of the phase $\kappa(t)$, for which the bound is $O(\log t)$ while the data of Computation 7.20 give a constant to 10^{-4} ; and the rate, $O(\log t/t)$ against the measured t^{-2} . With an entry inside the window the same holds with the effective symbol of Proposition 8.7, and part (ii) of the conjecture, the echo form, is Corollary 8.8(iii); at every support with a finite echo set the effective symbol is

the Schur complement of the echo tree, Proposition 8.13, and the echo form is Corollary 8.14(iii). Its content for the tail law is this: $\int_{|t|>T} |F|^2 \tilde{\sigma} dt = 8|A_0|^2 \int_T^\infty \cos^2(ta + \theta - \kappa/t) dt/t^2 (1 + o(1)) = 4|A_0|^2/T (1 + o(1))$, so for every prime-free critical point the tail energy of Corollary 7.13 equals $2|A_0|^2/(\pi T)$ at every height beyond the onset, and the tail law $-\lambda'/(\pi T)$ of Computation 7.14 is the single identity $2|A_0|^2 = -\lambda'$: with $|A_0| = C$, the boundary law. Item 2 of the program is therefore reduced to the flux identity, and to the echo form with primes.

Heuristic 7.25 (why the effective height is tied to the horizon). Below the horizon the zeros are sparser than the a/π degrees of freedom per unit height of a function of type a , and the minimizer vanishes at all of them: their share of the floor is 10^{-5} , 10^{-7} , 10^{-13} at $a = 0.6, 0.8, 1.0$. Above it, the excess of the zero density over the Nyquist density is $\log(t/T^*)/2\pi$, the same function of t/T^* for every a ; so, after scaling heights by the horizon, the minimizer faces the same problem at every support, which is why T_{eff}/T^* should tend to a constant, and why the profiles $\vartheta(T/T^*)$ should converge as a grows. The only way for a function of type a to be small at more points than it has freedom for is to steepen at the edge (the edge law and the Hadamard formula), and the edge amplitude feeds the tail at every height at once, at the rate $C^2/(\pi T)$; the minimizer balances the gain from hiding at one more zero just above the horizon against that cost, and the balance falls at a fixed multiple of T^* . The spare freedom below the horizon is worth $T^*/2\pi$ zeros, and the accumulated excess above it, $\int_{T^*}^T \log(t/T^*) dt/2\pi$, exhausts it at $T = eT^*$; the measured median heights are below that and rising with a . Two things enter that no counting supplies: positivity of every term, which is RH, and the absence of gaps between zeros wider than the resolution π/a above a few horizons, in which a shape could park its tail. The explanation is therefore conditional on where the zeros are; it says what Conjecture 7.1 asserts about them, not why the prime side must produce it.

The count in the heuristic has an exact form.

Lemma 7.26 (exhaustion height). *The number of zeros in $[-T, T]$ equals the Shannon number $2aT/\pi$ of the window $[-a, a]$ at bandwidth T , up to $O(\log T)$, exactly at $T = eT^*(a)$.*

Proof. $2N(T) = \frac{T}{\pi} (\log \frac{T}{2\pi} - 1) + O(\log T)$, and $\log \frac{T}{2\pi} - 1 = 2a$ at $T = 2\pi e^{2a+1} = eT^*$. $\square$

Below eT^* a function of type a has more freedom than there are zeros to avoid; above it, fewer. So the minimizer can be expected to hide from the zeros up to about eT^* and to be the pure edge tail beyond, which is the onset of the tail law; and its floor to sit in the zeros around eT^* . The data agree: the median height of the floor's mass is 2.39, 2.32, 2.65, 2.67, 2.74 horizons at $a = 0.6, \dots, 1.5$ (Computation 7.6), and $\vartheta(eT^*)$ read from the same tables is 0.43, 0.45, 0.47, 0.49, 0.50, while the law is within 10–15% at that height. Conjecture 7.1 then takes its sharpest conditional form.

Proposition 7.27 (sampling above the exhaustion height implies Conjecture 7.1). *Assume RH, let $\kappa \geq 1$, and suppose that for the minimizer f at every support $a \geq a_0$*

$$(S_\kappa) \quad \int_{|t|>\kappa T^*} |F(t)|^2 \rho(t) dt \leq M \sum_{|\gamma|>\kappa T^*} |F(\gamma)|^2 \quad (\text{the zeros above } \kappa T^* \text{ sample the tail of } F \text{ with constant } M), \text{ and}$$

$$(E_\kappa) \quad \int_{|t|>\kappa T^*} |F(t)|^2 \rho(t) dt \geq \frac{2C(a)^2}{\pi \kappa T^* M'} \quad (\text{the edge overlap of Lemma 7.11 holds at height } \kappa T^* \text{ within the factor } M'),$$

and that the boundary law $\lambda' = -2C^2$ holds. Then $\lambda' \geq -\pi \kappa M M' T^ \lambda$ on $[a_0, \infty)$: Conjecture 7.1 with $c = \pi \kappa M M'$.*

Proof. $\lambda = \sum_{\gamma} |F(\gamma)|^2 \geq \sum_{|\gamma| > \kappa T^*} |F(\gamma)|^2 \geq M^{-1} \int_{|t| > \kappa T^*} |F|^2 \rho \geq \frac{2C^2}{\pi \kappa M M' T^*}$, so $-\lambda' = 2C^2 \leq \pi \kappa M M' T^* \lambda$. $\square$

Both hypotheses are measured. (S_{κ}) with $\kappa = e$: the density integrals and the zero sums of Computation 7.6 agree to 7% at five horizons and to 1% at twenty, so $M \approx 1$; (E_e) : $T\vartheta(T)/T_{\text{eff}}$ at eT^* is 1.10–1.15 at every support, so $M' \approx 1$ with the inequality in the favourable direction. The measured constant, $c_0 = \Phi'/T^*$ between 3.3 and 3.9, is $\pi e \vartheta(eT^*)$ times that factor, against the bound $\pi e \approx 8.5$ of the proposition. What (S_{κ}) asks of the zeros is a sampling inequality on the half-line above eT^* for the high-frequency part of functions of type a with a logarithmic edge, with density $\rho \geq (2a+1)/2\pi$ against the Nyquist density a/π ; what (E_{κ}) asks of the minimizer is that its edge layer be no narrower than $1/(eT^*)$, i.e. that it hide from no zero above the exhaustion height. The first is a statement about the zeros alone, the second about the minimizer alone, and the count of Lemma 7.26 is the reason both should hold at a fixed multiple of the horizon.

The entries have a zero-side reading in the same picture. The tail of the floor carries $\cos^2(\gamma a) = \frac{1}{2}(1 + \cos 2\gamma a)$ over the zeros, so it contains the average of $\cos(2\gamma a)$ over the tail-weighted zeros, and by the explicit formula the zeros' rotations align exactly when $2a = \log n$.

Computation 7.28 (the zeros' rotation at the entries). `rh_alignment.py`, first 6000 zeros: $R(a) = \sum_{\gamma} \cos(2\gamma a) w(\gamma) / \sum_{\gamma} w(\gamma)$ with the tail weight $w = \gamma^{-2} \log^{-1}(\gamma/2\pi)$, on $0.3 \leq a \leq 1.3$ in steps of 0.0005. Its mean absolute value is 0.27 and its 95th percentile 0.53; at the entries of the primes it is -0.53 (a_2), -0.62 (a_3 , the minimum over the whole range, at $a = 0.5495$), -0.57 (a_5), -0.56 (a_7), -0.44 (a_{11}), each minimum within 0.0005 of a_n ; at the prime powers 4, 8, 9, whose weights $\Lambda(n)n^{-1/2} = 0.35, 0.25, 0.37$ are below the noise of this truncation, it is $+0.10, -0.08, +0.22$. Around a_3 the curve is a cusp: with the zeros above $3T^*$ only and weight γ^{-2} , R is $-0.097, -0.158, -0.226, -0.161, -0.103$ at offsets $-0.004, -0.002, 0, +0.002, +0.004$, linear flanks meeting at the entry. A cusp in the tail mass is a jump in λ' : this is the kink of Proposition 4.1 seen from the zeros, and with the edge law in place of a jump it is the soft kink of (5). The relay, the soft kink, the sawtooth of Computation 4.4 and the tail law are one mechanism: the zeros rotate with the window, align at the primes, and the alignment is felt through the tail that the edge feeds.

The sampling hypothesis (S_{κ}) has an exact form: the explicit formula turns the defect of the zeros against their density into a prime sum, and the prime sum lives on the entries, where the autocorrelation of the minimizer is singular.

Proposition 7.29 (the sampling defect is the prime sum at the entries). *Assume RH. Let $f \in L^2(-a, a)$ be odd with $P = \langle f, s \rangle$, and for $T > 0$ let $\chi_T(t) = 1 - e^{-(t/T)^{2m}}$ with m even ($m = 2$ unless said otherwise), an entire even high-pass at height T , and*

$$\check{h}_T(x) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T(t) e^{itx} dt,$$

the autocorrelation of f with its structure below the scale $1/T$ removed. Then

$$\sum_{\gamma} |F(\gamma)|^2 \chi_T(\gamma) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T(t) \Psi_{\infty}(t) dt - 2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \check{h}_T(\log n) - 2P^2(1 - e^{-(2T)^{-2m}}), \quad (17)$$

Proof. Weil's explicit formula in the normalization of (1) reads, for an even h analytic in $|\text{Im } t| < \frac{1}{2} + \varepsilon$ and $O(|t|^{-1-\varepsilon})$ there, $\sum_{\gamma} h(\gamma) = \frac{1}{2\pi} \int h \Psi_{\infty} - \sum_n \Lambda(n) n^{-1/2} (\check{h}(\log n) + \check{h}(-\log n)) + h(i/2) +$

$h(-i/2)$, $\check{h}$ the inverse transform [1, 3]; for $h = |F|^2$ it is the identity $Q(f) = \sum_{\gamma} |F(\gamma)|^2$ of §7.1. Apply it to $h(t) = F(t)F(-t)\chi_T(t)$, which is entire and even, equals $|F|^2\chi_T$ on the line, and is $O(|t|^{-1-\varepsilon})$ in every strip, f having compact support and χ_T being bounded there (for fixed m , $\text{Re}(t/T)^{2m} > 0$ in a strip once $|\text{Re } t| \gg m$). For odd f , $F(\pm i/2) = \mp P$, so $h(\pm i/2) = -P^2\chi_T(i/2) = -P^2(1 - e^{-(2T)^{-2m}})$, $i^{2m} = 1$ for even m , and $\check{h}$ is even. $\square$

The first term on the right of (17) is the density integral of (S_{κ}) with the exact density $\Psi_{\infty}/2\pi$, and the last is negligible. So the sampling constant is $M_{\chi}(T) = 1 + (2\sum_n \Lambda(n)n^{-1/2}\check{h}_T(\log n) + O(P^2(2T)^{-2m}))/\sum_{\gamma} |F(\gamma)|^2\chi_T(\gamma)$: the zeros above a height sample the minimizer at their mean density up to the prime sum at the entries. For an entry m outside the window, $\check{h}_T(\log m) = (g * \check{\chi}_T)(\log m)$ is the autocorrelation smeared at scale $1/T$ to a point at distance $\log m - 2a$ from its support, which is negligible once $T(\log m - 2a)$ is large: this is the alignment of Computation 7.28, confined to a neighbourhood of width $1/T$ of each entry. For the entries inside the window the term is set by the echoes.

Corollary 7.30 (the analytic form of the sampling constant). *Under RH and the hypotheses of Proposition 8.7 (one entry n inside the window, $a_2 < a < a_3$), with $c = \Lambda(n)n^{-1/2}$,*

$$M_{\chi}(T) = 1 + 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T (1 + o(1)) = 1 + \frac{2c^2}{\sigma_{\infty}(T)^2} (1 + O(1/\log T)) \quad (T \rightarrow \infty),$$

$$\langle g \rangle_T = \frac{\int_0^{\infty} \chi_T g t^{-2} dt}{\int_0^{\infty} \chi_T t^{-2} dt} :$$

the defect of the sampling is twice the square of the echo's amplitude $\rho = c/\tilde{\sigma}_{\infty}$ of (12), averaged over the tail with the weight χ_T/t^2 of the edge, and tends to zero like $1/\log^2 T$. The average matters at the heights computed: $\langle \tilde{\sigma}_{\infty}^{-2} \rangle_T = \sigma_{\infty}(T)^{-2}(1 - 2/\sigma_{\infty}(T) + \dots)$ is 0.7 times $\sigma_{\infty}(T)^{-2}$ at $T = 10T^*$.

Proof. By Corollary 8.8(iii), $|F|^2 = \frac{4|A_0|^2}{t^2\tilde{\sigma}} [\cos \varphi + \rho \cos(\varphi - t \log n)]^2 (1 + o(1))$ with $\varphi = ta + \theta - \kappa/t$. In $\check{h}_T(\log n) = \frac{1}{2\pi} \int |F|^2 \chi_T \cos(t \log n) dt$ the cross term $2\rho \cos \varphi \cos(\varphi - t \log n) = \rho[\cos(t \log n) + \cos(2\varphi - t \log n)]$ gives, against $\cos(t \log n)$, the non-oscillating $\frac{1}{2}\rho$, the other products oscillating at the frequencies $2a$, $2a \pm \log n$ and $2a \pm 2 \log n$, all nonzero; so $\check{h}_T(\log n) = \frac{|A_0|^2}{\pi} \int_0^{\infty} \frac{\rho \chi_T}{t^2 \tilde{\sigma}} dt (1 + o(1))$ with $\rho/\tilde{\sigma} = c/\tilde{\sigma}_{\infty}^2 (1 + O(\tilde{\sigma}_{\infty}^{-2}))$, while the density integral is $\frac{1}{2\pi} \int \frac{4|A_0|^2 \cos^2 \varphi}{t^2 \tilde{\sigma}} \chi_T \Psi_{\infty} dt (1 + o(1)) = \frac{|A_0|^2}{\pi} \int_0^{\infty} \frac{\chi_T}{t^2} dt (1 + o(1))$, which is $|A_0|^2 I_4 / (\pi T)$ with $I_4 = \int_0^{\infty} (1 - e^{-s^4}) s^{-2} ds = \Gamma(3/4)$ for $m = 2$. The entries outside the window and the polar term are o of this, and (17) gives $M_{\chi} - 1 = 2c\check{h}_T(\log n)/(\text{density integral})(1 + o(1)) = 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T (1 + o(1))$; the expansion of the average is $\int_1^{\infty} du/(u^2(L + \log u)^2) = L^{-2}(1 - 2/L + O(L^{-2}))$ with $L = \log(T/2\pi) - \lambda$, for a cutoff sharp at T . $\square$

Computation 7.31 (the identity on the zeros). `rh_sampling_defect.py`, first 6000 zeros, logs in `data/`. On the minimizers with primes at $a = 0.4, 0.45, 0.5$ (one entry inside the window) and $a = 0.6$ (two entries, Computation 8.16), every term of (17) is evaluated at $T = \kappa T^*$ for $\kappa = 1, \dots, 20$, with the cutoffs $m = 2$ and $m = 4$: the zero sum (with the part above the last zero, 305–455 horizons, taken at the density), the density integral by quadrature on a grid of step 0.01 to 3000 and 0.05 to 1.2×10^5 , the prime sum over all $n < 400$, and the polar term. The identity holds at every height to 3×10^{-5} of the zero sum or better. The prime sum is the entries inside the window: at $a \leq 0.5$, $n = 2$ gives all of it to three digits from $\kappa = e$ on, and 92–97% at $\kappa = 1$, the rest the smearing to $n = 3$ at distance 0.1–0.3 outside. The sampling constant $M_{\chi} = D_{\chi}/Z_{\chi}$ with the cutoff $1 - e^{-(t/T)^8}$ against the averaged form $1 + 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T$ of Corollary 7.30 and against $1 + 2c^2/\sigma_{\infty}(T)^2$:

κ	$a = 0.4$			$a = 0.45$			$a = 0.5$		
	M_χ	$\langle \cdot \rangle_T$	at T	M_χ	$\langle \cdot \rangle_T$	at T	M_χ	$\langle \cdot \rangle_T$	at T
e	1.068	1.083	1.151	1.085	1.076	1.133	1.128	1.070	1.120
6	1.044	1.047	1.072	1.046	1.044	1.066	1.051	1.041	1.062
10	1.034	1.035	1.050	1.033	1.033	1.047	1.032	1.031	1.044
20	1.024	1.024	1.034	1.023	1.023	1.032	1.021	1.022	1.030

From six horizons on the defect is the echo's, averaged over the tail, to a few parts in a thousand, and the value at T overestimates $M_\chi - 1$ by a factor 1.4: the average matters. Below six horizons the cutoff still reaches the heights where the minimizer hides from the zeros, the more so the smaller λ ($\lambda = 2 \times 10^{-4}$ at $a = 0.5$), and M_χ carries that. With the quartic cutoff (`log sampling_defect.log`) the leakage is larger and, between six and twenty horizons, happens to compensate the difference between the average and the value at T , so that M_χ there agrees with $1 + 2c^2/\sigma_\infty(T)^2$ to a few parts in a thousand; the sharper cutoff separates the two. The sharp constant $M(\kappa)$ of Computation 7.14, 1.01, 1.09, 1.02 at $\kappa = e$ for the three supports, sees the same defect at the sharp edge of the tail.

The arithmetic of Conjecture 7.1 is now in one place. By (17) the sampling defect at every support is $2 \sum_{\log n < 2a} \Lambda(n)n^{-1/2} \check{h}_T(\log n)$ plus the alignment at the entries within $1/T$ of the window's edge, and (S_κ) with a bounded M asks that this prime sum be at most a fixed fraction of the tail at $T = \kappa T^*$, uniformly in a : a statement about the autocorrelation of the minimizer at the entries, that is, about the echoes, and no longer about the zeros. For one entry it is Corollary 7.30; for a window containing many entries the echoes form a finite tree and $\sum_{\log n < 2a} \Lambda(n)^2/n \sim 2a^2$ by Mertens, so the defect need not tend to zero, but it is bounded as long as the echoes keep their size c_n/σ_∞ . No gap condition on the zeros enters.

The two hypotheses of Proposition 7.27 can be replaced by one, about the minimizer alone. By (17) with $\Psi = \Psi_\infty - \sum_n 2\Lambda(n)n^{-1/2} \cos(t \log n)$, the zero sum with the cutoff is $\frac{1}{2\pi} \int |F|^2 \chi_T \Psi$ minus the polar term: the prime sum at the entries is the oscillating part of the symbol. So what the zeros see is the tail of $|F|^2$ against the full symbol, and the sharp form of Computation 7.18, the local envelope of $t^2(\Psi - \lambda)|F|^2$, is the whole story.

Proposition 7.32 (the sharp form as the single hypothesis). *Assume RH. Let $\kappa \geq 1$, and for a support a put $T = \kappa T^*$ and $\chi_T^{(m)}(t) = 1 - e^{-(t/T)^{2m}}$ with m the even integer in $[\frac{3}{2}T - 2, \frac{3}{2}T)$. Suppose that for every support $a \geq a_0$*

$$(T_\kappa) \quad \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T^{(m)}(t) (\Psi(t) - \lambda) dt \geq \frac{-\lambda'(a)}{\pi M'' T} \quad (\text{the sharp tail law holds at height } \kappa T^* \text{ within the factor } M'').$$

Then $\lambda' \geq -\pi \kappa M'' T^ \lambda - \varepsilon(a)$ on $[a_0, \infty)$ with $0 \leq \varepsilon(a) \leq M'' e^{-8T^*(a)}$, and Conjecture 7.1 holds with $c = \pi \kappa M''$ up to this term, which the Grönwall argument of Section 7 absorbs.*

Proof. For this m the cutoff is bounded by $e^{1.01}$ in the strip $|\operatorname{Im} t| \leq \frac{1}{2} + \varepsilon$, $\varepsilon < 0.02$: for $|\operatorname{Re} t| \leq T$, $|(t/T)^{2m}| \leq (1 + (0.52/T)^2)^{3T/2}$, and for $|\operatorname{Re} t| \geq T$, $2m|\arg t| \leq 3(\frac{1}{2} + \varepsilon) < \pi/2$, so $\operatorname{Re}(t/T)^{2m} > 0$. Hence $h = F(t)F(-t)\chi_T^{(m)}$ satisfies the hypotheses of the explicit formula, and (17) holds with the polar term $2P^2(1 - e^{-(2T)^{-2m}})$, which lies in $[0, 2P^2(2T)^{-2m}]$ since m is even. Under RH, $\lambda = \sum_\gamma |F(\gamma)|^2 \geq \sum_\gamma |F(\gamma)|^2 \chi_T^{(m)}(\gamma) = \frac{1}{2\pi} \int |F|^2 \chi_T^{(m)}(\Psi - \lambda) + \lambda \frac{1}{2\pi} \int |F|^2 \chi_T^{(m)} - 2P^2(1 - e^{-(2T)^{-2m}})$, and $\lambda \geq 0$; so by (T_κ) , $-\lambda' \leq \pi M'' T \lambda + 2\pi M'' T P^2(2T)^{-2m}$. The last term is $\varepsilon(a)$: $P^2 \leq \|s\|^2 = \sinh a - a$, and $2m \log(2T) \geq (3\kappa T^* - 4) \log(2\kappa T^*)$ gives $\varepsilon \leq M'' e^{-8T^*}$ for $T^* \geq 2\pi e^{0.6}$, that is $a_0 \geq 0.3$. In the Grönwall argument, $\lambda(a)e^{\frac{\varepsilon}{2}T^*(a)} \geq \lambda(a_0)e^{\frac{\varepsilon}{2}T^*(a_0)} - \int_{a_0}^a \varepsilon e^{\frac{\varepsilon}{2}T^*}$, and the integral is bounded by $M'' \int e^{-(8-c/2)T^*}$, negligible against the first term for $c < 16$. $\square$

Under the hypotheses of Corollary 8.8 or 8.14, (T_κ) holds at a fixed support for κ large, with $M'' \rightarrow 1$: the sharp form is exact in the limit, and by Corollary 8.14(iv) it is, in the finite tree, an identity of the echo amplitudes. What Conjecture 7.1 asks is that the height from which the sharp form holds within a fixed factor be a bounded multiple of the horizon, uniformly in a . The measured onset is a few horizons at every support computed (Computations 7.18, 7.14, 7.6: 0.1% from five horizons on the prime-free critical points, a few per cent with the primes, including $a = 1.0$ and 1.25, where the echo set is infinite), and the bound $c = \pi e M''$ at $\kappa = e$ is again 8.5 against the observed 3.3–3.9. No sampling constant and no gap condition on the zeros remain in the hypothesis: the zeros enter only through the identity (17).

Heuristic 7.33 (prolate lower bound). The number of zeros in $[-T^*, T^*]$ is $2N(T^*) \approx (T^*/\pi)(2a-1)$, short of the Shannon number $2aT^*/\pi$ of $[-a, a] \times [-T^*, T^*]$ by T^*/π . By the Landau–Pollak–Slepian theory [15, 16], with the asymptotics of the prolate spectrum in the plunge region [17, 18] (about $(2/\pi^2) \log c$ eigenvalues between ε and $1 - \varepsilon$ for $c = aT^*$), a function in PW_a constrained to vanish on all zeros below the horizon can still keep all but $\eta \gtrsim \exp(-\pi T^*/(\log(aT^*) + O(1)))$ of its energy below the horizon; under RH this bounds λ from above, i.e. $\Phi \gtrsim \pi T^*/\log(aT^*)$, which is $0.7 T^*$ at $a = 1$. The observed value is $\Phi(1) = 59.5 = 1.28 T^*$ (Computation 6.1); the zeros are about twice as expensive a constraint set as the count alone predicts, and the difference is arithmetic. Conjecture 7.1 is the matching upper bound.

8 The boundary law

The derivative of the floor should be a boundary quantity. Since Q does not depend on a , $\lambda(a)$ is the infimum of a fixed form over the growing family $L^2(-a, a)$, and its derivative is a domain variation. For the Dirichlet fractional Laplacian, Hadamard’s formula expresses such a derivative through the boundary trace u/dist^s of the eigenfunction [11]; for the logarithmic Laplacian the boundary trace is the amplitude C of Computation 3.4. A Pohozaev identity for L_Δ with the local boundary term $\int_{\partial\Omega} u^2 \ln(\delta^{-2})(x \cdot \nu)$ was claimed in [10], which its authors withdrew on 26 April 2025 “due to a crucial error in Proof of Pohozaev identity”; to our knowledge no correct version exists. We therefore state the law as a conjecture and give the evidence.

Conjecture 8.1 (boundary law). *Assume (H_∞) . For a not an entry, with $C_\pm(a) = \lim_{\delta \rightarrow 0} \sqrt{\log(1/\delta)} |f(\pm(a - \delta))|$ the edge amplitudes of the minimizer (assumed to exist),*

$$\lambda'(a) = -(C_+(a)^2 + C_-(a)^2) = -2C(a)^2.$$

Equivalently, with Proposition 5.3, the Weil form on a window satisfies the Pohozaev-type identity $\mathcal{D}(f) = 2a C(a)^2$ on its minimizer. Proposition 8.3 below proves it under the edge law with two derivatives, and Propositions 8.4 and 8.7 supply that law from (H_∞) for the archimedean form and for $a_2 < a < a_3$; Computation 8.10 is the evidence at the other supports.

The law is a theorem under the edge law with derivatives, the hypothesis of Proposition 7.8: the boundary term of the Pohozaev identity is the overlap, at the scale $1/T$, of the leak of the dilation generator through the low-pass with the singular force outside the edge, and the logarithms cancel there as they did in Lemma 7.11. We first sharpen Lemma 7.12.

Lemma 8.2 (the edge force to second order). *Under the hypothesis of Proposition 7.8, the remainder $\rho(\delta) = g_{\text{out}}(a + \delta) + CL(\delta)^{1/2} - g_{\text{reg}}(a)$ of Lemma 7.12 satisfies $\rho(\delta) = o(L(\delta)^{-1/2})$ and $\rho'(\delta) = o(\delta^{-1} L(\delta)^{-3/2})$ as $\delta \rightarrow 0^+$.*

Proof. By the formula in the proof of Lemma 7.12, $g_{\text{out}}(a + \delta) = -\int_0^{2a} f(a - u)k(u + \delta) du + (K_P f)(a + \delta)$, and the prime part is smooth near the edge (it is made of copies of the interior of f). Write $k(x) = \frac{1}{2x} + k_0(x)$ with k_0 smooth and bounded; the part with k_0 is C^1 in δ . In the singular part put $f(a - u) = (\varphi + \psi)(u)$ for $u < \delta_0$ and substitute $u = \delta w$:

$$-\frac{1}{2} \int_0^{\delta_0} \frac{\varphi(u)}{u + \delta} du = -\frac{C}{2} \int_0^{\delta_0/\delta} \frac{(L(\delta) - \log w)^{-1/2}}{1 + w} dw,$$

and $\int_0^W \frac{(L - \log w)^{-1/2}}{1 + w} dw = \int_1^W \frac{(L - \log w)^{-1/2}}{w} dw + \int_0^\infty \left[\frac{1}{1+w} - \frac{1_{w>1}}{w} \right] (L - \log w)^{-1/2} dw + O(L^{-1/2}/W)$. The first integral is $2L^{1/2} - 2L(\delta_0)^{1/2}$; in the second the bracket integrates to zero, so the expansion $(L - \log w)^{-1/2} = L^{-1/2} \left(1 + \frac{\log w}{2L} + O(L^{-2} \log^2 w) \right)$ leaves $O(L^{-3/2})$. Hence the φ -part is $-CL(\delta)^{1/2} + \text{const} + O(L^{-3/2})$, and the ψ -part, $-\frac{1}{2} \int \psi(u) du / (u + \delta)$, is $o\left(\int_\delta^{\delta_0} L(u)^{-3/2} du / u\right) + o(L^{-3/2}) = o(L(\delta)^{-1/2})$. Differentiating under the integral, $\frac{d}{d\delta} \left[-\frac{1}{2} \int \frac{(\varphi + \psi)(u)}{u + \delta} du \right] = \frac{1}{2} \int \frac{(\varphi + \psi)(u)}{(u + \delta)^2} du$, and the same substitution gives $\frac{C}{2} \delta^{-1} L^{-1/2} \int_0^\infty \frac{(1 - \log w/L)^{-1/2}}{(1 + w)^2} dw + o(\delta^{-1} L^{-3/2})$, where $\int_0^\infty \frac{\log w}{(1 + w)^2} dw = 0$ by $w \mapsto 1/w$; so the derivative is $\frac{C}{2} \delta^{-1} L^{-1/2} (1 + O(L^{-2})) + o(\delta^{-1} L^{-3/2})$, which is $\frac{d}{d\delta} [-CL^{1/2}] + o(\delta^{-1} L^{-3/2})$. $\square$

Proposition 8.3 (edge law implies boundary law). *Let f be the minimizer at a support a that is not an entry, and suppose it satisfies the hypothesis of Proposition 7.8, the edge law with two derivatives. Then, with B_T as in Proposition 7.15,*

$$\lim_{T \rightarrow \infty} B_T = aC^2, \quad \text{hence} \quad \mathcal{D}(f) = 2aC^2 \quad \text{and} \quad \lambda'(a) = -2C^2$$

at every point of differentiability of λ (Proposition 5.3): Conjecture 8.1 holds. More precisely, in the model $\psi \equiv 0$,

$$B_T = aC^2 J_1(L_T) - aC g_{\text{reg}}(a) L_T^{-1/2} (1 + O(1/L_T)) + O(1/T),$$

$$J_1(L) = -\frac{2}{\pi} \int_0^\infty \int_0^\infty \left[\frac{L + \log(1/v)}{L + \log(1/s)} \right]^{1/2} k'(s + v) ds dv = 1 + O(L^{-2}),$$

with $k(w) = \sin w/w$, $L_T = L(1/T)$ and $g_{\text{reg}}(a) = 2P \sinh(a/2)$.

Proof. One edge. g_{out} and $(xf)'$ are odd, so $B_T = 2B_T^+$ with $B_T^+ = \int_{\delta>0} ((xf)')_T(a + \delta) g_{\text{out}}(a + \delta) d\delta$.

The leak. Near the right edge write $f(a - \delta) = u(\delta) = (\varphi + \psi)(\delta)\chi(\delta)$ as in the proof of Proposition 7.8, so that $(xf)'(a - \delta) = -au'(\delta) + b(\delta)$ with $b = u + \delta u' = O(L^{-1/2})$ bounded; the rest of $(xf)'$ is C^2 with compact support in $(-a, a)$ except at the left edge, where it has the mirror singularity. For $\delta > 0$, $((xf)')_T(a + \delta) = \int (xf)'(y) D_T(a + \delta - y) dy$ with $D_T(s) = \sin(Ts)/(\pi s)$. The part of $(xf)'$ supported at distance at least $c > 0$ from the right edge has transform $O(t^{-1} \log^{-1/2} t)$ (Step 1 of that proof), and its leak at $a + \delta$ is the tail $\int_{|t|>T}$ of the inversion integral against $e^{-it(a+\delta-y)}$, $|a + \delta - y| \geq c$, which is $O(1/(cT \log^{1/2} T))$ uniformly in δ after one integration by parts; paired with $g_{\text{out}} \in L^1$ it tends to zero. The part b leaks like f itself, $O(L_T^{-1/2} \min(1, 1/(T\delta)))$, and its pairing with $-CL^{1/2}$ is $O(1/T)$ by Lemma 7.11, with $g_{\text{reg}} + \rho$ it is $O(L_T^{-1/2} \log T/T)$. The main part is $-au'$, whose leak is, by one integration by parts ($u(0) = 0$, u cut off smoothly at δ_0),

$$-a \int_0^{\delta_0} u'(\delta') D_T(\delta + \delta') d\delta' = a \ell'_T(\delta), \quad \ell_T(\delta) = \int_0^{\delta_0} u(\delta') D_T(\delta + \delta') d\delta',$$

the derivative of the leak of f of Lemma 7.11.

The pairing. Write $g_{\text{out}}(a + \delta) = -CL(\delta)^{1/2} + g_{\text{reg}}(a) + \rho(\delta)$ by Lemmas 7.12 and 8.2. (i) With the singular force, $a \int \ell'_T(\delta) (-CL(\delta)^{1/2}) d\delta = -aC \iint u(\delta') L(\delta)^{1/2} \partial_\delta D_T(\delta + \delta') d\delta' d\delta$. For $u = \varphi\chi$ the substitution $s = T\delta'$, $v = T\delta$ gives $L(\delta') = L_T + \log(1/s)$, $L(\delta) = L_T + \log(1/v)$, $\partial_\delta D_T(\delta + \delta') = T^2 k'(s+v)/\pi$, $d\delta' d\delta = ds dv/T^2$, and the double integral becomes $\frac{aC^2}{2} J_1(L_T)$, taken in the Abel sense and cut where the model ends, as in Lemma 7.11. Here $J_1(\infty) = -\frac{2}{\pi} \int_0^\infty \int_0^\infty k'(s+v) ds dv = \frac{2}{\pi} \int_0^\infty k = 1$, and the term of order $1/L$ vanishes for the reason given there: $k'(s+v)$ is symmetric in (s, v) and the logarithm of the ratio antisymmetric. Numerically $J_1(L) = 1.029, 1.016, 1.012, 1.008, 1.006, 1.004$ for $L = 3, \dots, 8$, about $1 + 0.27/L^2$ (`rh_sum_rule.py j1`). The ψ -part of u contributes $o(1)$: with $|\psi'(\delta')| \leq \varepsilon(\delta')\delta'^{-1}L^{-3/2}$, $\varepsilon \rightarrow 0$, the same substitution bounds it by $o(1)$ times $\int_0^\infty (L + \log(1/s))^{-3/2} ds/s = 2L^{-1/2}$, which balances the $L^{1/2}$ of the force. (ii) With the regular value, $a g_{\text{reg}} \int_0^\infty \ell'_T(\delta) d\delta = -a g_{\text{reg}} \ell_T(0^+) = -\frac{aC}{2} g_{\text{reg}} L_T^{-1/2} (1 + O(1/L_T))$, since $\ell_T(0^+) = \frac{C}{\pi} \int_0^\infty (L_T + \log(1/s))^{-1/2} \frac{\sin s}{s} ds + o(L_T^{-1/2})$. (iii) With the remainder, $a \int \ell'_T \rho = -a \int \ell_T \rho'$ (as $\rho(0^+) = 0$ and $\ell_T(\infty) = 0$), and $|\ell_T(\delta)| \leq CL_T^{-1/2} \min(1, 1/(T\delta))$ with $\rho' = o(\delta^{-1}L^{-3/2})$ gives $o(L_T^{-1})$.

Collecting, $B_T^+ = \frac{aC^2}{2} J_1(L_T) - \frac{aC}{2} g_{\text{reg}} L_T^{-1/2} (1 + O(1/L_T)) + o(1)$, so $B_T \rightarrow aC^2$; the rest is Proposition 7.15, which gives $\mathcal{D} = 2B_\infty$, and Proposition 5.3, which gives $-a\lambda' = \mathcal{D}$. The regular value is $g_{\text{reg}}(a) = 2P \sinh(a/2)$ by Lemma 7.12. $\square$

The rate is visible in the data of Computation 7.17: on the prime-free critical point at $a = 0.5$ ($C = 1.0615$, $\beta = -1.469$ fitted on the edge nodes, against $C = 1.0626$ and $\beta = -1.400$ of Proposition 8.4 below, the fitted β absorbing the $O(L^{-2})$ correction over the fit's window; $g_{\text{reg}} = 0.0679$), $aC^2 - B_T = 0.040, 0.014, 0.0135, 0.0125$ at 3, 5, 10, 20 horizons, where the leading term $aC g_{\text{reg}} L_T^{-1/2}$ is 0.027 to 0.019; at these heights L_T is only 1.8 to 3.7 and the expansion is not yet quantitative, but the sign and the order are those of the law. In Fourier terms the same term is the polar cross term of the edge, $-\frac{1}{\pi} \int_T^\infty 4aP \sinh(a/2) |A_0| \sin \theta(t) dt / (t\tilde{\sigma}^{1/2})$ with $\sin \theta \approx -\pi/(4 \log t)$, and $C = |A_0|$.

For the prime-free form the edge law with its two derivatives, and with its constants, follows from the representation.

Proposition 8.4 (the edge law from the representation). *Assume (H_∞) , and let f be the minimizer of $W_\infty - 2P^2$ at a support a with $\lambda = \lambda_\infty(a) > \Psi_\infty(0)$, A_0 its edge amplitude (Corollary 7.24). Then f satisfies the hypothesis of Proposition 7.8 with*

$$C = |A_0|, \quad \beta = -\gamma - \log(2\pi a) - \lambda,$$

γ Euler's constant, and with $\psi^{(k)}(\delta) = O(\delta^{-k} L(\delta)^{-5/2})$ for $k \leq 2$. Equivalently, with $\sigma_\infty = \Psi_\infty - \lambda$,

$$f(a - \delta) = |A_0| (\sigma_\infty(1/\delta) - \gamma)^{-1/2} (1 + O(L^{-2})) :$$

the profile of the edge is the inverse square root of the symbol at frequency $1/\delta$, shifted by Euler's constant.

Proof. The model. For $s > 0$ and χ smooth, 1 on $[0, \delta_0/2]$ and 0 beyond $\delta_0 < 1$, put $\Phi_s(t) = \int_0^{\delta_0} (\log(1/\delta))^{-s} \chi(\delta) e^{-it\delta} d\delta$. From the Mellin representation $(\log(1/\delta))^{-s} = \Gamma(s)^{-1} \int_0^\infty x^{s-1} \delta^x dx$ and $\int_0^\infty \delta^x e^{-it\delta} d\delta = \Gamma(1+x)(it)^{-1-x} = \frac{-i}{t} \Gamma(1+x) e^{-x\Lambda}$ with $\Lambda = \log t + i\pi/2$ ($t > 0$), the Mellin-transform method for Fourier integrals with a logarithmic endpoint singularity [13, Ch. 4] gives, as $t \rightarrow +\infty$,

$$\Phi_s(t) = \frac{-i}{t} \Lambda^{-s} \sum_{k \leq N} \frac{\Gamma(s+k)}{\Gamma(s) k!} \Gamma^{(k)}(1) \Lambda^{-k} + O(t^{-1} |\Lambda|^{-s-N}),$$

an expansion that may be differentiated in t term by term. For $s = \frac{1}{2}$, with $\Gamma'(1) = -\gamma$ and $\Gamma''(1) = \gamma^2 + \pi^2/6$,

$$\Phi_{1/2}(t) = \frac{-i}{t} \Lambda^{-1/2} \left[1 - \frac{\gamma}{2\Lambda} + \frac{3(\gamma^2 + \pi^2/6)}{8\Lambda^2} + O(\Lambda^{-3}) \right] = \frac{-i}{t} (\Lambda + \gamma)^{-1/2} \left[1 + \frac{\pi^2}{16\Lambda^2} + O(\Lambda^{-3}) \right],$$

and with $M = \log t + \gamma$ one has $(\Lambda + \gamma)^{-1/2} = (M + i\pi/2)^{-1/2} = M^{-1/2} e^{-i\pi/(4M)} (1 - \frac{\pi^2}{16M^2} + O(M^{-3}))$, so that

$$\Phi_{1/2}(t) = \frac{-i}{t} M^{-1/2} e^{-i\pi/(4M)} (1 + O(M^{-3})) :$$

the real corrections of order M^{-2} cancel. Scaling δ by $e^{\beta'}$, the transform of $C(\log(e^{\beta'}/\delta))^{-1/2} \chi(\delta)$ is $\frac{-iC}{t} M_{\beta'}^{-1/2} e^{-i\pi/(4M_{\beta'})} (1 + O(M^{-3}))$ with $M_{\beta'} = \log t + \beta' + \gamma$.

The factor. By Stirling's formula and the rational factor, $\tilde{\sigma}(t) = \ell(t)(1 + O(t^{-2}))$ with $\ell(t) = \log(t/2\pi) - \lambda$; and substituting $\tau = tx$ in (10), $\theta(t) = \frac{1}{\pi} \int_0^\infty \frac{u(tx) - u(t)}{1-x^2} dx$ with $u = \log \tilde{\sigma}$. For $tx \geq \tau_0$, $u(tx) - u(t) = \log(1 + \log x/\ell(t)) + O(t^{-2})$; expanding in powers of $\log x/\ell$, with $\int_0^\infty \frac{(\log x)^k}{1-x^2} dx = 0$ for even k (by $x \mapsto 1/x$) and $= -\pi^2/4, -\pi^4/8$ for $k = 1, 3$, while the range $x < \tau_0/t$ and the tails $|\log x| > \ell$ contribute $O(\log \log t/t)$,

$$\theta(t) = -\frac{\pi}{4\ell} - \frac{\pi^3}{24\ell^3} + O(\ell^{-5}) + O\left(\frac{\log \log t}{t}\right),$$

and the same representation differentiated under the integral gives $\theta^{(j)}$, $j \leq 3$, with the expansion differentiated term by term. Hence, differentially,

$$\frac{A_0}{t\sigma_-(t)} = \frac{-i|A_0|}{t} \ell^{-1/2} e^{-i\pi/(4\ell)} \left(1 + O(\ell^{-3}) + O\left(\frac{\log \log t}{t}\right) \right),$$

where $A_0 = -i|A_0|$ for a minimizer positive near the right edge.

Matching. Choose β' with $\beta' + \gamma = -\log 2\pi - \lambda$, so that $M_{\beta'} = \ell$: the transform $\hat{\varphi}$ of the model $\varphi(\delta) = |A_0|(\log(a/\delta) + \beta)^{-1/2} \chi(\delta)$, $\beta = \beta' - \log a = -\gamma - \log(2\pi a) - \lambda$, agrees with $A_0/(t\sigma_-)$ to relative order ℓ^{-3} . By Corollary 7.24,

$$D(t) := \tilde{F}(t) - \hat{\varphi}(t) = \frac{A(t) - A_0}{t\sigma_-(t)} + \left[\frac{A_0}{t\sigma_-(t)} - \hat{\varphi}(t) \right] + \frac{e^{-2ita}\Phi_-(t)}{\tilde{\sigma}(t)}.$$

(i) $A(t) - A_0 = O(\log t/t)$ with $(A - A_0)^{(j)} = O(\log t/t^{1+j})$, $j \leq 3$: its pieces in the proof of Corollary 7.24 are a rational function, t times the Riesz projection of a smooth function with derivatives $O(t^{-2-j})$, and t times the transform of a C^∞ function on $u > 0$ with all derivatives integrable, and in each case the derivative of $tP_\pm[\cdot]$ minus its limit loses one power of t (for the projection, by the Hilbert-transform estimate of that proof applied to the derivatives); so the first piece is $O(t^{-2} \log t)$ with derivatives $O(t^{-2-j} \log t)$. (ii) The bracket is $O(t^{-1} \ell^{-7/2})$ with derivatives $O(t^{-1-j} \ell^{-7/2})$, $j \leq 3$. (iii) The third piece is the transform of $\text{far} * (1/\tilde{\sigma})$ with far supported in $u \leq -2a$, and the kernel of $1/\tilde{\sigma}$ is C^∞ and exponentially decaying away from the origin ($\tilde{\sigma}$ is zero-free in a strip around the line, as in Lemma 7.22), so its inverse transform is C^∞ on $(-2a + \varepsilon, 0]$.

The remainder. For $D \in C^3$ with $D^{(j)}(t) = O(t^{-1-j} \Lambda(t))$, $j \leq 3$, Λ slowly varying, the inverse transform ψ satisfies $\psi^{(k)}(\delta) = O(\delta^{-k} \Lambda(1/\delta))$ for $k = 1, 2$: split $\int (it)^k D e^{it\delta} dt$ at $|t| = 1/\delta$, bound the inner part by $\int_{1/\delta}^{1/\delta} t^{k-1} \Lambda$, and integrate the outer part by parts three times, the boundary terms being $(t^k D)^{(m)}(1/\delta) \delta^{-m-1}$, $m \leq 2$, and the integral $\delta^{-3} \int_{1/\delta}^\infty |(t^k D)^{(m)}|$. Applied to (ii) with

$\Lambda = \ell^{-7/2}$, to (i) with $\Lambda = t^{-1} \log t$ (where the same splitting gives $O(\delta^{1-k} \log^2(1/\delta))$), and with (iii) smooth, the inverse transform $\psi = f(a - \delta) - \varphi(\delta)$ of D satisfies $\psi^{(k)} = O(\delta^{-k} L^{-7/2}) + O(\delta^{1-k} \log^2(1/\delta))$ for $k = 1, 2$. Since $f(a^-) = 0$ by Theorem 3.2 and $\varphi(0^+) = 0$, $\psi(\delta) = \int_0^\delta \psi' = O(L(\delta)^{-5/2})$. Finally f is C^∞ inside the window, as $(K_\infty - \lambda)f = 2Ps$ there and the symbol is elliptic at infinity. This is the hypothesis of Proposition 7.8 with the stated C , β and remainder; the second form of the statement is $\log(a/\delta) + \beta = \sigma_\infty(1/\delta) - \gamma + O(\delta^2)$. $\square$

Corollary 8.5 (the boundary law and the tail law for the prime-free form). *Assume (H_∞) . For the minimizer of $W_\infty - 2P^2$ at a support a with $\lambda_\infty(a) > \Psi_\infty(0)$, at every point of differentiability of λ_∞ ,*

$$\lambda'_\infty(a) = -2|A_0|^2, \quad \lim_{T \rightarrow \infty} T \cdot \frac{1}{2\pi} \int_{|t| > T} |F(t)|^2 (\Psi_\infty(t) - \lambda) dt = \frac{2|A_0|^2}{\pi} = -\frac{\lambda'_\infty(a)}{\pi} :$$

the analytic tail law of Computation 7.14 is a theorem, and Conjecture 7.19(i) holds with $|A|^2 = -\lambda'/2$.

Proof. Propositions 5.3, 7.15 and 8.3 hold for the prime-free form with the same proofs (K_∞ , Ψ_∞ and $W_\infty - 2P^2$ in place of K , Ψ and Q ; the entries play no role). Proposition 8.4 supplies the hypothesis of Proposition 8.3 with $C = |A_0|$, which gives the first identity, and the tail energy is $2|A_0|^2/(\pi T)(1 + o(1))$ by Corollary 7.24, as computed after it. $\square$

Computation 8.6 (the constants of the edge). `rh_edge_constants.py`, log in `data/`. On the prime-free edge-FEM critical points at $a = 0.3, 0.4, 0.5, 0.6$ ($\lambda = 0.2226, -0.0780, -0.3222, -0.5359$; $\lambda' = -3.429, -2.666, -2.258, -2.039$ from the dilation engine) the ratio of the profile $f(a - \delta)$ to the prediction $(-\lambda'/2)^{1/2} (\log(a/\delta) + \beta)^{-1/2}$ of Proposition 8.4 and Corollary 8.5, with $\beta = -\gamma - \log(2\pi a) - \lambda = -1.434, -1.421, -1.400, -1.368$ and nothing fitted, is 1.001–1.003 at every node between $\delta = 10^{-10}$ and 10^{-5} at all four supports; it rises to 1.01 at $\delta = 10^{-3}$, where L is only 6, and to 1.004 at the last nodes near 10^{-12} . On the mesh refined below 10^{-6} (Computation 7.20) the ratio at $a = 0.5$ is 1.0003, 1.0005, 1.0009 at $\delta = 10^{-10}, 10^{-8}, 10^{-6}$, so the residual is the discretization. The amplitude C of Computation 3.4, fitted there, is $(-\lambda'/2)^{1/2}$, and β is the symbol's: $\log(a/\delta) + \beta = \sigma_\infty(1/\delta) - \gamma$.

The entries inside the window are a finite perturbation of this structure, and for one entry everything goes through with the same constants.

Proposition 8.7 (the minimizer with an entry inside the window). *Assume (H_∞) . Let f be the minimizer at a support a with exactly one entry n inside the window and $a < d < 2a$, $d = \log n$ (that is, $a_2 < a < a_3$, $n = 2$), $c = \Lambda(n)n^{-1/2}$, $\lambda > \Psi_\infty(0)$. Let $\tilde{\sigma}_\infty$ be $\Psi_\infty - \lambda$ with its real zero divided out and k the kernel of $1/\tilde{\sigma}_\infty$, and put $L = (-a, a - d)$, $M = (a - d, d - a)$, $R = (d - a, a)$, $f_X = f\mathbf{1}_X$. Then*

$$f(a+u) = c(k * f_R(a + \cdot + d))(u) + r(u), \quad r \in C^1 \text{ near } u = -d, \quad r''(u) = O(|u+d|^{-1}(\log 1/|u+d|)^{-5/2}), \quad (18)$$

the echo of the right edge at $a - d$; and, with $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - c^2/\tilde{\sigma}_\infty$, $\tilde{\sigma}_{\text{eff}}$ its positive part after dividing out the finitely many real zeros and $\sigma_{\text{eff}, \pm}$ its factors (Lemma 7.21),

$$\tilde{F} = \frac{P_- [\hat{H}'_- / \sigma_{\text{eff}, +}]}{\sigma_{\text{eff}, -}} \quad \text{on } \mathbb{R}, \quad (19)$$

where $\hat{H}'_-$ is in $H^1(\mathbb{R})$, analytic in the lower half-plane, and is the transform of a source made of the polar step and a smooth part near the edge, and of sources supported at distance at least $\delta_0 = \min(d, 2a - d)$ to the left of it.

Proof. On the window $Kf = \lambda f + 2Ps$ and $K = K_\infty + K_P$ with $K_P f = -\sum_m c_m(f(\cdot - \log m) + f(\cdot + \log m))$; the shifts by $\log m > 2a$ move the window off itself, and for $m = n$, $f(y - d) = f_L(y - d)$ on R , $f(y + d) = f_R(y + d)$ on L , both vanish on M . Hence on the whole line

$$K_\infty f = \lambda f + 2Ps \mathbf{1}_{[-a, a]} + c f_L(\cdot - d) + c f_R(\cdot + d) + g_\infty \mathbf{1}_{|y| > a}, \quad g_\infty = K_\infty f \text{ outside,}$$

and after transforming and multiplying by e^{-ita} , $\sigma_\infty \tilde{F} = \tilde{H} + \tilde{G}_+$ with $\tilde{G}_+$ the archimedean force to the right of a and $\tilde{H}$ the transform of the “-” sources: the polar term, the archimedean force to the left of $-a$, and the two copies, supported in $u \in (d - 2a, 0)$ and $u \in (-2a, -d)$ respectively ($u = y - a$). All sources are in $L^1 \cap L^2$ with $uh \in L^2$, so Proposition 7.23 applies verbatim: $\tilde{F} = P_-[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-}$ with $\hat{H}_-$ the sources corrected at the real zero.

The echo. The copy on L has transform $c e^{-itd} \tilde{F}_R$, $\tilde{F}_R = e^{-ita} \widehat{f_R}$, and its contribution to $\tilde{F}$ is $P_-[c e^{-itd} \tilde{F}_R/\sigma_{\infty,+}]/\sigma_{\infty,-} = c e^{-itd} \tilde{F}_R/\tilde{\sigma}_\infty - P_+[c e^{-itd} \tilde{F}_R/\sigma_{\infty,+}]/\sigma_{\infty,-}$. The first term is the transform of $ck * f_R(a + \cdot + d)$; the second is the transform of a function smooth on $u < 0$: by Lemma 7.22 the kernel of $1/\sigma_{\infty,+}$ is smooth on $[d, \infty)$, so $P_+[\cdot]$ is the transform of a smooth function on $u > 0$, and $1/\sigma_{\infty,-}$ then spreads it to $u < 0$ through its kernel evaluated away from the origin. The remaining terms of $\tilde{F}$ are the polar step, the far force (smooth near $-d$), the copy on R (supported in $u \in (d - 2a, 0)$, whose contribution is smooth away from its support’s endpoints), and the rational corrections of Proposition 7.23, which are weighted primitives of the sources, $-ie^{\pm it\lambda u} \int_{-\infty}^u h(u') e^{\mp it\lambda u'} du'$: one derivative smoother than the sources, with the same singular support and a step at the edge. All are smooth near $u = -d$ except the correction of the copy on L , which is C^1 there with the second derivative of the primitive of the edge profile, $O(|u + d|^{-1} L^{-3/2})$, and $O(|u + d|^{-1} L^{-5/2})$ after the convolution with k , which gains a logarithm; this is (18), with k smooth away from the origin and $f_R(a + \cdot + d)$ supported in $u \leq -d$.

The return. The copy on R near the edge is $c f_L(a + u - d) = c f(a + (u - d))$ for $u \in (d - 2a, 0)$, and by (18) at $u - d$, $f(a + (u - d)) = c(k * f_R(a + \cdot))(u) + r(u - d)$; so the copy on R equals $c^2(k * f_R(a + \cdot))(u) \mathbf{1}_{u < 0}$ plus $cr(u - d) \mathbf{1}_{u < 0}$, a function C^1 up to the edge with a step there (value $cr(-d)$) and a transform $O(t^{-2}(\log t)^{-3/2})$ beyond the step, plus, at its left end $u = d - 2a$, the copy of the left edge. Its transform is therefore $c^2 P_-[\tilde{F}_R/\tilde{\sigma}_\infty]$ plus a step plus a far-type source. Since $\tilde{F}_R = \tilde{F} - \tilde{F}_L - \tilde{F}_M$ and f_L, f_M are supported at distance at least $2a - d$ from the edge, $c^2 P_-[\tilde{F}_R/\tilde{\sigma}_\infty] = c^2 P_-[\tilde{F}/\tilde{\sigma}_\infty] - c^2 P_-[(\tilde{F}_L + \tilde{F}_M)/\tilde{\sigma}_\infty]$, the second term the transform of a source smooth near the edge. Write $c^2 P_-[\tilde{F}/\tilde{\sigma}_\infty] = c^2 \tilde{F}/\tilde{\sigma}_\infty - c^2 P_+[\tilde{F}/\tilde{\sigma}_\infty]$ and move the first term to the left of $\tilde{\sigma}_\infty \tilde{F} = \hat{H}_- + \hat{G}_+$: this is $\sigma_{\text{eff}} \tilde{F} = \hat{H}'_- + \hat{G}'_+$ with $\hat{H}'_- = \hat{H}_- - c^2 P_-[\tilde{F}/\tilde{\sigma}_\infty]$ and $\hat{G}'_+ = \hat{G}_+ + c^2 P_+[\tilde{F}/\tilde{\sigma}_\infty]$, a “-” and a “+” function with the properties required in the proof of Proposition 7.23 ($1/\tilde{\sigma}_\infty$ has an integrable kernel, so $f * k \in L^1$). The symbol σ_{eff} is real, even, real-analytic, $\sim \log t$, with finitely many real zeros; dividing them out and applying the Liouville argument of that proof gives (19), and $\hat{H}'_-$ has the stated structure. $\square$

Corollary 8.8 (the edge, the echo and the tail law with an entry inside the window). *Under the hypotheses of Proposition 8.7: (i) the edge of the transform has the form of Corollary 7.24 with σ_{eff} in place of σ , $A(t) \rightarrow A_0$ purely imaginary; (ii) f satisfies the hypothesis of Proposition 7.8 at the edges, with the same constants as in Proposition 8.4, $C = |A_0|$ and $\beta = -\gamma - \log(2\pi a) - \lambda$, and is smooth on the window except at the edges and at the two echo points $\pm(a - d)$, where it is continuous with the singularity of $k * (\text{edge profile})$; (iii) the echo form (12) holds, with $\rho = c/\tilde{\sigma}_\infty$;*

(iv) $\lambda'(a) = -2|A_0|^2$ at every point of differentiability (Conjecture 8.1 at these supports); (v) under RH, $\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\lambda'(a)/\pi$: Conjecture 7.7 holds for $a_2 < a < a_3$.

Proof. (i) is the proof of Corollary 7.24 applied to (19): the near sources of $\hat{H}'_-$ are the steps $-2iP \sinh(a/2) - icf(a-d)$ and a smooth part, the far sources sit at distance at least δ_0 , and Lemma 7.22 holds for σ_{eff} (a smooth symbol, $\sigma_{\text{eff}} = \ell - c^2/\ell + O(t^{-2})$, zero-free in a strip once its real zeros are divided out), so the far feedback is again $-ir(0^+) + O(1/t)$ and the symmetry argument gives A_0 purely imaginary. (ii) The proof of Proposition 8.4 goes through with σ_{eff} : $\theta_{\text{eff}} = \theta_\infty + \frac{1}{2}\mathcal{H}[\log(1-c^2/\ell^2)] = -\pi/(4\ell) + O(\ell^{-3})$, since the added term is even and $O(\ell^{-2})$ with the expansion of that proof giving $\pi c^2/(2\ell^3)$ at the first order, and $\tilde{\sigma}_{\text{eff}}^{-1/2} = \ell^{-1/2}(1+c^2/(2\ell^2)+O(\ell^{-3}))$; so the edge term matches the model $|A_0|(\log(a/\delta)+\beta)^{-1/2}$ with the same β up to a relative $c^2/(2\ell^2)$, whose inverse transform is $\frac{1}{2}c^2|A_0|(L+\beta)^{-5/2}(1+O(1/L))$ by the Mellin expansion with $s = \frac{5}{2}$, inside the remainder class; the far-type sources contribute smooth functions near the edge. The regularity off the edges is interior regularity on L, M, R (the forcing $cf_L(\cdot-d)$ on R is smooth inside R), and the structure at $\pm(a-d)$ is (18) and its mirror. (iii) The copy on L contributes $ce^{-itd}\tilde{F}_R/\tilde{\sigma}_\infty$ to $\tilde{F}$ up to smooth feedback, by the proof above, and $\tilde{F}_R = A(t)/(t\sigma_{\text{eff},-}) + \dots$ by (i); with $F = 2i\text{Im}(F_R + F_e)$ this is (12). (iv) Lemma 8.2 holds with the copies, which are continuous across the edge with the cusp of the echo, $O(L^{-3/2})$ with derivative $O(\delta^{-1}L^{-5/2})$, inside the remainder class; and in the proof of Proposition 8.3 the singularities of $(xf)'$ at the echo points, at distance at least δ_0 from the edge, leak like the left edge does. (v) Proposition 7.8 applies with one change in its Step 1: f_0 now carries the two echo cusps, whose transforms are $O(t^{-1}L^{-3/2})$ with the phases $e^{\pm it(a-d)}$; their squares are $O(t^{-2}L^{-3})$, and their cross terms with the edges oscillate at the frequencies d and $2a-d$, both nonzero, which Step 2 handles exactly as it handles $\cos(2ta)$. Hence $\lim T \sum_{|\gamma| > T} |F(\gamma)|^2 = 2|A_0|^2/\pi = -\lambda'/\pi$ by (iv). $\square$

Computation 8.9 (the constants of the edge with the prime inside the window). `rh_edge_constants.py`, log in `data/`. On the minimizers with primes at $a = 0.4, 0.45, 0.5$ ($\lambda = 0.01471, 0.002405, 0.000194$; $\lambda' = -0.4659, -0.1013, -0.01124$ from the dilation engine; the entry 2 inside the window, $0.347 < a < 0.549$) the ratio of $f(a-\delta)$ to $(-\lambda'/2)^{1/2}(\log(a/\delta)+\beta)^{-1/2}$ with $\beta = -\gamma - \log(2\pi a) - \lambda$ ($-1.514, -1.619, -1.722$), nothing fitted, is 1.000–1.002 at every node between $\delta = 10^{-12}$ and 10^{-6} at the three supports, 1.004–1.008 at 10^{-4} , and 1.03–1.05 at 10^{-3} , where L is 6 and the relative correction $c^2/(2L^2)$ of the symbol, 0.3%, is joined by the corrections of the asymptotic regime. The alternative $\beta - 2c$, which a first-order return of the echo would produce, is off by 2–4% over the whole range. So the prime inside the window changes neither constant: the edge amplitude is $(-\lambda'/2)^{1/2}$ and the shift is the symbol's.

Computation 8.10 (the boundary law). C from the edge FEM (two-stage mesh to $10^{-12}a$, fit of $1/f^2$ against $\log(a/\delta)$ on $14 \leq t \leq 20$, residual $\leq 5 \times 10^{-5}$, mesh bias 10^{-3} to 10^{-2} in C) and λ'_{48} from `rh_dilation.py` (Hellmann–Feynman, ball arithmetic, 48 modes; the truncation error of λ'_{48} is below 10^{-4} at these supports by the K -stability of Computation 4.4); data in `boundary_law.csv` and `data/`, reproduced by `rh_hadamard.py`:

a	primes	λ (FEM)	C	β	λ'_{48}	$r = -\lambda'/2C^2$
0.3	no	+0.2225566	1.30891	-1.49	-3.42881	1.0007
0.4	no	-0.0780502	1.15425	-1.48	-2.66640	1.0007
0.5	no	-0.3222425	1.06222	-1.45	-2.25815	1.0007
0.6	no	-0.5358860	1.00932	-1.42	-2.03880	1.0007
0.4	yes	1.470510×10^{-2}	0.48266	-1.57	-0.465947	1.0001
0.45	yes	2.404578×10^{-3}	0.22494	-1.67	-0.101283	1.0009
0.5	yes	1.936801×10^{-4}	0.07493	-1.78	-1.123556×10^{-2}	1.0005
0.549	yes	1.491447×10^{-5}	0.02412	-1.87	-1.143529×10^{-3}	0.983

Seven cases give $r = 1$ to better than 10^{-3} , across supports where the prime-free floor is positive and negative and with the primes present; the prime-free offset 7×10^{-4} is the mesh bias of C on that family, and the full-form offsets ($1, 9, 5 \times 10^{-4}$) are within the stated bias. The last case measures C at $a_3 = 0.54931$ and λ' at 0.549, inside the sawtooth where Φ' changes by about 9% per 0.001 (Computation 4.4); its 1.7% is within the combined error of the amplitude (10^{-2} in C , hence 2% in C^2) and the offset in a . The evidence is confined to $a \leq 0.6$ (heights below 22), far from the regime of Conjecture 7.1; extending it needs a multiprecision edge engine.

An exactly solvable case: for one half of the Dirichlet form of L_Δ on $(-a, a)$, the scaling identity $L_\Delta(u(\cdot/a)) = (L_\Delta u)(\cdot/a) - 2 \log a u(\cdot/a)$ gives $\lambda(a) = \lambda(1) - \log a$, so the law predicts $C(a)^2 = 1/(2a)$. The prime-free form differs from this by the bounded corrections (the polar term and $\operatorname{Re} \psi - \log |t|$); its amplitudes give $2aC^2 = 1.028, 1.066, 1.129, 1.223$ at $a = 0.3, \dots, 0.6$ and its derivatives $-a\lambda' = 1.029, 1.067, 1.129, 1.223$, both near 1 and equal to each other, which is the content of the table restated, not an independent test.

Remark 8.11 (consequence of Conjecture 8.1). $\Phi'(a) = 2C(a)^2/\lambda(a)$, and Conjecture 7.1 becomes

$$C(a)^2 \leq \frac{c}{2} T^*(a) \lambda(a) \quad (a \geq a_0) :$$

the edge amplitude of the minimizer is bounded by its energy times the horizon. This is the form in which the conjecture is a statement about one function at one place. The relay of Computation 4.4 reads: as a approaches an entry from below, λ falls toward the deleted form's failure point while C stays of the same order, and $\Phi' = 2C^2/\lambda$ rises until the entry rescues the floor. Under the Riemann Hypothesis, $C(a)$ is the boundary trace of the band-limited function that best hides from the zeros above the horizon.

8.1 The echo tree

Proposition 8.7 treats one entry and one echo. At a general support the entries copy the window onto itself in several ways, the echoes have echoes, and a copy of an echo can return to the edge along several chains. The structure is combinatorial, and it is finite as long as the walks that the entries generate inside the window are confined.

Let $D_a = \{\log m : \Lambda(m) \neq 0, \log m < 2a\}$ be the entries inside the window and $c_d = \Lambda(m)m^{-1/2}$ for $d = \log m$. A *walk* is a finite sequence of points of $[-a, a]$ whose consecutive points differ by $\pm d$ with $d \in D_a$. The *echo set* S is the set of points visited by walks starting at a or $-a$, and $S^\circ = S \cap (-a, a)$; $S = -S$. The walks are *confined* at a , (F_a) , if the set of points visited by walks starting anywhere in $[-a, a]$ is finite. Write $S = \{s_0 = -a < s_1 < \dots < s_N = a\}$, $I_j = (s_{j-1}, s_j)$, $f_j = f \mathbf{1}_{I_j}$. On S° let $C_{pq} = c_{|p-q|}$ if $|p-q| \in D_a$ and 0 otherwise (the weighted adjacency of the echo points), $b_p = c_{a-p}$ if $a-p \in D_a$ and 0 otherwise, $\bar{b}_p = b_{-p}$ (the couplings to the two edges),

and let C_a be the matrix on $S^\circ \cup \{a\}$ obtained from C by adjoining the row and column b ; C and C_a are real symmetric.

Lemma 8.12 (the entries move the pieces onto pieces). *For $d \in D_a$ and each j , each of $I_j + d$ and $I_j - d$ is either disjoint from $(-a, a)$ or equal to some $I_{j'}$.*

Proof. If $s_{j-1} + d \geq a$, the interval $I_j + d$ lies to the right of the window. Otherwise $s_{j-1} + d \in [-a, a)$ is one step from a point of S , so it belongs to S ; if also $s_j + d \leq a$, then $s_j + d \in S$ for the same reason, and no point q of S lies strictly between them, since $q - d$ would be a point of S in I_j ; so $I_j + d = I_{j'}$. The remaining case $s_{j-1} + d < a < s_j + d$ is impossible: then $a - d \in I_j$, while $a - d \in S$ because $d < 2a$. The same holds for $-d$. $\square$

Proposition 8.13 (the echo tree). *Assume (H_∞) . Let f be the minimizer at a support a that is not an entry, with $\lambda > \Psi_\infty(0)$ and the walks confined, (F_a) . Let $\tilde{\sigma}_\infty$ and k be as in Proposition 8.7, and for $p \in S$ let $\Phi_p(t) = \int \chi(v) f(p+v) e^{itv} dv$, with one fixed $\chi \in C_c^\infty$ equal to 1 near 0 and supported in $(-\delta_1, \delta_1)$, δ_1 half the least gap of S ; for $p = \pm a$ this is the transform of f near the edge, and $\Phi_{-a} = -\Phi_a$. Then:*

(i) $f \in C^\infty((-a, a) \setminus S^\circ)$.

(ii) For $p \in S^\circ$, near $v = 0$,

$$f(p+v) = \sum_{q \in S, |p-q| \in D_a} c_{|p-q|} (k * f(q+\cdot))(v) + r_p(v), \quad (20)$$

where $f(q+\cdot)$ is the window function shifted by q and r_p is C^1 near 0 with $r_p''(v) = O(|v|^{-1}(\log 1/|v|)^{-5/2})$. In Fourier form,

$$(\tilde{\sigma}_\infty(t)I - C)\Phi_{S^\circ}(t) = b\Phi_a(t) + \bar{b}\Phi_{-a}(t) + \Psi(t), \quad \Psi(t) = O(t^{-2}(\log t)^{-3/2}). \quad (21)$$

(iii) Let $R = b^T(\tilde{\sigma}_\infty I - C)^{-1}b$ and $\bar{R} = \bar{b}^T(\tilde{\sigma}_\infty I - C)^{-1}b$, the returns of the right edge to itself and to the left edge through the tree, and

$$\sigma_{\text{eff}} = \tilde{\sigma}_\infty - R = \frac{\det(\tilde{\sigma}_\infty I - C_a)}{\det(\tilde{\sigma}_\infty I - C)}, \quad (22)$$

a rational function of $\tilde{\sigma}_\infty(t)$ whose real zeros and poles are the heights at which $\tilde{\sigma}_\infty$ equals an eigenvalue of C_a , respectively of C : finitely many, none above the height where $\tilde{\sigma}_\infty > \|C_a\|$, and $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2/\tilde{\sigma}_\infty + O(\tilde{\sigma}_\infty^{-2})$. With $\tilde{\sigma}_{\text{eff}} > 0$ the symbol with its real zeros and poles divided out by rational factors and $\sigma_{\text{eff}, \pm}$ its factors (Lemma 7.21),

$$\tilde{F} = \frac{P_-[\hat{H}''/\sigma_{\text{eff},+}]}{\sigma_{\text{eff},-}} \quad \text{on } \mathbb{R}, \quad (23)$$

where $\hat{H}'' \in H^1(\mathbb{R})$ is analytic in the lower half-plane and is the transform of a source made of steps at the edge, a part smooth near the edge, a part whose transform is $O(t^{-2}(\log t)^{-3/2})$, the mirror chains $P_-[\bar{R}\Phi_{-a}]$, and sources supported at distance at least $\delta_0 = a - \max S^\circ$ to the left of the edge.

Proof. The equation on the line. As in the proof of Proposition 8.7, with $g_\infty = K_\infty f$ outside the window,

$$(K_\infty - \lambda)f = 2Ps \mathbf{1}_{[-a,a]} + \sum_{d \in D_a} c_d \sum_j (f_j(\cdot - d) + f_j(\cdot + d)) \mathbf{1}_{(-a,a)} + g_\infty \mathbf{1}_{|y|>a},$$

and by Lemma 8.12 every copy $f_j(\cdot \mp d) \mathbf{1}_{(-a,a)}$ is either zero or the whole shifted piece, landing on some $I_{j'}$. All sources are in $L^1 \cap L^2$ with $yh \in L^2$, so Proposition 7.23 applies at the right edge with $\hat{H}_-$ the sources to the left of a corrected at $\pm t\lambda$: $\tilde{F} = P_-[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-} = \hat{H}_-/\tilde{\sigma}_\infty - P_+[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-}$, and on the window the second term is the transform of a smooth function (Lemma 7.22: $P_+[\cdot]$ is the transform of a function on $u > 0$, and the kernel of $1/\sigma_{\infty,-}$ is smooth away from the origin). So on $(-a, a)$, $f = k * \hat{h}_- + (\text{smooth})$, where $\hat{h}_-$ is the source with its corrections, which are weighted primitives of the sources, $-ie^{\pm it\lambda u} \int_{-\infty}^u h(u') e^{\mp it\lambda u'} du'$ on $u < 0$: one derivative smoother than h , with the same singular support, and a step at the edge. Since k is smooth away from the origin, $k * g$ is smooth near a point where g is smooth, for $g \in L^1$.

Regularity, (i). Let $p \notin S$ and let $O(p)$ be the set of points visited by walks from p : finite by (F_a) and disjoint from S , since a walk from p to $\pm a$ reversed is a walk from $\pm a$ to p . The sources near $q \in O(p)$ are the polar term, smooth inside the window, the copies landing on intervals adjacent to q , which are $c_d f(q \mp d + \cdot)$ with $q \mp d \in O(p)$ (a copy landing on an interval not adjacent to q is supported away from q), the far force, and the corrections; so the vector $(f(q + \cdot))_{q \in O(p)}$ solves near 0 the system $(K_\infty - \lambda)f(q + \cdot) = \sum_{q' \in O(p), |q-q'| \in D_a} c_{|q-q'|} (1 + \mathcal{M})f(q' + \cdot) + (\text{smooth})$, with $\mathcal{M}$ the primitive of the corrections, an operator of order -1 . The commutator of K_∞ with a cutoff has a bounded kernel ($J_\Gamma(s)(\chi(y) - \chi(y - s))$ is bounded near $s = 0$), so the usual bootstrap, one derivative per step on a shrinking neighbourhood and for all $q \in O(p)$ at once, gives $f \in C^\infty$ near p .

The local relations, (ii). For $p \in S^\circ$ the same list of sources near p holds with $q \in S$: the copies landing on the two intervals adjacent to p are $c_{|p-q|} f(q + \cdot)$ for $q = p \mp d \in S$ (one-sided when $q = \pm a$), the other copies, the polar term, the far force and the far corrections are smooth near p , and the corrections of the adjacent copies are C^1 at p with the second derivative of the primitive of an echo, $O(|v|^{-1} L^{-3/2})$ before and $O(|v|^{-1} L^{-5/2})$ after the convolution with k , which gains a logarithm; this is (20). Transforming (20) against χ : $\chi \cdot (k * f(q + \cdot))$ has the transform $\Phi_q/\tilde{\sigma}_\infty$ up to the transform of a smooth compactly supported function (the error of the localization, $\chi(k * (\chi f(q + \cdot))) - k * (\chi f(q + \cdot))$, is smooth since k is smooth away from the origin), and χr_p has a transform $O(t^{-2}(\log t)^{-3/2})$; multiplying by $\tilde{\sigma}_\infty$ gives (21), where Ψ collects $\tilde{\sigma}_\infty \widehat{\chi r_p}$ and the rapidly decaying errors.

The edge, (iii). The sources landing on $I_N = (s_{N-1}, a)$ are $c_d f_{j(d)}(\cdot - d)$ with $I_{j(d)} = I_N - d$, that is, $c_d f(a - d + u)$ for $u \in (s_{N-1} - a, 0)$, for each $d \in D_a$; their transforms are $c_d P_-[\Phi_{a-d}]$ up to far sources (the cut at $s_{N-1} - a$, at distance δ_0) and smooth ones. By (21), wherever $\tilde{\sigma}_\infty(t)I - C$ is invertible, $\sum_d c_d \Phi_{a-d} = b^T \Phi_{S^\circ} = R \Phi_a + \bar{R} \Phi_{-a} + b^T (\tilde{\sigma}_\infty I - C)^{-1} \Psi$. Here $\Phi_a = \tilde{F} + (\text{far source})$ by the choice of χ , and $P_-[R\tilde{F}] = R\tilde{F} - P_+[R\tilde{F}]$; moving $R\tilde{F}$ to the left of $\tilde{\sigma}_\infty \tilde{F} = \hat{H}_- + \hat{G}_+$ gives $\sigma_{\text{eff}} \tilde{F} = \hat{H}'_- + \hat{G}'_+$ with $\hat{H}'_- = \hat{H}_- - R\tilde{F} + P_-[R \Phi_a + \bar{R} \Phi_{-a} + b^T (\tilde{\sigma}_\infty I - C)^{-1} \Psi] - (\text{the copies on } I_N)$ and $\hat{G}'_+ = \hat{G}_+ + P_+[R\tilde{F}]$. The identity (22) is the Schur complement of the entry (a, a) of $\tilde{\sigma}_\infty I - C_a$, and since $\tilde{\sigma}_\infty$ is real-analytic with $\tilde{\sigma}_\infty \rightarrow \infty$, its level sets at the eigenvalues of C and C_a are finite. At a real zero $\pm t_j$ of σ_{eff} the right-hand side vanishes, and the rational correction of Proposition 7.23 applies; at a real pole $\pm s_k$ (an eigenvalue of C reached by $\tilde{\sigma}_\infty$: a standing wave of the interior tree) both $\sigma_{\text{eff}} \tilde{F}$ and $\hat{H}'_-$ have a simple pole, with equal residues, and dividing by the rational factor $(t^2 + s_k^2)/(t^2 - s_k^2)$ of $\tilde{\sigma}_{\text{eff}}/\sigma_{\text{eff}}$ removes them; the Liouville function of the proof of Proposition 7.23 is then entire off $\pm s_k$ with at most simple poles there, hence a rational function of the form

$\sum_k c_k^\pm/(z \mp s_k)$ by its growth, and the $c_k^\pm$ are fixed by the regularity of $\tilde{F}$ at $\pm s_k$; these terms are transforms of one-sided exponentials, smooth near the edge up to a step. Collecting, $\hat{H}_-''$ is $\hat{H}_-'$ with these rational corrections, and its sources are as stated: the polar step and the steps $c_d r_{a-d}(0^-)$ of the smooth parts of the copies, smooth parts, the corrections of order $t^{-2}(\log t)^{-3/2}$, the mirror chains, and the far sources. Lemma 7.21 applies to $\tilde{\sigma}_{\text{eff}}$, which is smooth, positive, and $\tilde{\sigma}_\infty(1 + O(\tilde{\sigma}_\infty^{-2}))$ at infinity with bounded log-derivative, and Lemma 7.22 to its factors, since σ_{eff} is analytic in the strip of $\tilde{\sigma}_\infty$ with $R = O(1/\log |t|)$ there; the Liouville argument of Proposition 7.23 then gives (23). $\square$

Corollary 8.14 (the edge, the echoes and the tail law at a support with a finite tree). *Under the hypotheses of Proposition 8.13, with $\rho = (\tilde{\sigma}_\infty I - C)^{-1}b$ (the echo amplitudes) and $E(t) = 1 + \sum_{p \in S^\circ} \rho_p(t) e^{-it(a-p)}$:*

- (i) *the edge of the transform has the form of Corollary 7.24 with σ_{eff} in place of σ , $A(t) \rightarrow A_0$ purely imaginary;*
- (ii) *f satisfies the edge law with derivatives with the constants of Proposition 8.4, $C = |A_0|$ and $\beta = -\gamma - \log(2\pi a) - \lambda$, and is smooth on the window except at the edges and at the points of S° , where it has the cusps of (20);*
- (iii) *the echo form:*

$$S(t) = -\frac{|A_0| |E(t)| \cos(ta + \theta_{\text{eff}}(t) + \arg E(t) - \kappa(t)/t)}{t \tilde{\sigma}_{\text{eff}}(t)^{1/2}} + O(t^{-2}(\log t)^{-3/2}),$$

with θ_{eff} the Hilbert phase of $\tilde{\sigma}_{\text{eff}}$ and $\kappa(t) = O(\log t)$ real; the chains of the tree are the terms of the Neumann series $\rho = \sum_{\ell \geq 0} C^\ell b / \tilde{\sigma}_\infty^{\ell+1}$, convergent where $\tilde{\sigma}_\infty > \|C\|$, and R is a finite continued fraction along a chain;

- (iv) *the sharp form is an identity of the tree: with $\sigma_a = \tilde{\sigma}_\infty - \sum_{d \in D_a} 2c_d \cos(\tau d)$ and t frozen in $\tilde{\sigma}_\infty$ and ρ ,*

$$\langle \sigma_a(\tau) |E(\tau)|^2 \rangle_\tau = \sigma_{\text{eff}}(t), \quad \langle \sigma_a(\tau) e^{2i\tau a} E(\tau)^2 \rangle_\tau = -\bar{R}(t)$$

(means over $\tau \in \mathbb{R}$); hence the local envelope of $t^2(\Psi - \lambda)|F(t)|^2$ over the oscillations is $2|A_0|^2(1 - \bar{R}/\tilde{\sigma}_{\text{eff}} + O(\theta_{\text{eff}}^2))(1 + o(1))$, and $\bar{R} \equiv 0$ when the trees of the two edges are disjoint;

- (v) *$\lambda'(a) = -2|A_0|^2$ at every point of differentiability, and under RH, $\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\lambda'(a)/\pi$: Conjecture 7.7 holds at a ;*
- (vi) *under RH, when the trees of the two edges are disjoint, the sampling constant of Corollary 7.30 is*

$$M_\chi(T) - 1 = \frac{R + \tilde{\sigma}_\infty(|\rho|^2 - 1)}{\sigma_{\text{eff}}} \Big|_{t=T} (1 + o(1)) = \frac{2|b|^2}{\tilde{\sigma}_\infty(T)^2} (1 + O(1/\tilde{\sigma}_\infty)), \quad |\rho|^2 := 1 + \sum_p \rho_p^2,$$

$$\text{and } |b|^2 = \sum_{\log m < 2a} \Lambda(m)^2/m.$$

Proof. (i), (ii), (v) are the proofs of Corollary 8.8(i), (ii), (iv), (v) with σ_{eff} of (22): the near sources are steps and smooth parts, the far feedback is $-ir(0^+) + O(1/t)$, the corrections of order $t^{-2}(\log t)^{-3/2}$ contribute $O(t^{-1}(\log t)^{-3/2})$ to $A(t)$, inside $A_0 + O(\log t/t)$; $\tilde{\sigma}_{\text{eff}}^{-1/2} = \ell^{-1/2}(1 + |b|^2/(2\ell^2) + O(\ell^{-3}))$ and $\theta_{\text{eff}} = -\pi/(4\ell) + O(\ell^{-3})$, so the profile matches the model with the same β up to a relative $|b|^2/(2\ell^2)$, inside the remainder class; the cusps at the finitely many points of

S° , at distance at least δ_0 from the edges, have transforms $O(t^{-1}L^{-3/2})$ with the phases e^{itp} , their cross terms with the edges oscillate at the nonzero frequencies $a \mp p$, and Step 2 of Proposition 7.8 handles them as it handles $\cos(2ta)$. (iii) By (21), $\Phi_{S^\circ} = \rho \Phi_a + \bar{\rho} \Phi_{-a} + O(t^{-2}(\log t)^{-3/2})$ with $\bar{\rho} = (\tilde{\sigma}_\infty I - C)^{-1} \bar{b}$, $\bar{\rho}_p = \rho_{-p}$ by the symmetry $S = -S$, and $F = \sum_{p \in S} e^{itp} \Phi_p$ up to the transform of a smooth function; so $F = e^{ita} \Phi_a E(t) + e^{-ita} \Phi_{-a} \overline{E(t)} + \dots = 2i \operatorname{Im}(e^{ita} \Phi_a E) + \dots$, and $\Phi_a = \tilde{F} + (\text{far}) = A(t)/(t\sigma_{\text{eff},-}) + \dots$ by (i). The Neumann series and the continued fraction are the expansion of the resolvent and the Schur complement of a path. (iv) Write $E = \sum_{p \in S^\circ \cup \{a\}} e_p e^{-i\tau(a-p)}$ with $e_a = 1$, $e_p = \rho_p$. The mean of $|E|^2$ is $|\rho|^2$ and the mean of $2\cos(\tau d)|E|^2$ is $\sum_{|p-q|=d} e_p e_q$, so the mean of $\sigma_a |E|^2$ is $\tilde{\sigma}_\infty |\rho|^2 - e^T C_a e = \tilde{\sigma}_\infty |\rho|^2 - 2b^T \rho - \rho^T C \rho$, and $\rho^T C \rho = \tilde{\sigma}_\infty (|\rho|^2 - 1) - b^T \rho$ by $(\tilde{\sigma}_\infty - C)\rho = b$; the sum is $\tilde{\sigma}_\infty - b^T \rho = \sigma_{\text{eff}}$. For the mirror term, $e^{2i\tau a} E^2 = \sum_{p,q} e_p e_q e^{i\tau(p+q)}$, whose mean against σ_a is $\tilde{\sigma}_\infty \sum_{p \in S^\circ} \rho_p \rho_{-p} - \sum_p e_p (C_S e)_{-p}$, with C_S the adjacency on S and $e_{-a} = 0$; by $(\tilde{\sigma}_\infty - C)\rho = b$, $(C_S e)_{-p} = \tilde{\sigma}_\infty \rho_{-p}$ for $p \in S^\circ$ and $(C_S e)_{-a} = \bar{b}^T \rho = \bar{R}$, which leaves $-\bar{R}$. With $|F|^2 = 2|\Phi_a|^2 |E|^2 - 2\operatorname{Re}(e^{2ita} \Phi_a^2 E^2)$ and $\Phi_a^2 = -|A_0|^2 e^{2i\theta_{\text{eff}} - 2i\kappa/t} / (t^2 \tilde{\sigma}_{\text{eff}}) (1 + o(1))$, the envelope follows; $\bar{R} \equiv 0$ if no chain of C joins the two trees. (vi) In (17), $\frac{1}{2\pi} \int |F|^2 \chi_T \Psi_\infty = (\tilde{\sigma}_\infty + \lambda) |\rho|^2 I_T (1 + o(1))$ and $2 \sum_n \Lambda(n) n^{-1/2} \tilde{h}_T(\log n) = e^T C_a e I_T (1 + o(1))$ with $I_T = \frac{1}{2\pi} \int 2|\Phi_a|^2 \chi_T$, the oscillating terms averaging out and the mirror cross terms having no mean when the trees are disjoint; so $M_\chi - 1 = e^T C_a e / (\tilde{\sigma}_\infty |\rho|^2 - e^T C_a e) (1 + o(1))$, and $e^T C_a e = R + \tilde{\sigma}_\infty (|\rho|^2 - 1)$ by the computation in (iv). $\square$

Computation 8.15 (the echo set). `rh_echo_tree.py`, log in `data/`. The echo set S and the orbits of walks from 200 interior points, with a cap of 5000 points, for $0.35 \leq a \leq 0.84$:

a	entries	$ S^\circ $	chain	orbit	$\ C\ $	$\ C_a\ $	trees of $\pm a$
0.35–0.53	2	2	1	2	0	0.490	disjoint
0.55–0.68	2, 3	6	2	4	0.490	0.901	disjoint
0.70–0.75	2, 3, 4	12	3	7	0.901	1.172	disjoint
0.775–0.80	2, 3, 4	22	6	12	1.172	1.219	disjoint
0.825	2, 3, 4, 5	148	30	75	1.471	1.471	disjoint
0.8375	2, 3, 4, 5	426	49				
0.8422	2, 3, 4, 5	3434	482				
0.8430	2, 3, 4, 5	8854	1572				

(chain: the longest shortest chain from the edge; orbit: the largest orbit of a walk from an interior point.) The walks are confined up to $a_\infty = 0.8430$ (between 0.84297 and 0.84336) and not beyond: with the entries 2, 3, 4, 5 in a window of length $2a > 1.686$ the net moves $\log 3 - \log 2$, $\log 4 - \log 3$, $\log 5 - \log 4$ can be composed indefinitely without leaving the window, and the echo set becomes dense. Below a_∞ the tree is finite; its size grows from 2 points for one entry, through 6 (one echo of an echo, $a - \log 3 + \log 2$) for $a_3 < a < a_4$, to thousands near a_∞ . At every support of the table the trees of the two edges are disjoint, so $\bar{R} \equiv 0$ and the sharp form of Corollary 8.14(iv) holds without the mirror correction. The resonances: $\min \tilde{\sigma}_\infty = 1.003$ (at $t = 6.6$, just above $t_\lambda = 6.29$, for $\lambda \approx 0$), so σ_{eff} has no real zeros or poles for $a < a_4$ ($\|C_a\| = 0.90$), real zeros but no poles for $0.70 < a < 0.77$, and both beyond; all of them lie below the horizon ($\tilde{\sigma}_\infty(T^*) = 2a - \lambda$). Where the walks deconfine, the symbol of the window is negative on sets reaching far above the horizon (the last sign change at $33T^*$ for $a = 1.0$ and beyond $150T^*$ for $a = 1.25$ in Computation 7.20): the infinite tree has a band, and the finite tree does not.

Computation 8.16 (the tree at $a = 0.6$: the second echo). Edge FEM at $a = 0.6$ with the entries 2 and 3 (486 nodes, the mesh of Computation 8.10, $\lambda = 5.9459 \times 10^{-7}$; $\lambda_{64} = 5.9583 \times 10^{-7}$ and $\lambda'_{64} = -4.1545 \times 10^{-5}$ from the dilation engine; $T^* = 20.86$). The tree: $S^\circ = \{\pm 0.0931, \pm 0.1945, \pm 0.4986\}$,

the right edge's chain $a \rightarrow a - \log 3 = -0.4986 \rightarrow a - \log 3 + \log 2 = 0.1945$ and its branch $a \rightarrow a - \log 2 = -0.0931$, $b = (c_3, c_2, 0)$ on $(-0.4986, -0.0931, 0.1945)$, C with the single link c_2 between -0.4986 and 0.1945 ; so $R = c_2^2/\tilde{\sigma}_\infty + c_3^2/(\tilde{\sigma}_\infty - c_2^2/\tilde{\sigma}_\infty)$ and the second echo has the amplitude $\rho_{0.1945} = c_2 c_3/(\tilde{\sigma}_\infty^2 - c_2^2)$: 0.040, 0.026, 0.018 at $5, 10, 20 T^*$. (i) *The constants.* With nothing fitted, $f/[(-\lambda'/2)^{1/2}(L + \beta)^{-1/2}]$, $\beta = -1.904$, is 1.0003–1.003 on $10^{-12} < \delta < 10^{-6}$ and 1.013 at 10^{-4} : the two entries change neither constant. (ii) *The nulls and the pointwise form*, 463 nulls on $3.5\text{--}120 T^*$: with the tree's phase $\theta_{\text{eff}} + \arg E$ the residual $r + \vartheta - \kappa/t$ has rms 0.0114 on $5\text{--}80 T^*$ (300 nulls), against 0.0162 for the first generation alone ($\rho_p = b_p/\tilde{\sigma}_\infty$, $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2/\tilde{\sigma}_\infty$), 0.036 for the window's symbol and 0.41 for no shift; at $5, 7, 10, 15, 20, 30 T^*$ the tree leaves $-0.002, -0.006, -0.006, -0.004, -0.003, -0.0002$ where the first generation leaves $-0.039, +0.019, +0.010, +0.017, -0.011, +0.008$. The relative L^2 distance of S from the tree form over one period, $|A| = (-\lambda'/2)^{1/2}$, is 0.038, 0.0088, 0.0034 at $5, 10, 20 T^*$, against 0.053, 0.028, 0.020 for the first generation and 0.20, 0.097, 0.055 for the window's symbol: the first generation's error is the missing second echo, 0.04, 0.026, 0.018. Beyond $35 T^*$ the interior mesh ($1/\text{spacing} \approx 34 T^*$) limits S and all forms degrade alike (0.011 at $50 T^*$, 0.11 at 100). (iii) *The sampling defect.* The identity (17) holds at every height to 10^{-5} of the zero sum. With $\lambda = 6 \times 10^{-7}$ the quartic cutoff is too soft, its weight $(t/T)^4$ below the horizon letting through the density integral there (which is 10^{-3} at $\kappa = 1$ against a zero sum of 6×10^{-7}), so the cutoff $1 - e^{-(t/T)^8}$ is used (Proposition 7.29 with $m = 4$). Its M_χ against the tree's prediction of Corollary 8.14(vi) averaged over the tail, $1 + \langle e^T C_a e / \tilde{\sigma}_\infty \rangle_T / \langle \sigma_{\text{eff}} / \tilde{\sigma}_\infty \rangle_T$, is 1.110/1.106, 1.076/1.079, 1.050/1.057 at $\kappa = 6, 10, 20$ (first order $2|b|^2 \langle \tilde{\sigma}_\infty^{-2} \rangle_T$: 1.098, 1.075, 1.054; the value at T instead of the average, 1.160, 1.113, 1.077, is off by a factor 0.7 in $M_\chi - 1$); at $\kappa = e$ the cutoff still lets through the density integral below the horizon, and $M_\chi = 1.68$ against 1.19. The sharp constant $M(\kappa)$ of Computation 7.14 is 1.14, 1.12, 1.064, 1.049 at $\kappa = e, 6, 10, 20$.

8.2 Beyond the finite tree: the window's symbol and the band

Above a_∞ the echo set is dense and the tree has no finite matrix. The representation that survives is the one with the symbol of the window, which has only two sources, the polar term and the force outside the far edge; the copies are inside the symbol. It is exact at every support, and what it does not give is the asymptotics of its edge amplitude below the band of the echo lattice, where the symbol changes sign.

Proposition 8.17 (the window's symbol at every support). *Assume (H_∞) . Let f be the minimizer at a support a that is not an entry, $\sigma_a = \Psi_a - \lambda$ with $\Psi_a = \Psi_\infty - \sum_{d \in D_a} 2c_d \cos(td)$ the symbol of the window, $\pm t_1, \dots, \pm t_N$ its real zeros, finitely many, all below the last sign change t_s , $\tilde{\sigma}_a = \sigma_a \prod_j (t^2 + t_j^2)/(t^2 - t_j^2) > 0$, and $g_{\text{out}} = K_a f$ outside the window, K_a the operator with symbol Ψ_a . Then:*

- (i) *Lemma 7.21 holds for $\tilde{\sigma}_a$ with the bounds on θ and θ' replaced by $O(\log \log(e + |t|))$: $\sigma_{a,\pm} = \tilde{\sigma}_a^{1/2} e^{\pm i\theta_a}$ are the boundary values of the outer functions, $c \leq |\sigma_{a,\pm}(z)| \leq C(\log(e + |z|))^{1/2}$, and $1/\sigma_{a,\pm}$ are bounded on the line with derivative $O(\log \log(e + |t|))$.*
- (ii) *With $\tilde{H}$ the transform of $e^{-ita}(2Ps\mathbf{1}_{[-a,a]} + g_{\text{out}}\mathbf{1}_{y < -a})$ and $\hat{H}_-$ its correction at the $\pm t_j$ as in Proposition 7.23,*

$$\tilde{F} = \frac{P_-[\hat{H}_-/\sigma_{a,+}]}{\sigma_{a,-}} \quad \text{on } \mathbb{R}, \quad (24)$$

and, with Φ_- as in Corollary 7.24 built from g_{out} , $\tilde{F} = A_a(t)/(t\sigma_{a,-}) + e^{-2ita}\Phi_-/\tilde{\sigma}_a$ with A_a given by the formula of (15) with $\sigma_{a,\pm}$, $A_a(-t) = \overline{A_a(t)}$.

(iii) At a support with a finite tree, (24) and (23) represent the same $\tilde{F}$: the echoes of the right edge inside the window are the first terms of the expansion of $1/\sigma_{a,-}$ in the lattice frequencies, and the terms of that expansion which would place an echo beyond the far edge are cancelled by the far source.

Proof. σ_a is real-analytic, even, and tends to $+\infty$, so its real zeros are finitely many and lie below t_s ; $\tilde{\sigma}_a$ is even, positive, C^2 , bounded below, and $O(\log t)$. (i) In the proof of Lemma 7.21 the hypothesis $(\log \tilde{\sigma})' \in L^2$ serves only to bound $\theta' = \frac{1}{2}\mathcal{H}[(\log \tilde{\sigma})']$. Write $\log \tilde{\sigma}_a = \log \tilde{\sigma}_\infty + u$ with $\tilde{\sigma}_\infty$ the archimedean symbol with its real zero divided out; u is C^2 with u', u'' bounded ($\tilde{\sigma}_a \geq c_1$ and σ'_a, σ''_a are bounded), and $u = O(\log \log(e + |t|))$ since $c_1 \leq \tilde{\sigma}_a \leq C\tilde{\sigma}_\infty$. The Hilbert transform of u' is bounded on $|\tau - t| < 1$ by $\sup |u''|$, and on $|\tau - t| > 1$ an integration by parts gives $u(t \pm 1) + \int_{|\tau-t|>1} u(\tau)(t - \tau)^{-2} d\tau = O(\log \log(e + |t|))$; the same integration by parts in (10) bounds $\mathcal{H}[u]$. The modulus bounds of the lemma use only $\log \tilde{\sigma}_a$, and hold. (ii) On the window $K_a f = \lambda f + 2Ps$, since the copies $f(\cdot \mp \log m)$ with $\log m > 2a$ vanish there; so on the line $K_a f = \lambda f \mathbf{1}_{[-a,a]} + 2Ps \mathbf{1}_{[-a,a]} + g_{\text{out}} \mathbf{1}_{|y|>a}$, where $g_{\text{out}} = K_\infty f + \sum_{d \in D_a} c_d (\text{copies landing outside})$ is in $L^1 \cap L^2$ with $yg_{\text{out}} \in L^2$, and $\sigma_a \tilde{F} = \tilde{H} + \tilde{G}_+$. The proof of Proposition 7.23 applies with $\sigma_{a,\pm}$: the rational correction at the N pairs of zeros, the H^1 property of $\hat{H}_-/\sigma_{a,+}$ (from (i)), and the Liouville argument with the bounds of (i). The edge form is (15) read with $\sigma_{a,\pm}$, and the symmetry is as before. (iii) Both representations are exact for the same function, and $\tilde{F}$ is supported in $(-2a, 0)$; for one entry, with $\tilde{\sigma}_\infty$ frozen at s the factorization $s - 2c \cos \tau = \sigma_{\text{fp}}(1 - \rho_{\text{fp}} e^{i\tau})(1 - \rho_{\text{fp}} e^{-i\tau})$, $\sigma_{\text{fp}} + c^2/\sigma_{\text{fp}} = s$, $\rho_{\text{fp}} = c/\sigma_{\text{fp}}$, is exact for $s \geq 2c$, and $1/\sigma_{a,-} = \sigma_{\text{fp}}^{-1/2} \sum_k \rho_{\text{fp}}^k e^{-ik\tau}$ is the infinite chain of echoes at $a - kd$, of which the window holds $k = 0, 1$. $\square$

What (ii) does not give is $A_a(t) \rightarrow A_0$. In the proof of Corollary 7.24 the far feedback $-tP_+[e^{-2ita}\Phi_-/\sigma_+]$ was $-ir(0^+) + O(1/t)$ because the kernel of $1/\sigma_+$ is smooth beyond the origin (Lemma 7.22); the kernel of $1/\sigma_{a,+}$ is singular at the points $\sum_d n_d d$ of the lattice, because $\log \sigma_{a,+}$ contains all the harmonics of the entries, and the feedback of the far edge through it is the family of its echoes landing beyond the right edge, an almost-periodic function of t whose amplitudes decay only like inverse powers of $\log t$. In the finite tree this feedback is $\tilde{R}\Phi_{-a}$, and it is absent when the trees of the two edges are disjoint; in the window's symbol it is always present, together with the rational corrections of the N zeros, which are $O(\sum_j t_j^2/t^2)$ and, with N zeros up to t_s , are not small below $t \approx N^{1/2}t_s$. So (24) is exact everywhere and asymptotic only far above the last sign change; the finite tree is what makes the asymptotics start at the horizon.

Lemma 8.18 (the periodic limit of the window's symbol). *Let ν be the law of*

$$X = \sum_{p \leq e^{2a}} \sum_{k \geq 1, p^k < e^{2a}} 2c_p^k \cos(k\tau_p)$$

with independent uniform phases τ_p , one per prime, so that $\text{supp } \nu = [-X_{\max}, X_{\max}]$ with $X_{\max} = \sum_{d \in D_a} 2c_d$. For every continuous φ and every s ,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \varphi\left(s - \sum_{d \in D_a} 2c_d \cos(td)\right) dt = \int \varphi(s - x) d\nu(x);$$

in particular the mean of $\log |s - \sum_d 2c_d \cos(td)|$ over t is the logarithmic potential $U_\nu(s) = \int \log |s - x| d\nu(x)$, a continuous function of s . Put $G(s) = e^{U_\nu(s)}$, the geometric mean of the window's symbol at the frozen height s . Then

$$G(s) = s - \sum_{d \in D_a} \frac{c_d^2}{s} + O(s^{-3}) \quad (s \rightarrow \infty),$$

equal to the effective symbol of the finite tree to this order; and for one entry

$$G(s) = \frac{1}{2}(s + \sqrt{s^2 - 4c^2}) \quad (s \geq 2c), \quad G(s) = c \quad (|s| \leq 2c),$$

the former being the fixed point of the infinite continued fraction $\sigma \mapsto s - c^2/\sigma$.

Proof. The numbers $\log p$ for distinct primes are linearly independent over $\mathbb{Q}$, so $(t \log p \bmod 2\pi)_p$ is equidistributed on the torus (Weyl), the prime powers riding on the phases of their primes; this gives the limit. The law of $\sum_k 2c_{p^k} \cos(k\tau_p)$ for one prime is the push-forward of $d\tau/2\pi$ by a trigonometric polynomial, absolutely continuous with inverse-square-root singularities, and ν is the convolution of these laws over the primes; so $\log |\cdot|$ is locally integrable against ν and U_ν is continuous. For $s > X_{\max}$, $U_\nu(s) = \log s + \mathbb{E} \log(1 - X/s) = \log s - \mathbb{E}[X^2]/(2s^2) + O(s^{-3})$ with $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = \sum_d 2c_d^2$. For one entry ν is the arcsine law on $[-2c, 2c]$, whose logarithmic potential is $\log \frac{1}{2}(|s| + \sqrt{s^2 - 4c^2})$ outside and $\log c$ inside the support. $\square$

The band of the echo lattice is $\{t : \tilde{\sigma}_\infty(t) < X_{\max}\}$, the heights at which the periodic limit of the window's symbol changes sign: up to $47T^*$ at $a = 1$, $412T^*$ at $a = 1.25$ and $2 \times 10^4 T^*$ at $a = 1.5$, while the actual last sign changes are at 33, 259 and $397T^*$ (Computation 8.19). Below a_∞ the band is empty or lies below the horizon ($X_{\max} = 2.25$ at $a = 0.6$, $\tilde{\sigma}_\infty(T^*) = 2a$), which is the other face of the confinement: a finite tree has resonances, not a band. In the band the geometric mean of the symbol is $G(\tilde{\sigma}_\infty)$, finite and smooth, so the outer factor of the window's symbol has modulus $G^{1/2}$ on average, and $|F|^2$ cannot follow $1/|\sigma_a|$ where $\sigma_a < 0$ without the envelope of $t^2\sigma_a|F|^2$ losing twice the measure of the negative set. Whether the minimizer loses it, or hides from the negative set as it hides from the zeros, is what Computation 8.19 measures on the K -mode minimizers of Computation 7.6, the only ones computed in the band.

Computation 8.19 (the sharp form in the band). `rh_band.py`, logs `band_a*.log` in `data/`. On the K -mode minimizers of Computation 7.6 at $a = 1.0, 1.25, 1.5$ ($K = 80, 140, 230$; $\lambda = 1.5 \times 10^{-26}$, 3.4×10^{-51} , 9.2×10^{-93} ; λ' from the dilation engine; F in closed form through spherical Bessel functions at 80 digits, reproducing the zero sums of Computation 7.6 to all digits), the local envelope of $t^2(\Psi - \lambda)|F|^2$ over the window of width $4T^*$ centred at κT^* , divided by $-\lambda'$; the fraction of the window on which the window's symbol σ_a is negative; and the part of the envelope carried by the negative set:

κ	$a = 1.0$			$a = 1.25$			$a = 1.5$		
	envelope	$\sigma_a < 0$	on it	envelope	$\sigma_a < 0$	on it	envelope	$\sigma_a < 0$	on it
3	1.708	0.049	-0.172	2.151	0.060	-0.217	3.156	0.084	-1.40
5	1.071	0.026	-0.018	1.216	0.045	-0.016	1.084	0.061	-0.025
7	1.046	0.019	-0.004	1.049	0.031	-0.008	1.027	0.048	-0.015
10	1.001	0.011	-0.000	1.012	0.019	-0.004	1.010	0.034	-0.009
15	0.987	0.008	-0.001	1.013	0.012	-0.001	0.998	0.026	-0.005
20	0.968	0.005	-0.000	0.996	0.010	-0.001	0.997	0.018	-0.003
30	0.989	0	0	0.997	0.007	-0.000	0.976	0.012	-0.002
50	1.041	0	0	0.990	0.003	-0.000	0.994	0.008	-0.001

From five horizons on the envelope is $-\lambda'$ within 2–8%, and within 1–3% from seven, at all three supports, inside the region where the symbol of the window changes sign (its last sign change is at 33, 259 and $397T^*$). The negative set occupies 2.6–6% of the heights at $5T^*$ and carries -0.02 of the envelope; the correlation coefficient of $t^2|F|^2$ with $1/|\sigma_a|$ over the window is below 0.04 in modulus at every height from $5T^*$ on. So $|F|^2$ is not large where the symbol is small: the minimizer does not follow the window's symbol pointwise, the amplitude $A_a(t)$ of Proposition 8.17 compensating

its sign changes, and the law is carried by the mean, as in the finite tree. The envelope against the archimedean symbol alone, $\langle t^2(\Psi_\infty - \lambda)|F|^2 \rangle / (-\lambda')$, is 1.24, 1.35, 1.18 at $5T^*$ and 1.05, 1.02, 1.01 at $10T^*$: the prime sum at the entries, the sampling defect of Proposition 7.29, is 15–30% of the tail at five horizons and a few per cent at ten. Below five horizons the windows reach the mass of the floor (medians 2.4 – $2.7T^*$ in Computation 7.6) and the envelope is not a tail quantity. The K -mode minimizers resolve the edge to $\delta \approx e^{-2}a/K^2$, heights of order 10^3T^* , and their tails were validated on the zeros in Computation 7.6.

The band that matters is not the periodic limit's. The walks are confined to the window even when the echo set is dense: from a point p only the steps $\pm d$ with $p \pm d \in [-a, a]$ are allowed, and from most points these are a small fraction of the $2|D_a|$ steps of the lattice. The echo graph is the quasi-periodic structure cut out of $\mathbb{Z}^{\pi(e^{2a})}$ by the window, and its adjacency is a far smaller operator than the periodic limit's.

Proposition 8.20 (the confined lattice). *Let a be any support that is not an entry, S its echo set, finite or not, and C the weighted adjacency on S° , $C_{pq} = c_{|p-q|}$ for $|p-q| \in D_a$. Let $D_{\max} = \max_{p \in S^\circ} \sum_{q \sim p} C_{pq}$ be the largest weighted degree. Then C is a bounded self-adjoint operator on $\ell^2(S^\circ)$ with*

$$\rho(C) = \|C\| \leq D_{\max} \leq \sum_{d \in D_a, d \leq a} 2c_d + \sum_{d \in D_a, a < d < 2a} c_d < X_{\max} = \sum_{d \in D_a} 2c_d,$$

and $\rho(C)$ is the supremum of the top eigenvalues of the adjacencies of the finite subgraphs. In the other direction, let Σ be the full echo lattice of the window, the set of all points $\pm a + \sum_p m_p \log p \in [-a, a]$ ($m_p \in \mathbb{Z}$, p prime, $p < e^{2a}$), of which S is the component of $\pm a$ under the allowed steps, with the same adjacency; then

$$\rho(C_\Sigma) \geq \bar{\rho}(a) := \sum_{d \in D_a} 2c_d \left(1 - \frac{d}{2a}\right), \quad \bar{\rho}(a) = \frac{4e^a - 4a - 4}{a} (1 + o(1)) \quad (a \rightarrow \infty),$$

the mean weighted degree of the lattice. The formal echo expansion of Corollary 8.14(iii), $\rho = \sum_{\ell \geq 0} C^\ell b / \tilde{\sigma}_\infty^{\ell+1}$, converges in $\ell^2(S^\circ)$ at every height where $\tilde{\sigma}_\infty(t) > \rho(C)$, and the return $R = b^T(\tilde{\sigma}_\infty - C)^{-1}b$ and the effective symbol $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - R$ are real-analytic functions of t there, with $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2 / \tilde{\sigma}_\infty + O(\tilde{\sigma}_\infty^{-2})$. The band of the confined lattice is $\{t : \tilde{\sigma}_\infty(t) \leq \rho(C)\}$; it ends at $t_{\text{band}} = 2\pi e^{\rho(C)+\lambda}(1 + o(1))$.

Proof. A symmetric matrix with non-negative entries is bounded by its largest row sum (Schur's test), which is $D_{\max}$. A step $\pm d$ from p counts in the degree of p only if $p \pm d \in [-a, a]$; both directions are possible only if $d \leq a - |p| \leq a$, so an entry $d > a$ contributes at most c_d and an entry $d \leq a$ at most $2c_d$, which is the middle bound, and it is strictly below $X_{\max}$ because some entry lies in $(a, 2a)$ at every support $a \geq a_2$. For a bounded self-adjoint operator with non-negative entries the norm is the supremum of $\langle v, Cv \rangle$ over finitely supported unit vectors, which is the supremum of the top eigenvalues of the finite truncations, increasing with the truncation. For the lower bound take the points of Σ with exponent vectors in the cube $|m_p| \leq L$; by Weyl's equidistribution their projections $\pm a + \sum_p m_p \log p$ are uniformly distributed modulo any interval, so a fixed fraction of the $(2L+1)^\pi$ vectors (π the number of primes) lands in the window with projections uniformly distributed in $[-a, a]$, and all but $O(L^{\pi-1})$ of their allowed steps stay in the cube; the Rayleigh quotient of the normalized indicator of this set is the mean over it of the weighted degree, and the weighted degree of a point at p is $\sum_d c_d(\mathbf{1}_{p+d \leq a} + \mathbf{1}_{p-d \geq -a})$, whose mean over p uniform in $[-a, a]$ is $\sum_d 2c_d(1 - d/2a)$. The asymptotic form is the prime number theorem

in the form $\sum_{\log m < X} \Lambda(m) m^{-1/2} (X - \log m) = 4e^{X/2} - 2X - 4 + o(e^{X/2})$. The Neumann series converges in operator norm for $\tilde{\sigma}_\infty > \|C\|$, and the resolvent is analytic in $\tilde{\sigma}_\infty$ there, hence in t , with the expansion of its first terms. $\square$

What the proposition does not do is extend Proposition 8.13 to a dense echo set: the local relations (20) were proved with the other singular points at a positive distance, and the representation with the effective symbol needs them. Beyond a_∞ the exact statement is Proposition 8.17, the echo expansion is formal, and the band of the confined lattice is where it ceases to converge. It is nevertheless the quantity to compare with the onset: the tail law of the zeros was measured from about five horizons at $a = 1.0, 1.25, 1.5$ (Computation 7.6), where the periodic band reaches 47, 412 and 2×10^4 horizons, and Computation 8.21 finds the confined band ending at one to three.

Computation 8.21 (the band of the confined lattice). `rh_lattice_band.py`, `logs lattice_band*.log` in `data/`. The echo set by breadth-first search from $\pm a$ in the exponent-vector representation, truncated to the first N points by chain length, and the top eigenvalue ρ_N of the truncated adjacency by power iteration; $D_{\max}$ over the points found; the mean degree $\bar{\rho}(a)$ of Proposition 8.20; and the band tops in units of the horizon, $t_{\text{band}}/T^* = e^{\rho-2a}$:

a	entries	ρ_{500}	ρ_{2000}	ρ_{8000}	ρ_{40000}	$\bar{\rho}(a)$	$D_{\max}$	$X_{\max}$	confined t_{band}/T^*	periodic
0.8	3	1.219 (all 24 points)				1.05	1.47	2.94	0.68	4
0.9	4	1.659	1.659	1.659		1.41	2.19	4.38	0.87	13
1.0	5	1.920	1.936	1.941		1.75	2.93	5.85	0.94	47
1.1	7	2.275	2.293	2.298		2.15	3.54	7.08	1.10	131
1.25	8	2.663	2.773	2.862	2.912	2.80	4.26	8.52	≥ 1.5	412
1.5	12	3.209	3.446	3.622	3.855	4.09	6.51	13.0	≥ 2.4	2×10^4

The chains reached by 40000 points are 13 links long at $a = 1.25$ and 7 at $a = 1.5$, where ρ_N is still rising; the mean degree $\bar{\rho}$ is a lower bound for the lattice, which the truncations approach from below at $a = 1.5$ and exceed by 10% at $a \leq 1.25$, so $\rho(C)$ is within a few per cent of $\bar{\rho}$ at these supports: the confined lattice is nearly regular. Its spectral radius is a third of the periodic band top and two thirds of the largest degree. The band of the confined lattice ends at 0.9, 1.5 and about 3.0 horizons at $a = 1.0, 1.25, 1.5$ (the last from $\bar{\rho}$), against the periodic band top at 47, 412 and 2×10^4 horizons and the last sign change of the window's symbol at 33, 259 and 397. The onset of the tail law of the zeros is five horizons at the three supports (Computation 7.6: $T\vartheta/T_{\text{eff}} = 1.03, 1.04, 1.015$ at $5T^*$) and of the sharp form five to seven (Computation 8.19): two to five times the confined band top, as it is a few horizons above the horizon for the finite trees, whose band lies below it.

The prediction. The lower bound $\bar{\rho}(a) \sim 4e^a/a$ grows exponentially while $\tilde{\sigma}_\infty(\kappa T^*) = 2a + \log \kappa$ grows linearly, so the band of the confined lattice outruns the horizon: $e^{\bar{\rho}(a)-2a}$ is 3.0, 34 and 2041 at $a = 1.5, 2$ and 2.5 . Up to $a = 1.5$ the band ends within three horizons and the tail law begins at five; at $a = 2$ the band reaches some thirty-five horizons and at $a = 2.5$ two thousand. If the onset of the sharp form follows the band, as it does at every support computed, the effective height of the zeros that carry the tail will outrun the horizon from $a \approx 2$ on, the universality of the profiles $\vartheta(T/T^*)$ of Computation 7.6 will break there, and the route of Proposition 7.32 to Conjecture 7.1, which needs the sharp form at a fixed multiple κ of the horizon, will need κ growing like $e^{\bar{\rho}(a)-2a}$; if instead the onset stays at a few horizons, the sharp form holds inside the band, where the echo expansion diverges, and the mechanism that carries the law is not the convergence of the echoes. The floor's mass itself, at 1–3 horizons, lies inside the band at every support from $a = 1$ on. The decisive measurement is a minimizer at $a = 2$ ($T^* = 343$, about 500 modes at 1500 bits, or an edge FEM with the entries up to 53), and the quantity to read is the height at which $T\vartheta(T)$ becomes constant: a few horizons, or tens. No support computed so far decides it.

9 Relation to the literature, and acknowledgement

Suzuki [4] organizes the Weil form on $[-a, a]$ as a family of self-adjoint operators and conjectures that an operator with the zeros as spectrum is their limit as $a \rightarrow \infty$; Conjecture 7.1 is a differential inequality for the ground state of that family. Zhu [6] certifies both parity sectors at $a = 0.8$; his Landau–Widom decay law is fitted to the even-sector floor, calibrated on $L \leq 1.2$ and validated at $L = 1.4, \dots, 2.0$, and his exploratory 950-mode value at $L = 1.19$, between 10^{-48} and 10^{-46} , is for the full space of test functions, i.e. the even floor; the odd floors of Section 6 interpolate to $\lambda(1.19) \approx 3 \times 10^{-44}$, and the even floor is already three orders below the odd one at $a = 0.8$ (Theorem 1.3), so the two are consistent. Comparisons between his law and the odd Φ' are cross-sector and are labelled so. Connes–Consani [7] treat the range $a \leq \frac{1}{2} \log 2$ where only the archimedean place acts; the wall of Proposition 3.1 is that place acting alone at the edge for every a . The atlas [20] contains an exact-rational certificate of the odd floor to $a = 3/8$ with constant $1/200$, inside Zhu’s range, and the first application of Theorem 3.2 to this operator. The folds paper [19] computed the floors to $a = 1.1$, the deleted-form failure points and the symbol’s negative set, and described the relay as a sawtooth, which Computation 4.4 measures; the Möbius companion [21] treats the Green-energy side of the same program.

Computation 7.5 is the converse use of the explicit formula: the form was built from the primes and the archimedean symbol, and evaluating the minimizer on the actual zeros returns the floor and its derivative; with Computations 5.9 and the Fourier-side check the identity has now been measured from the matrices, from the symbol and from the zeros. An independent referee report on the first version of this paper identified the gaps corrected here: the identity had been asserted for the exact form on numerical grounds (now Theorem 5.2 for the K -mode form and Proposition 5.3 for the exact one), Conjecture 7.1 had been phrased in terms of Φ and so presupposed $\lambda > 0$, the abstract overstated several computations, the resolution threshold had been misattributed, the scaling identity carried a sign error, (5) lacked the constant $1 - \log 2$, and Computation 6.1 misdescribed its data; the report’s own recomputations of every table from the data files are gratefully acknowledged.

Reproducibility

All scripts and data are in [22]: `rh_weil_odd.py`, `rh_weil_arb.py` (python-flint 0.9.0), `rh_edge_fem.py` (numpy 2), `rh_soft_kink.py`, `rh_floor_grid.py`, `rh_arb_grid.py`, `rh_dilation.py` (python3 `rh_dilation.py` a `K` [`bits`] [`noprimes`]), `rh_hadamard.py`, `rh_fourier_check.py` (the Fourier-side check of Computation 5.9), `rh_bv_check.py`, `rh_zero_side.py` (Computation 7.5, with the zeros in `data/zeros_6000.txt`), `rh_tail_law.py` and `rh_alignment.py` (Computations 7.6 and 7.28), `rh_sum_rule.py` and `rh_cutoff_identity.py` (Computations 7.10, 7.17 and 7.18), `rh_wiener_hopf.py` (Computation 7.20), `rh_echo_tree.py` (Computation 8.15), `rh_sampling_defect.py` (Computation 7.31), `rh_band.py` (Computation 8.19), `rh_lattice_band.py` (Computation 8.21), and the data `floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`, `boundary_law.csv`, `relay_scan.csv` and the folder `data/` (edge-FEM outputs and dilation logs behind Computation 8.10).

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